

CurveMole User Manual

Version 1.2.0

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Updated for CurveMole 1.2.0

GPL-3.0-or-later

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CurveMole is a desktop-first, scriptable application for fitting one-dimensional scientific curves. The stable API baseline is CurveMole 1.2.0 Stable. This release edition documents the released scientific engine and application. The Markdown file is the authoritative source. The LaTeX source and PDF edition are generated automatically from it and must carry the same version number.

Stable edition for CurveMole 1.2.0. This release includes the workflow features documented in Appendices F/G and the scientific and data-integrity corrections summarized in Appendices I/J. Earlier executables can differ; keep the matching manual and automation references with your analysis.

Scientific corrections (release, 2026-11-02). The seven findings in the latest scientific audit are corrected in this edition: parameter scaling, nonlinear uncertainty derivatives, convolution alignment, compensated summation, profile confidence regions and saved execution provenance. Appendix J describes the changes and their limits. Use the documents attached to your exact release build; these corrections do not apply retroactively to earlier executables.

Useful links:

- CurveMole repository
- Latest release and desktop downloads
- Issue tracker
- Version DOI, Concept DOI, RRID and citation
- Quick Start
- Python automation audit
- YAML automation audit

1. About this manual

1.1 Intended audience

This manual is for researchers and technical users who need to fit spectra or other x-y data while retaining control over data preparation, model construction, constraints, uncertainty, and export. Typical data include:

- infrared and Raman spectra;
- powder diffraction patterns;
- kinetic traces;
- chromatographic profiles;
- temperature, pressure, or field-dependent signals;
- any other one-dimensional curve that can be represented by numeric x and y columns.

No programming knowledge is required for the graphical workflow. Separate chapters cover the command-line interface, YAML workflows, Python API, formula builder, and plugins.

1.2 Version policy for the manual

In a release edition, the title and version marker identify the CurveMole release being documented, rather than an independent document revision. A testing edition retains the stable version metadata and identifies its additional behavior in the testing notice and minimum-version labels.

Testing editions retain this stable baseline in the title. A testing notice and explicit minimum-version labels distinguish upcoming beta additions; the beta release assets identify the exact package build containing them.

The release automation performs four checks:

1. the application version and both manual version declarations must agree;
2. all local links in the Markdown source must resolve;
3. Pandoc must generate a standalone LaTeX document without warnings;
4. XeLaTeX must compile the generated source into a valid PDF.

The release assets use versioned names so that a downloaded manual can always be associated with the correct executable.

1.3 Terminology and notation

Menu commands are written as **File > Import data**. Buttons and field names are shown in **bold**. Commands, filenames, identifiers, and parameter paths use `monospace` text.

CurveMole uses the following residual convention in displays and exports:

$$r_i = y_i - f(x_i).$$

The optimizer minimizes the opposite sign, $f(x_i) - y_i$, after applying uncertainty and spectrum scaling. The sign has no effect on a least-squares minimum but matters when reading exported residuals.

1.4 Scientific responsibility

CurveMole can optimize a mathematically defined model, but it cannot decide whether that model is physically appropriate. A small residual or a high descriptive R^2 does not by itself establish uniqueness, chemical meaning, or predictive validity. In particular:

- do not add components solely to improve a scalar fit statistic;
- use bounds only when they have a defensible meaning;
- inspect correlations and active bounds;
- check whether residuals contain structure;
- report the exact CurveMole version, model, constraints, weighting, and uncertainty method used.

For this release edition, also read Appendix J before interpreting transformed errors, nonlinear links, convolution centres or profile intervals. Solver success and an assessment of **OK** do not detect every demonstrated numerical error.

2. Core concepts

CurveMole keeps the scientific engine independent from the graphical interface. The desktop application, command-line interface, YAML workflows, and Python API all operate on the same domain objects.

2.1 Project

A project is the complete analysis workspace. It can contain data, models, masks, transformations, fit results, custom formulas, history, and interface state. Projects are saved as `.fitproj` files. Restoring a workspace preserves its saved state. Replaying the original optimization also requires its initial model and numerical environment; these are not fully captured by every fit record (Appendix J.7).

2.2 Series and curves

A **series** groups related curves. A **curve** contains one x array and one y array, with optional `sigma_x`, `sigma_y`, or point weights. Importing multiple y columns from a common file produces multiple curves in one imported series.

Original numeric arrays are stored as immutable 64-bit floating-point values. Data Calculator operations create reversible transformations instead of overwriting the original arrays.

2.3 Model, component, and parameter

Each curve has its own model. A model is an ordered sequence of components such as a background and several peaks. Every component has:

- a function, for example `gaussian` or `linear`;
- a user-facing name;
- a composition operator;
- a set of parameters;
- an enabled or disabled state;
- optional metadata, such as polynomial order or spline x nodes.

A parameter has a current value, optional lower and upper bounds, a fixed state, and an optional expression link. Standard errors and confidence limits are populated after a successful fit when they are available.

2.4 Active, selected, and visible curves

These states are deliberately distinct:

- **Active curve:** the single curve whose model and handles are being edited.
- **Selected curves:** one or more curves chosen for multi-curve operations.
- **Visible curves:** curves drawn in Overlay or Waterfall display modes.

Activating a curve does not automatically select every visible curve. Before a fit, mask transfer, or multi-curve calculation, check the target shown in the relevant dialog.

2.5 Fit plan and result

A fit plan specifies the curves, fit mode, solver settings, per-spectrum weights, and whether numerical contributions should be equalized. A fit result records the optimized parameters, statistics, residual arrays, covariance, correlation, warnings, timestamp, and solver status.

3. Installation and first launch

3.1 Reference requirements

The current reference configuration is:

Item	Minimum or supported target
Processor	x86-64 or Apple Silicon on macOS; 4 CPU cores recommended
Memory	8 GB RAM
Storage	SSD recommended
Display	1920 x 1080 recommended
Windows	Windows 10 22H2 legacy compatibility or Windows 11, 64-bit
macOS	macOS 13 Ventura or newer, Intel or Apple Silicon
Linux	Ubuntu 22.04 LTS, Debian 12, or a comparable newer distribution
Python	Python 3.12 or newer for Python installations

A dedicated GPU is not required. Large multi-curve fits and resampling analyses may require more memory and time than the reference configuration.

3.2 Desktop packages

Open the latest release and select the package for your system.

Platform	Release asset	Normal launch method
Linux x86-64	`CurveMole-VERSION-linux-x86_64.AppImage`	Mark executable and run
Windows x86-64	`CurveMole-VERSION-windows-x86_64-setup.exe`	Run the per-user installer
Windows portable	`CurveMole-VERSION-windows-x86_64-portable.zip`	Extract and run `CurveMole.exe`
macOS Apple Silicon	`CurveMole-VERSION-macos-arm64.dmg`	Drag CurveMole to Applications
macOS Intel	`CurveMole-VERSION-macos-x86_64.dmg`	Drag CurveMole to Applications

> **Caution Windows and macOS packages are unsigned.** CurveMole is a non-profit project; these packages do not use developer-paid signing or notarization. The operating system may require approval on first launch.

Download from the official GitHub release page and verify `SHA256SUMS.txt`. On Windows, if SmartScreen displays **Windows protected your PC**, choose **More info > Run anyway**. On macOS, drag CurveMole from the DMG to Applications and try opening it once. If blocked, open **System Settings > Privacy & Security**, choose **Open Anyway**, then **Open**, and authenticate if asked.

3.3 Linux AppImage

In a terminal opened in the download directory:

```
chmod +x CurveMole-1.2.0-linux-x86_64.AppImage
./CurveMole-1.2.0-linux-x86_64.AppImage
```

If the system cannot mount AppImages through FUSE, use extraction mode:

```
APPIMAGE_EXTRACT_AND_RUN=1 ./CurveMole-1.2.0-linux-x86_64.AppImage
```

Some minimal Linux installations may need graphical runtime libraries supplied by the distribution. On Debian or Ubuntu, the commonly required packages are:

```
sudo apt install libegl1 libgl1 libxkbcommon-x11-0 libxcb-cursor0
```

Choose **Tools > Integrate CurveMole with the desktop** to copy a new AppImage to a stable per-user location, add CurveMole to the application menu, install its icon, and register `.fitproj` projects. No administrator privileges are required. If a working manual `curvemole.desktop` launcher already exists, CurveMole backs it up once and adopts its command rather than creating a duplicate application entry.

The Windows installer similarly creates a Start-menu entry, offers an optional desktop shortcut, and registers `.fitproj`. On macOS the application appears in Applications and Launchpad. On every platform, a registered project can be opened by double-clicking it or choosing **Open with CurveMole**.

3.4 Verifying SHA-256 checksums

Every release includes `SHA256SUMS.txt`. On Linux:

```
sha256sum -c SHA256SUMS.txt --ignore-missing
```

On macOS:

```
shasum -a 256 CurveMole-1.2.0-macos-arm64.dmg
```

On Windows PowerShell:

```
Get-FileHash .\CurveMole-1.2.0-windows-x86_64-setup.exe -Algorithm SHA256
```

Compare the reported value with the corresponding line in `SHA256SUMS.txt`.

3.5 Installing the Python package

Use an isolated environment. From a downloaded wheel:

```
python3 -m venv ~/.venv/curvemole
source ~/.venv/curvemole/bin/activate
python -m pip install ./curvemole-1.2.0-py3-none-any.whl
curvemole gui
```

On Windows PowerShell, activation is:

```
py -m venv $env:USERPROFILE\.venv\curvemole
& $env:USERPROFILE\.venv\curvemole\Scripts\Activate.ps1
python -m pip install .\curvemole-1.2.0-py3-none-any.whl
curvemole gui
```

3.6 Running from a source checkout

Clone the repository and install its locked development environment:

```
git clone https://github.com/SebRoLENS/curvemole.git
cd curvemole
uv sync --locked --group dev
uv run curvemole gui
```

For an editable installation using `pip`:

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install -e .
curvemole gui
```

3.7 Confirming the installed version

```
curvemole --version
```

The About dialog also displays the version. A project stores the application version that created it, and exports include the application version in machine-readable metadata.

3.8 Version badge and updates

Choose **Help > Update channel...** to select **Stable** (approved releases for normal use) or **Beta** (the `testing` branch). Beta receives bug fixes sooner and may receive new features earlier. It may update very frequently and introduce new bugs. The selection is saved and can be changed at any time. A new beta installation defaults to Beta unless a channel preference was already saved; stable installations default to Stable.

Switching from Beta to Stable stops beta updates. CurveMole keeps the installed version until an equal or newer stable release is available; it does not downgrade. Going back to an older version is not recommended, but advanced users who know what they are doing can download and install any version from the GitHub repository. The channel dialog includes a link to the download page. The badge continues to show **Beta** while a beta package is installed, even when the selected channel is Stable.

The status bar contains a clickable version badge. CurveMole checks the latest GitHub release for the selected channel at startup and once per hour while the application remains open. Stable checks exclude prereleases; Beta checks consider both stable and beta releases in version order. The badge is:

- green when the installed release is current;
- yellow when a bug-fix update (`x.x.y`) is available;
- red when a feature (`x.y.x`) or major (`y.x.x`) update is available;
- grey while checking or when the check cannot be completed.

An automatic notification is shown only once for each newly detected release. Click the badge or choose **Help > Check for updates** to check again, inspect the release notes, or reopen the update dialog.

The notification identifies the available version and its release type. For a version written as `major.minor.patch`, the first changed number determines the type:

First changed number	Release type	Example
Patch (third)	Bugfix	`1.0.0` to `1.0.1`
Minor (second)	Feature release	`1.0.1` to `1.1.0`

First changed number	Release type	Example
Major (first)	Major update	`1.1.0` to `2.0.0`

Linux AppImage and Windows desktop packages can normally use **Update now**. CurveMole downloads the matching release asset and verifies the SHA-256 digest supplied by GitHub. Linux replaces the AppImage, Windows portable installations update from the portable ZIP, and installed Windows applications launch the verified installer. The application directory must be writable and no fit or uncertainty task may be running. Existing Windows portable users are offered a one-time migration to the installer; projects, settings and plugins remain in their existing user directories. Python, source, macOS, unsupported architectures, and protected installations must be updated manually from the release page.

4. A complete first fit

This tutorial uses `examples/gaussian.csv` from the source archive. It contains x, y, and `sigma_y` columns.

4.1 Import the example

1. Start CurveMole.
2. Choose **File > Import data** or press **Ctrl+I**.
3. Select `examples/gaussian.csv`.
4. In the mapping dialog, confirm comma delimiter, decimal point, and header.
5. Select x as **X column** and check y under **Y column(s)**.
6. Select `sigma_y` as the uncertainty type and column.
7. Press **OK**.

The curve appears in the left tree and becomes active. The central plot displays the data, and the model panel on the right identifies the active curve.

4.2 Add a background

1. Press the **+** button in **Model and parameters**, or press **Ctrl++**.
2. Select **Constant** and enable **Add as background**.
3. Keep composition set to `add`.
4. Press **OK**.
5. In the parameter table, enter an initial `offset` near the visual baseline.

4.3 Add a Gaussian peak graphically

1. Add another component and select **Gaussian**. Leave **Add as background** and **Initialize from manually selected points on the graph** unchecked.
2. Press **OK**. The pointer changes to a crosshair.
3. Click at the approximate peak center.
4. Keep the mouse button pressed and drag horizontally from the center to either half-height edge.
5. Release the mouse button.

CurveMole uses twice the dragged distance as the initial FWHM. It estimates a positive peak height from the data minus the existing model, converts this height to the area parameterization, and adds the component as one undoable action.

If placement is incorrect, press **Ctrl+Z** and repeat. Press **Esc** during placement to cancel without adding the component.

4.4 Adjust the model before fitting

Select the Gaussian in the component list. Three green handles appear:

- drag the central target to change center and height;
- drag either vertical side handle to change FWHM;
- edit exact values or bounds in the parameter table.

For this example, a positive lower bound on `sigma` already exists intrinsically. You may enter a tighter scientifically motivated range, but it is not required.

4.5 Mask an interval if needed

Mask editing is active only while the graphical **Mask** toolbar button is enabled:

1. click **Mask**; the button becomes grey while masking is active;
2. in Single view it acts on the displayed spectrum; in Overlay/Waterfall choose the active spectrum or all visible spectra in the scope prompt;
3. right-drag to mask an interval, left-drag to restore it;
4. click **Mask** again to return to normal navigation.

The example does not normally require masking.

4.6 Run the fit

1. Press **Ctrl+F** or choose **Fit > Fit**.
2. Use **Single / independent** mode.
3. Check that the example curve is enabled in the table.
4. Keep **Local least squares (automatic)** and `linear` loss.
5. Press **OK**.

After a successful fit, CurveMole commits the optimized parameter values back into the displayed model and immediately redraws the model sum, individual components, and residual panel. Parameter standard errors appear in the `±1 sigma` column when covariance is available. Open **View > Diagnostics** to inspect residual warnings.

4.7 Save and export

Save the complete workspace with **File > Save project as**. Use a `.fitproj` extension.

Then choose **File > Export analysis bundle**. All available output types are selected by default, and the dialog can create a named subfolder to keep the complete export together. The bundle includes numeric spectra, function parameters, figures, reports, Python analysis tables, reusable models, a project copy, and a README that describes the saved files. Clear individual output choices when a smaller bundle is needed. The **File** menu also offers clearly labelled **Quick export** commands for spectra, function parameters, and a Matplotlib figure of the current plot view; these focused exports are also included in the complete bundle.

5. Main window and controls

5.1 Series and curves tree

The left dock contains a search field and a tree with **Visible**, **Series / Curve**, and **State** columns.

- Click a curve to activate it.
- When switching spectra, the function selected in Model & Parameters stays at the same position in the function list, regardless of its name or type. If the new spectrum has fewer functions, its last function is selected; an empty model clears the parameter table. Show all functions retains its project-wide selection.
- Use Ctrl-click or Shift-click for a multi-selection, or use **Select all** and **Deselect all** above the tree.
- Use **Remove selected** to delete one or more accidentally imported curves. Removal is undoable.
- Use **New series** to create an empty named series. Empty series are retained when the project is saved.
- Right-click one or more selected spectra and choose **Move selected to series** to reorganize them.
- Right-click selected spectra and choose **Move selected up/down** to change their order inside a series. Multiple adjacent spectra move as a stable group.
- Right-click a series and choose **Merge series into** to append all its spectra to another series and remove the source series.
- Moving, merging, creating, and reordering series are all Undo/Redo operations and do not invalidate fitted parameters.
- Use the visibility checkbox to include or exclude it from Overlay and Waterfall displays. Hover over the checkbox for an explanation. Hiding a spectrum retains its data and fit; hidden spectra are excluded from fit plans. Highlighting rows separately selects spectra for operations. Hidden spectrum names appear muted; Up/Down navigation skips them. Click a hidden spectrum to inspect it, or check its visibility box to include it in navigation again.
- Double-click an editable name, or right-click and choose **Rename data...** or **Rename series...**. Series names must remain unique.
- Right-click a series and choose **Delete series...** to remove it together with all its curves, models and results. A populated series requires confirmation; Undo restores the complete series. Original files on disk are retained.
- Right-click a **curve** and choose **Choose data colour...** to set its colour. Red is reserved for the fitted Model sum and cannot be assigned to a spectrum.
- Right-click a **series** and choose **Series palette** to recolour the complete series with one of the built-in non-red palettes. Palette and individual colours are saved in the project.
- Use the search field to filter curves by name without removing them.

Curve states are:

State	Meaning
Not fitted	No successful fit is associated with the current curve state
Ready	The curve is prepared for fitting
Running	A fit involving the curve is in progress
Fitted	The last fit completed successfully
Modified/outdated	Data or model changed after the last successful fit
Failed	The latest fit attempt failed

Uncertainty tags are added to the fit state for the method currently selected in **Uncertainty Analysis > Method**. A failed replica, an adaptive stability limit or a timeout adds a warning icon beside the spectrum and an amber status. Hover over the State column for the reason and counts. Switching the method updates these tags without refitting or deleting results. It is the same universal method selected through the **Analysis - / +** header in **Model and parameters**: changing either control updates both panels and the spectrum tags. See Section 12.

5.2 Plot workspace

The central workspace contains the data/model plot and an optional residual plot. The controls use two compact rows with palette-aware selection states. They wrap when dock resizing leaves less width; long coordinate readings stay on one line with the complete reading available in a tooltip.

Controls above it provide:

- **Display:** Single, Overlay, or Waterfall;
- **All series / Active series / Selected:** one-click scope for Overlay and Waterfall;
- **X offset** and **Y offset:** display-only spacing, shown only in Waterfall;
- **Residuals:** show or hide the linked residual panel;
- **Autoscale:** automatically frame the experimental data when the active spectrum changes;
- **All points / Unmasked points:** choose whether automatic framing includes masked data;
- **Plot style:** draw imported data as Lines, Points, or Lines + points;
- **View options (display only):** a separate toolbar group between background commands and Mask. Open its menu to inspect original data, change background display, or include/exclude recorded operations in the preview for the current spectrum or all spectra. It changes only the graph: data are not modified and operations are not undone.

Choose **View > Plot appearance...** for separate data, function and residual styles, line widths, symbols, colours and grid settings. These display settings are stored with the project.

With Autoscale off, switching spectra preserves the view. **View all** and **View unmasked** explicitly frame the corresponding experimental points; model extrapolations, labels and handles do not affect these limits.

The main toolbar contains the graphical **Mask** button. Its scope is chosen on entry in Overlay/Waterfall; mouse buttons determine the operation.

Offsets never modify data and are removed when converting pointer positions back to scientific coordinates. Fitting and numeric data export use the underlying current numeric coordinates. Current-view figure exports preserve the selected display positions, including waterfall offsets.

The compact coordinate readout displays the pointer position on one line. Hover over it to see the full values and the nearest finite point from the active curve. Overlay and Waterfall views use adaptive peak-preserving downsampling and clip drawing to the visible range for dense spectra. Single view keeps the full display resolution. Descending x arrays may be reversed only in the temporary display copy so that fast clipping remains available. Fitting, project storage, and numeric export use full-resolution data, rather than that display copy. Fits use current transformed observations with the selected masks and fit ranges; numeric exports can optionally keep only unmasked rows. Projects retain immutable originals as well as reversible transformations.

Model functions initially receive names such as **Voigt1**, **Voigt2**, and **Gaussian1**. Right-click a function and choose **Rename function...** to assign any non-empty name, including Unicode. The name belongs to that component of that spectrum; internal identifiers and

parameter links do not change. Renaming supports Undo/Redo, preserves fitted status, and survives saving/reopening, duplication and copying. Plot labels, notebook entries, uncertainty tables and exports use the name. Their labels are shown above each function maximum by default and are shifted vertically when necessary to avoid overlap. Right-click the main plot and toggle **Show component labels** to hide or show these labels. After every completed fit, CurveMole explicitly applies the returned optimized parameters to the displayed model and redraws the plot while preserving the current zoom. The **Model sum is always red**; imported spectra and selectable series palettes exclude red so a data curve can never be confused with the fitted sum.

5.3 Navigation and axes

Normal plot navigation is available when neither placement nor Mask mode is active. Right double-click frames all experimental data, including masked points. Left double-click frames only unmasked experimental points. These gestures are independent of the **Autoscale** checkbox and its **All points / Unmasked points** selector. Hold the right button and drag to pan the spectrum; a single right-click opens the plot context menu after the double-click interval. Display controls wrap onto additional rows when dock resizing narrows the plot. Axes always retain the original data units. Numeric tick labels switch to scientific notation only when their decimal representation is too wide for the available space; this never changes spectrum values or adds an SI unit prefix. The mouse wheel remains available during peak placement, spline placement, Quick Fit, and live fit refreshes. A refresh preserves the current viewport instead of resetting a zoom chosen by the user. When the plot has keyboard focus, Up and Down activate the previous or next checked spectrum in project order. Up/Down in the curve tree also skips unchecked spectra.

Use **View > Axes** for:

- automatic range;
- logarithmic x or y;
- reversed x or y;
- locked plot view.

Reversing an axis is only a view setting. Logarithmic axes do not transform the data used by the fit. Values incompatible with a log display may not be visible even though they remain stored.

5.4 Model and parameters dock

The right dock normally lists components for the active curve. Enable **Show all functions** to list every model function in the project, grouped by series and spectrum. Group headings cannot be selected as functions. The list supports Ctrl-click and Shift-click multi-selection, including selections across spectra. Each function has an inline composition menu beside its name: **add**, **subtract**, **multiply**, **divide** or **convolve**. Click it to change that function's role; the change is undoable and marks only its spectrum's fitted result as outdated. For every model, insertion and list order do not determine the calculation. **Calculation order** explicitly orders factors and convolutions and previews the resulting expression. **Reorder** sorts only the displayed list and automatic names. The Up/Down list controls change presentation only. Exported parameter rows and component trace columns follow this displayed order and use the current function names. Their values and uncertainties remain associated with the same functions, including models with multiplication, division or convolution.

Select all selects every displayed function. **Select all except backgrounds** selects only functions that are not marked as background and clears any previous background selection. In Show all functions mode these actions span the project. Selection is separate

from enabling: a checked function contributes to the model and fit; an unchecked function remains stored but is excluded. Hover directly over its checkbox for an explanation. Changing a checkbox on a selected function applies the enabled state to the whole selection. **Mark as background** classifies the selected functions; it does not subtract them from the data.

The controls below the list are:

Control	Action
+	Add a component
Duplicate	Duplicate the selected component and its settings
Up / Down	Change component order
Delete	Delete the component after confirmation
Reorder	Sort the function list and renumber automatic names using a parameter value
Advanced reorder rules	Choose the numbering parameter separately for each function type
Copy fit	Copy selected model information to other curves
Copy fit to next	Copy to the next curve in the source series, then open it
Copy parameter	Copy one parameter from one source function to chosen project-wide targets

Click the **Analysis - / +** column heading to open **Displayed uncertainty**, a menu of methods recorded for at least one spectrum. The checked entry is the method currently displayed. There is no separate selector above the parameter table. The sole available method is selected automatically. With several methods, choose the one to show for every spectrum: spectra without it show a dash instead of another method's errors. The choice survives switching spectra, Show all functions, saving and reopening. With no recorded analysis, clicking the selector explains how to run one and offers a link to **Uncertainty Analysis**. It remains available when several functions are selected, even while the parameter table is hidden.

Bulk actions on selected functions can delete, duplicate, reorder, enable or disable, mark or unmark backgrounds, and fix or free their parameters as applicable. Each bulk edit is one Undo/Redo operation. Uncheck a component in the list to disable it without deleting it. Disabled components are not evaluated or fitted.

Deleting a function preserves the roles and parameter values of surviving functions, including Multiply, Divide and Convolve. Undo restores the original functions and composition. If remaining parameters link to a function being deleted, deletion is cancelled and the affected links are listed; remove those links or constraints before deleting the function.

Copy parameter is enabled only when exactly one source function is selected. Select a row in the parameter table to make that parameter the default, open the copy window, and explicitly check the target functions. The window shows the source spectrum, function, parameter, and numeric value. By default only the value is copied; optional controls also copy bounds, fixed/free state, and the parameter link/relation. Targets without the parameter or whose retained bounds reject the source value are skipped and reported. Choose **All** in the parameter list to copy every source parameter in one edit, using the selected value, bounds, fixed/free and link fields. A target must contain every source parameter; extra target parameters stay unchanged. A missing parameter or a bounds conflict skips the entire target function. The complete copy is undoable.

5.5 Quick actions

The main toolbar includes a bordered **Functions** group with a shared title, a full-size **Add function** icon button, a function selector, the **Background** and **Initialize with points** options, and a full-size **Function details** button. All these controls sit inside the same frame. Both icon buttons use the same icon size as the other toolbar buttons; the two-row controls keep the group compact. The selector contains the complete current registry: built-ins, project formulas, reusable user formulas, and loaded plugin functions. CurveMole remembers the last selection across sessions and also updates it when a function is added through the ordinary Add dialog.

The **Background** and **Initialize with points** checkboxes below the selector configure the next function insertion. The choices are remembered for each function only after explicitly changing them here. Both start off, including for Linear and Cubic spline; the regular Add dialog's choices are independent. With **Initialize with points** checked, peaks use centre/FWHM click-drag placement. With it cleared, each **Add** click inserts one automatically estimated peak. Other functions whose point initialization is off are added immediately. During point placement, right-click or press Return/Enter to finish, just like **Finish**. Point placement still requires its minimum number of points. Section 8 describes point placement, initial locking and graphical controls.

The toolbar also provides the most common project, model, fitting, background, and calculator actions as icons. Hover over an icon to see its action name.

5.6 Dockable tools

The **View** and **Tools** menus show or hide these docks:

- **Worksheet:** read-only view of current x, y, uncertainty, weight, and mask state;
- **Diagnostics:** residual summary and warnings;
- **Log:** operational messages and errors;
- **Data Calculator:** reversible scalar and curve-to-curve transformations;
- **Function Builder:** safe mathematical custom functions;
- **Uncertainty Analysis:** Monte Carlo, bootstrap, and profile calculations;
- **Laboratory notebook:** project notes and searchable object descriptions;
- plugin panels contributed by installed plugins.

Opening a tool places it in the shared tab group beside **Model and parameters** and selects its tab. Closing or hiding a tab preserves the panel and its current state; opening it again returns to the same panel. Tabs can still be moved or floated. When many tabs are open, use the arrows to scroll through their titles. Titles longer than 25 characters end in an ellipsis; hover over one to read its full title. CurveMole remembers window geometry, dock arrangement, and theme. Use **View > Reset layout** to restore the standard tabbed layout.

5.7 Themes

View > Theme provides System, Light, and Dark themes. Theme choice affects only presentation. It does not change plot data, exported numeric values, or fitting.

5.8 Drag and drop

Drop a `.fitproj` file onto the window to open it. Drop any other local file to begin content-aware data import, regardless of its extension. If a drop contains both a project and data files, the first project is opened.

6. Importing data

6.1 Supported text formats

The graphical importer shows `.txt`, `.dat`, `.csv`, and `.tsv` as common choices but can open any text-based file whose contents form a valid numeric table. For example, spectrometer files named `.xy`, `.xye`, or with a vendor-specific suffix can be selected through **All files** or dropped onto the window. The extension is not used as proof of validity. Files must contain at least two columns and at least two rows after parsing.

CurveMole detects common delimiters:

- whitespace;
- comma;
- semicolon;
- tab;
- vertical bar.

It also proposes decimal point or decimal comma and whether the first data row is a header. Always inspect the preview before accepting the mapping.

6.2 Comments, blank lines, and encoding

Blank lines are ignored. Lines whose first non-space character is `#`, `;`, `%`, or `!` are treated as comments by the default importer. UTF-8 with an optional byte order mark is the default encoding.

CurveMole also detects leading instrument metadata or preamble lines before the rectangular table. **Ignore first rows** shows the proposed number of physical lines to skip and can be changed from 0 to 1,000,000. Skipping occurs before delimiter, decimal, and header interpretation; therefore a column-name row after the preamble can still be used as the first data-row header. The accepted value is stored in the curve import metadata for reproducibility.

The semicolon can therefore be ambiguous: a line beginning with semicolon is a comment, while semicolons inside table rows can be delimiters. Check the preview for locale-specific exports.

Before confirming files, the file chooser already includes a small spectrum graph. Highlight a file to preview it; select X and Y from the adjacent dropdowns (first and second column by default). Reading happens in the background. The graph uses the same peak-preserving adaptive renderer as the main plot and full-data axis limits. Choices are remembered per file and proposed in the subsequent import mapping dialog. With multiple files, Apply this mapping to all files deliberately uses the mapping confirmed for the first file. Files requiring custom parsing can still be opened when the quick preview is unavailable, then configured in the mapping dialog.

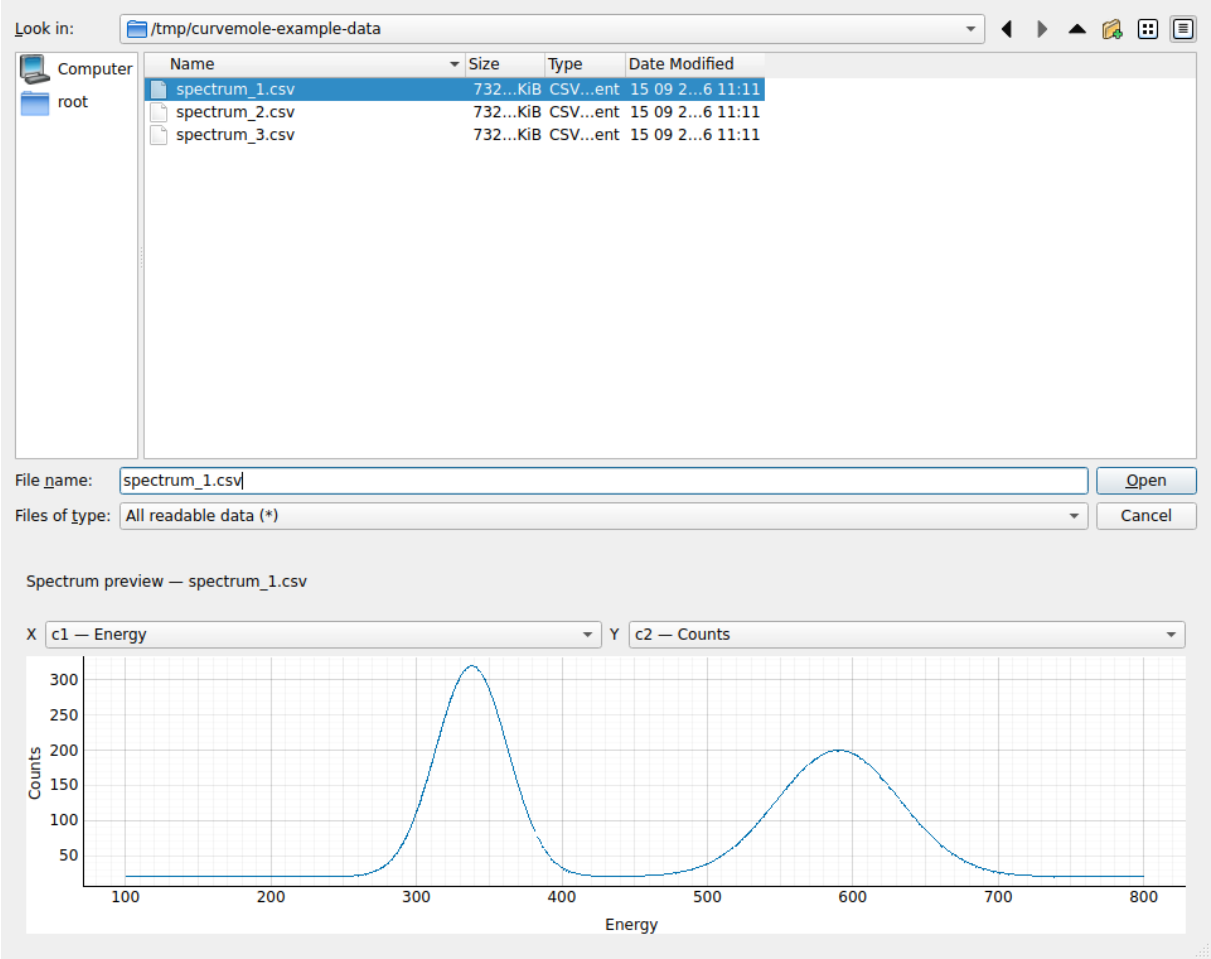


Figure 1: Preview while selecting files, before opening the import mapping dialog.

6.3 Column mapping

Destination lets you create a new series or select an existing series by name. A new series requires a unique non-empty name. Selecting an existing series appends new spectra there while preserving its identity, existing spectra, notes and palette. The destination applies to every selected file in that import; later per-file mapping dialogs keep it fixed. Cancelling a later mapping or failing to parse any file leaves the destination unchanged. All files are parsed before any are added.

Ordinary imports select one X and one active Y. All original columns remain available in the imported curve. For files with more than two columns, the **as separated files** checkbox appears directly below the Y selection. Hover over it for an explanation. Enabling it imports independent spectra in the same series, named after their Y headers (or `Y - column N` without headers); it does not create files on disk.

Choose the structure explicitly:

- **Shared X:** check multiple Y columns, all using the selected X. **Select all Y columns** selects every column except X; deselect any error or auxiliary columns before importing.
- **Repeated groups:** choose the first column, number of Ys per X, and error type per Y, then click **Apply pattern**. One Y per X produces `XYXY...`; two produces `XYYXYY...`. With symmetric errors these become `XYE...` or `XYEYE...`; asymmetric errors use `YE+E-` for each Y. Review or edit the resulting associations. Incomplete trailing spectra are excluded, and the unassigned column numbers are shown below the table.
- **Custom:** add or remove spectrum rows and choose any X/Y associations, including nonadjacent columns and irregular groups. Switching from repeated groups to Custom retains the generated rows for editing.

Column names never infer the structure, axes, or error columns. Each table row has its own error/weight type, error columns, and confidence level. When an error type is enabled, its selector initially uses the column immediately to the right of Y; asymmetric errors select the next two columns as error + and error -, respectively. These defaults can be changed. Missing columns or errors overlapping an assigned X/Y must be resolved before importing. Error types initially remain **None**. The same adjacent-column defaults apply to ordinary imports when errors are enabled.

Optional columns are:

- `sigma_x`;
- `sigma_y`;
- Y error (confidence interval), or a pair of asymmetric Y error columns;
- generic weight;
- variance;
- inverse variance.

Only one of `sigma_y`, confidence-interval errors, generic weight, variance, or inverse variance can be selected for a given spectrum. Variance is converted to `sigma_y` by square root. `sigma_x` is stored in the project but is not used by the version 1.2.0 optimizer.

For confidence intervals, choose **Y error (confidence interval)** and one symmetric half-width column, or check **Asymmetric error** and select **Y error +** and **Y error -** separately. Both columns contain positive magnitudes, not signed errors or interval endpoints: the interval is `[y - error -, y + error +]`. **Data error confidence** declares its central coverage (default 95%). This is independent of the confidence selected for fitted parameter intervals. Existing `sigma_y` imports continue to mean one standard deviation, regardless of this field. The import preview shows the supplied errors at their original confidence.

The spectrum preview below the numeric table updates as you change X/Y columns, uncertainty mappings, delimiter, decimal separator, header or skipped rows. It plots the complete file, reusing the main graph's peak-preserving downsampling and clipping rules. Parsed data are cached while changing columns. Descending X is reversed for display;

nonmonotonic X is not clipped/downsampled. The view bounds always come from full-resolution finite X/Y pairs. With no selected Y or an invalid mapping, the preview clears and explains the problem. For batch import, it previews the file named in the mapping dialog.

6.4 Batch import

When importing multiple files, **Apply this mapping to all files in this batch** reuses the first file's parsing and column mapping. Enable it only if all files truly share a layout. CurveMole does not silently guess a new mapping for an incompatible file in the same batch.

Separated-spectrum mappings reuse column positions, so files with the same layout can have different headers. Each spectrum keeps the Y name from its own file. Ordinary batch mappings continue to use the selected column names.

Files are imported in natural filename order: `scan_2` precedes `scan_10`, regardless of the order returned by the file chooser.

Calibrated WinSpec SPE files

CurveMole also reads binary WinSpec SPE 2.x files containing calibrated one-dimensional acquisitions. The first column is wavelength in nm, followed by one intensity column per frame. Select the frame through the Y-column mapping. SPE 3.x, uncalibrated axes, images and multiple-ROI acquisitions are unsupported; export those acquisitions to calibrated x-y text before importing.

Automatic folder import

Choose **File > Automatic folder import...** to add arriving files to the current project. Select an existing folder, an optional case-insensitive **Filename contains** filter, the one-based X/Y columns and a **Wait for stable file** delay (2 seconds by default). Monitoring starts only when you press **Start**.

- **Also import matching files already in the folder** includes existing files; otherwise the session starts with newly arriving files.
- **Import changed files again as NEW spectra** retains the previous acquisition and adds a new one when file contents change.
- **Show newest imported spectrum** controls whether successful imports take focus; it can be changed while monitoring runs.
- **Automatic workflow** selects Import only or a loaded plugin's import processor. Configure the plugin first; processing runs on the new acquisition in the background before it is added to the live project.

Only files directly in the folder are scanned. Symlinks, hidden/temporary files and projects are ignored, and automatic inputs are limited to 256 MiB. Identical content at the same path is not imported twice within a session. Each successful acquisition is one Undo operation. Failed files appear in the log; correct them or the settings and choose **Retry failed files**. Imports wait while fitting, a desktop automation or a modal dialog is active. Closing the monitor panel leaves it running; choose **Stop** or **File > Stop automatic import** to stop. Changing projects ends the session, and reopening CurveMole does not start monitoring automatically.

6.5 Invalid numeric cells

Non-numeric cells are converted to missing numeric values. Rows are retained in the stored arrays but excluded from fitting when x or y is not finite. A row is also excluded if:

- `sigma_y` is non-finite or not strictly positive;
- a point weight is non-finite or negative;
- it is masked;
- it lies outside active fit ranges defined through the core API.

CurveMole does not reorder rows, average duplicate x values, or interpolate missing values during import.

6.6 Uncertainties and weights

If absolute `sigma_y` values are supplied, point residuals are scaled as:

$$r_i^{(w)} = \frac{f(x_i) - y_i}{\sigma_{y,i}}.$$

If inverse variances $w_i = 1/\sigma_i^2$ are supplied, CurveMole multiplies residuals by $\sqrt{w_i}$. A generic weight column directly multiplies residuals by its value. This distinction is intentional and is recorded in the curve metadata.

A zero weight is retained in the dataset, but when weights are the active Y-error representation it is excluded from fitted rows, N and degrees of freedom. Turning off data-error weighting retains such rows when X/Y are finite. A zero or negative `sigma_y` is invalid and excludes the row while that error weighting is enabled. Inverse-variance weights follow Y-uncertainty propagation: multiplying Y by `a` divides those weights by `a**2`. Arbitrary direct residual multipliers retain their declared multiplier semantics.

For a confidence level C, interval half-widths are converted to approximate Gaussian scales by dividing each side by $z = \Phi^{-1}((1 + C)/2)$. At every evaluation, a prediction above the measured value uses the upper scale; a prediction below uses the lower scale. Equal half-widths reproduce symmetric Gaussian weighting. The error arrays and confidence are saved and exported; plotted bars keep the supplied half-widths, rather than these converted scales.

Fit > Advanced algorithm options > Use data errors / weights in the fit is on by default. Turn it off to fit without imported Y errors or point weights. Masks, fit ranges and spectrum-level weights remain in effect. Rows invalid only because of their error or point weight can then participate. The display of bars and the noise model of Monte Carlo remain independent of this choice.

This is an approximation: two interval half-widths and a confidence level do not specify the full measurement distribution. The scales describe a Gaussian side on each side of the reported value; Monte Carlo assigns equal probability to the two sides, reproducing the supplied central interval. Covariance is a local Jacobian approximation, and asymmetrical parameter intervals should be assessed using resampling or a suitably calibrated profile. Correlation between measured points and X uncertainty are not incorporated into this Y-residual fit.

Use **Error bars** above the main graph to show or hide the bars. Autoscaling includes their endpoints, and bars follow logarithmic, waterfall and background views. Scalar and curve scaling propagate widths and exchange upper/lower sides for negative factors. Custom and column formulas use local derivative propagation; for strongly nonlinear transformations this is only a local approximation.

7. Curves, masks, and transformations

7.1 Immutable originals

CurveMole preserves imported x, y, uncertainty, and weight arrays. Displayed working arrays are reconstructed by applying a transformation history to the originals. This design makes data preparation reversible and auditable.

7.2 Masking points and intervals

Masked points remain stored but are excluded from fitting. They are drawn as faded markers, and masked intervals are shaded.

Enable the graphical **Mask** button: right-drag masks an interval and left-drag unmasks it. A right/left click performs the equivalent operation on a point. The button becomes grey while active. Turn it off to navigate without changing exclusions. Changing display mode also exits masking.

7.3 Mask targets

Target	Effect
Active	Changes only the active curve
All visible	Changes every currently visible curve

Single view always uses the active spectrum. Overlay and Waterfall prompt for Active or All visible when entering Mask mode; cancelling leaves masking off.

For interval masks, the same numeric x interval is applied to every target. For a single-point mask transferred from the active curve, CurveMole locates the nearest x value on other targets and applies it only if it lies within the configured tolerance.

Set this tolerance through **Data > Mask transfer tolerance**. A value of zero requires an exact matching x value on transferred point masks.

7.4 Undo and mask history

A graphical mask action is placed on the application Undo stack. Press **Ctrl+Z** to undo it or **Ctrl+Y** to redo it. The project also stores mask arrays and numeric mask intervals.

Fits, including Quick Fit and sequential fitting, also support Undo/Redo. Undo restores the pre-fit models, parameter values, fit results, and sequential pause state; Redo restores the completed fit without running the solver again. Cancelled or failed worker operations restore the previous state. The Undo stack retains the latest 20 operations (fits and other edits combined); input data arrays are not duplicated for fit history. Undo/Redo are disabled while a worker is running.

Quick Fit is available before opening Fit...: its first run uses the default independent-fit settings on the selected curves, falling back to the active curve. Later runs retain the configured solver, loss and algorithm settings but always use **Single/Independent**, even after a sequential or global fit. Open Fit... to use another fitting mode.

7.5 Data Calculator

Open **Tools > Data Calculator**. The following scalar operations are available:

- add to or subtract from y;
- multiply or divide y;
- shift x;
- scale x;
- normalize y by maximum absolute value;
- normalize y by signed integrated area.

Every calculator operation (including restoring data) offers **Active curve** or **Choose multiple spectra...**. The picker initially shows the active series; **All series** shows every spectrum grouped under colored series headings. Use checkboxes or Select all shown. Hidden spectra are excluded. Cancel changes nothing. A batch is checked in full before any spectrum is modified; one Undo restores the whole batch. Normalization also scales `sigma_y` consistently. Dividing by zero and normalizing a zero or invalid signal are rejected.

Advanced column formulas

The note below Formula reminds you to use Python syntax for mathematical operators.

The green **Advanced - column formula** entry is first in the operation list, followed by the green **Custom formula** for a single plotted axis. New imports retain every source column as numeric data in the project, including columns not plotted. Original column order and row alignment are preserved. `c1`, `c2`, `c3`, etc. refer to one-based source columns; `x` and `y` refer to the current plotted axes. **Insert into formula** lists the available codes with original column names; choose an entry, then press **Insert** to add its code at the cursor. **Write result to** chooses X, Y or any retained column for the result. The hint below it explains that Apply writes the formula result into this column. Merely selecting an input column does not edit the formula or change its destination. A live summary such as `Y - Counts <- c2 / c3**2` makes the direction explicit. **Spectra to modify** separately chooses which spectra receive the operation.

For a file with Energy, Counts and Monitor columns, plotting c1 against c2:

1. choose **Advanced - column formula**;
2. choose **Y - Counts** as the destination;
3. insert `c2` and `c3` from the dropdown to enter `c2 / c3**2`;
4. choose Active curve or multiple spectra, then Apply non-destructively.

You can also enter `y / c3^2` (^ and ** both mean exponentiation in Advanced), `c2 - c4`, or `sqrt(c5)`. A header may be referenced as `${Monitor counts}`; codes remain unambiguous for unusual names. Only restricted mathematical expressions are accepted, never Python execution. Scalar results are broadcast to every row; missing columns and non-finite output (including division by zero) reject the batch. Masks do not change row alignment: formulas operate on all rows.

Writing to a plotted source column updates that axis. Writing to an unplotted column leaves the graph unchanged and makes the new values available to later formulas. Normal calculator operations on X/Y are also reflected in their mapped columns. Calculations are local to each spectrum, even when spectra originated in the same file. In a multi-spectrum batch `cN` always means position N in each file; check that the files have compatible layouts. Formulas updating X or Y propagate supplied uncertainty using the local derivative with respect to that target axis; inverse-variance Y weights follow the same rule. This does not jointly propagate errors in other columns or correlations between inputs. Changing an auxiliary uncertainty column does not remap the error arrays originally selected for fitting. Nonlinear formula derivatives use local, scale-aware perturbations and Richardson extrapolation. For `z=y**3`, the audited tiny-value example now agrees with `sigma_z=abs(3*y**2)*sigma_y`.

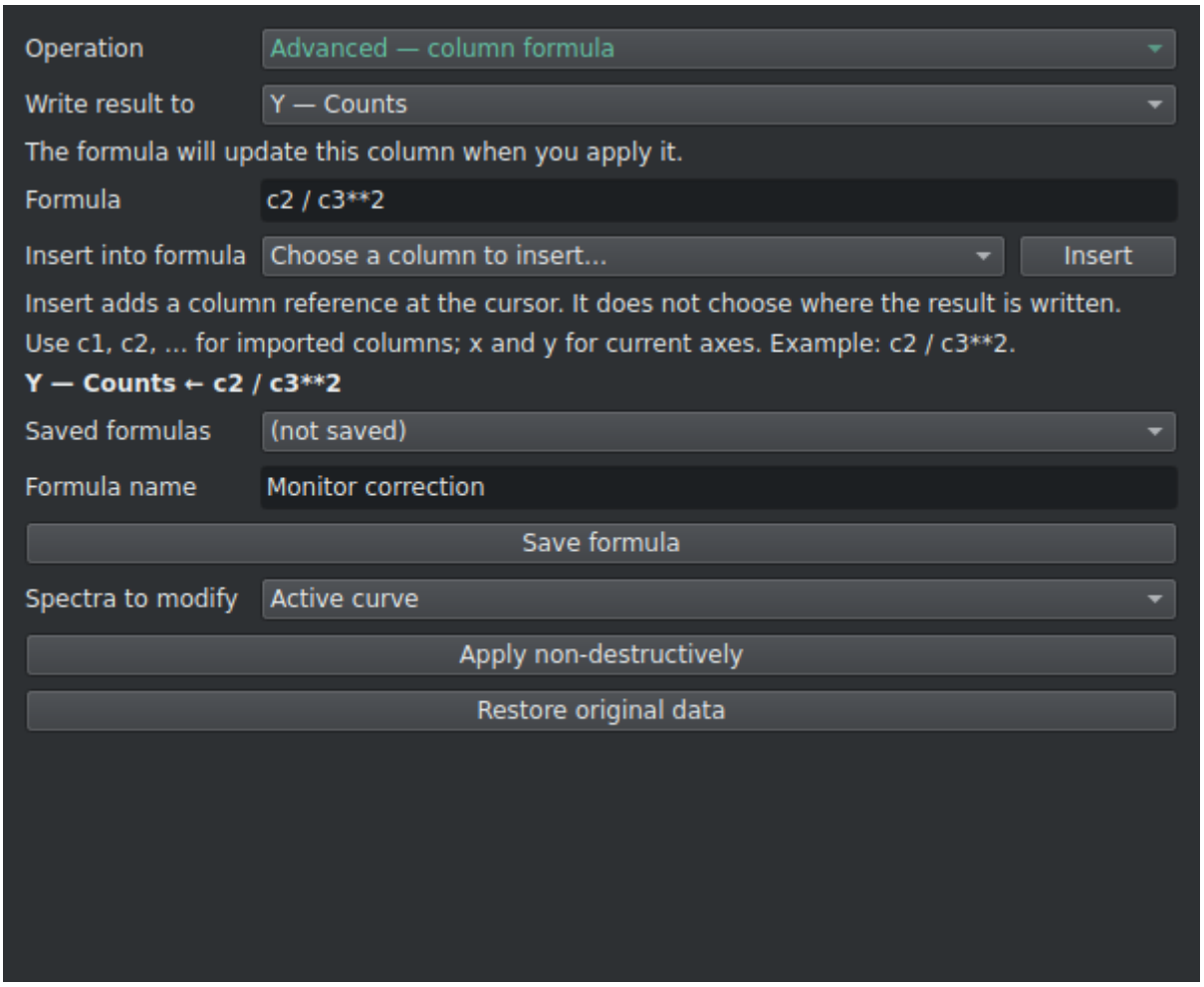


Figure 1: Advanced calculator with column insertion and destination controls in dark mode.

Propagation remains a first-order local approximation; input correlations and uncertainties in other columns are not jointly propagated.

Name and **Save formula** to retain the expression and destination in the project, then recall it through Saved formulas. Retained columns, formulas and their Undo/Redo history survive `.fitproj` saving without the original source file. Projects created before this feature contain only their previously stored axes: re-import the source to recover additional columns.

7.6 Curve-to-curve calculations

The calculator can add, subtract, multiply, or divide by another curve. The operand is aligned to the target x grid by:

- linear interpolation;
- nearest-neighbor interpolation;
- cubic spline interpolation.

Extrapolation is disabled by default. Outside the operand range, aligned values become missing and will therefore be excluded from later fitting. Enabling extrapolation is an explicit advanced choice. Cubic interpolation requires at least four unique finite operand x values.

The aligned operand array is stored with the transformation so that reopening the project does not depend on a later change to the source curve.

7.7 Restoring original data

Restore original data removes all currently applied transformations from the selected target while retaining them on the redo stack. The imported arrays are not re-read from disk. The whole action is also integrated with application Undo.

7.8 Worksheet

The Worksheet is a read-only inspection view. It shows up to 100,000 rows for the active curve with columns for x, y, `sigma_y`, weight, and mask state. It is not a spreadsheet editor and does not modify source arrays.

7.9 Editing individual data points

Choose **Data > Edit data points...** to edit measured points directly in the graph. The editing controls appear above the plot. Only the active, displayed dataset is edited; the original source coordinates remain immutable.

1. Click a point to select it. Its current X/Y coordinates appear in the editing bar.
2. Choose **X**, **Y**, or **XY** to constrain the data coordinates that may change.
3. Drag the point and release to commit one undoable move. During dragging, the graph previews the new position without changing the saved dataset.
4. Use **Delete selected point**, **Delete**, or **Backspace** to remove it. At least two points must remain. Deletion removes the row from current data, fitting, worksheets and numeric exports, keeping associated errors, weights, masks and auxiliary columns aligned.
5. Choose **Done** or press **Esc** to leave editing. Esc also discards an unfinished drag; previously completed moves and deletions remain applied.

Right-drag pans and the wheel zooms while editing. Point selection uses the actual samples, including in dense plots. Logarithmic axes, Waterfall offsets and visual background subtraction are accounted for; the editing bar shows the underlying data coordinates. X/Y constraints refer to those data coordinates.

Moves and deletions use the global **Undo/Redo** commands. They are stored in the project as reversible transformations and can also be removed with **Restore original data** in the Data Calculator. A changed fitted dataset is marked as modified, so refit it to update results and uncertainties. Entering function placement, changing the active dataset or starting a calculation ends point editing. Read-only projects cannot be edited.

8. Building models

All entries in the function library are fitting functions. Constant, linear, polynomial, cubic-spline, peak-shaped, and custom formulas are not separated into a special background-function class. Background is instead a property of a model component. Select a component in **Model and parameters** and enable **Mark as background** when that component represents the experimental baseline or background.

Data > Subtract background... uses these component-level marks. If no component is marked, CurveMole first asks which model functions should be designated as background. If background components already exist, it asks which of the marked components should be subtracted. The subtraction uses the current resolved/fitted parameter values, is reversible with Undo, applies over the complete data array, and disables the subtracted components afterwards to prevent double-counting.

The **Subtracted / Not subtracted** labels follow the subtraction history of each spectrum. Copying a fit preserves the destination's existing subtraction; it does not subtract the source background from the destination's data. The saved subtraction retains the parameter values used at that time. Editing or copying later parameter values does not recalculate it. A duplicated function starts with its own history and is not marked as subtracted merely because its source was. On opening older projects, CurveMole can reconnect a recorded subtraction to a uniquely identified builtin background whose internal ID changed. This restores the label and function-state restoration without changing the measured data, the stored subtraction array or the current fitted parameter values. If the old history is ambiguous, the label reads **Subtraction recorded - function uncertain**; **Revert background** still uses the actual recorded array. Confirming Revert removes the subtraction, so **Not subtracted** is then the expected status.

The same dialog can apply the operation to every eligible spectrum, using the enabled functions already marked as background in each model. **Data > Revert background...** lists only spectra changed by this dedicated subtraction command and restores the selected ones, including the relevant function states. **View options > Background > Remove fitted background in this view** instead removes the enabled marked background only from the plot; it never changes curve arrays, fit inputs, or exported values.

The **View options (display only)** toolbar button opens a menu with **Current spectrum** and **All spectra** scopes. The menu explicitly states that data are not modified and operations are not undone. Scope selects the targets for the next display change; it does not change views by itself. All spectra includes hidden spectra, which use the chosen view when displayed later. Per-spectrum choices follow spectrum navigation.

- **Preview stored data - all operations included** closes display filters for the scope.
- **Preview original data - without processing or masks** shows the imported X/Y values, including visually deleted points, without displaying the effects of masks or processing. The stored data, masks and processing history remain unchanged.
- **Show data before background subtraction** omits recorded background subtractions and replays the remaining operations in order. It also works after the background functions have been disabled by a real subtraction.
- **Operations included in preview** allows individual display exclusions on the current spectrum. In All spectra scope, entries are grouped by operation type with a count such as **Y multiplication - 3/5 spectra**. Unchecking a type excludes every occurrence on spectra that contain it; spectra without that type are unaffected. Different numeric parameters do not prevent grouping. Mixed inclusion states are labelled. Unchecking an operation only hides its effect in the preview; it does not undo it.

All exclusions replay the retained recorded operations in their original order, using their recorded parameters and array operands. A later operation can depend on rows removed by an earlier one. An incompatible combination is rejected for the entire scope, with an explanation; the last valid view remains visible. Recorded Y replacements remain absolute replacements when replayed.

Models, residuals and plugin annotations are hidden for reconstructed/original data because their fits refer to the stored processed data. Error bars and framing use the displayed arrays. The banner always states **Display only** and **Data unchanged; no operations undone**. **Close all previews** returns every spectrum to its stored display; **Close preview for this scope** affects only the chosen scope. Starting an edit, mask operation, graphical placement or fit closes reconstructed previews and uses the stored data and processing. Background-only visual subtraction remains compatible with the normal fit overlay. These preview choices are temporary and do not modify processing history, Undo/Redo, exports or the saved project's scientific data.

During graphical cubic-spline placement, nodes may be placed anywhere in plot coordinates, including outside the x/y extent of the measured data. Adding nodes does not autorange the graph. Left-drag continues to pan and the mouse wheel continues to zoom while placement is active.

8.1 Component composition

New models use **global function roles**, independent of insertion and display order.

Operator	Global role
`add`	Include the function in the starting sum
`subtract`	Subtract the function from the starting sum
`multiply`	Multiply the whole intermediate model by the function
`divide`	Divide the whole intermediate model by the function
`convolve`	Convolve the whole intermediate model with the function

For example, two Add peaks G1 and G2 plus a Multiply function M produce `(G1 + G2) * M`, regardless of which function was added first. Additive peaks and backgrounds normally use Add. Background marking requires Add or Subtract; unmark a background before changing it to a factor or kernel.

Calculation order lists individual Multiply, Divide and Convolve functions. Move them up/down or drag them to choose application order, from top to bottom. Its expression preview distinguishes `convolve((G1 + G2) * M, R)` from `convolve(G1 + G2, R) * M`; those calculations can give different results. Only OK applies the choice; Cancel discards it. Changes support Undo/Redo and persist in projects, model files and copied structures. Without an explicit choice the default is multiplication, then division, then convolution, independent of insertion order.

With no Add/Subtract terms, multiplication and division start from 1; a convolution without a preceding signal starts with its first kernel. One Multiply function is therefore itself, one Divide function is its reciprocal, and one Convolve function is itself. An empty or fully disabled model is zero. Consecutive convolutions use full FFT convolution with uniform x-spacing scaling and crop once back to the input length, so inserting the same kernels in another order does not change edge handling. Convolution uses full FFT integration with absolute uniform X spacing and a single centered crop for consecutive kernels. A finite, strictly monotonic, uniform grid is required at every scale. For even-length signals, kernels are sampled on an odd grid with an exact central lag, removing the half-sample shift; ascending and descending axes give the same physical center. Kernel coordinates retain the input-domain midpoint convention. Choose a sufficiently wide, finely sampled window: finite-window truncation and discretization remain approximations. Fitting evaluates the

full current grid before selecting residual rows. Marked additive backgrounds receive the same global operations as the full model.

Add/Subtract terms use compensated summation in construction order independent of random component IDs. The cancellation example `1e16+1-1e16` now retains 1. Compensation reduces rounding loss; it is not arbitrary-precision arithmetic.

The same global roles apply to every project, model file and workflow, including files saved by earlier versions. There is no separate ordered-composition mode or first-function restriction. A model previously calculated in insertion order can produce a different result with global roles: review its expression in Calculation order and refit after opening it.

Division by a component that reaches zero and other non-finite model evaluations are rejected by the fitter.

8.2 Built-in peak parameterization

Every built-in peak uses signed integrated `area`, not peak height, as its intensity parameter. Negative areas produce negative peaks. This common parameterization makes areas comparable between different line shapes.

Gaussian

Parameters: `area`, `center`, `sigma`.

$$G(x) = \frac{A}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x - x_0}{\sigma}\right)^2\right].$$

The derived FWHM is $2.354820045\,\sigma$. `sigma` is intrinsically positive.

Lorentzian

Parameters: `area`, `center`, `gamma`.

$$L(x) = \frac{A\gamma}{\pi[(x - x_0)^2 + \gamma^2]}.$$

`gamma` is the half width at half maximum, so FWHM is 2γ. `gamma` is intrinsically positive.

Voigt

Parameters: `area`, `center`, `sigma`, `gamma`.

The Voigt component is the area-normalized convolution of a Gaussian with standard deviation `sigma` and a Lorentzian with HWHM `gamma`. CurveMole evaluates it with the SciPy Voigt profile. Both width parameters are intrinsically positive.

The displayed FWHM is the approximation:

$$\text{FWHM} \approx 0.5346(2\gamma) + \sqrt{0.2166(2\gamma)^2 + (2.354820045\,\sigma)^2}.$$

Pseudo-Voigt

Parameters: `area`, `center`, `fwhm`, `eta`.

The profile is an area-normalized linear mixture with a common FWHM:

$$pV(x) = A[(1-\eta)G_{\text{unit}}(x) + \eta L_{\text{unit}}(x)].$$

`fwhm` is intrinsically positive and `eta` is intrinsically bounded to $[0, 1]$. `eta=0` is Gaussian and `eta=1` is Lorentzian.

8.3 Built-in functions commonly used as backgrounds

The library names are **Constant**, **Linear**, **Polynomial** and **Cubic spline**. They become background components only when **Add as background** in the Add dialog or **Mark as background** in the model panel is enabled.

Constant

Parameter: `offset`.

$$B(x) = c.$$

Linear

Parameters: `intercept`, `slope`.

$$B(x) = c_0 + c_1 x.$$

Polynomial

Choose an order from 0 to 50 when adding the component. Parameters are `c0`, `c1`, and so on through the chosen order:

$$B(x) = \sum_{k=0}^{n} c_k x^k.$$

High polynomial orders are numerically sensitive when x values are large or poorly scaled. Prefer a low order or rescale x with the Data Calculator when scientifically appropriate.

Cubic spline

Spline x nodes are metadata, rather than optimizer parameters. Their y values (`y0`, `y1`, and so on) are normal parameters that may be fitted, fixed, bounded, or linked. Two nodes produce linear interpolation. Three or more nodes produce a natural cubic spline, with extrapolation outside the node range.

New graphically placed spline y nodes are fixed by default so an intentionally drawn baseline does not move when fitting begins. Uncheck **Fixed** for individual nodes, or use **Lock all** / **Unlock all** below the parameter table for every parameter in the selected function. The spline remains continuous through masked intervals even though masked measurements do not contribute to the objective function.

8.4 Functions: selection, insertion and details

Choose a registered function from the selector inside the **Functions** toolbar group and press its **Add** button. The list includes built-in peaks and backgrounds, generic functions, project-contained formulas, reusable formulas, and trusted plugin functions. The last choice is remembered. Below the selector, check **Background** to mark the added function as background and **Initialize with points** to initialize it from graph points. Both options default to off.

- a peak with **Initialize with points** checked allows repeated centre/FWHM click-drags until **Finish**, right-click or Return/Enter ends placement; its parameters remain free for fitting;
- a peak with that option cleared is added immediately, one per **Add** click. CurveMole estimates centre, height and width from the most prominent positive or negative peak in the finite, unmasked data minus the current model. Repeated clicks estimate the remaining residual peaks. Short or flat data use a finite fallback estimate; all estimates can be edited before fitting;
- a cubic spline with **Initialize with points** enabled opens point-by-point node placement;
- Linear and other functions use manual-point placement only when explicitly enabled using **Initialize with points**;
- non-peak functions with point initialization off are added with their initial parameters; a spline in this mode uses an initial node grid with free y parameters.

The arrow beside **Add function** chooses how the new function combines with the existing model: **Add (+)**, **Subtract (-)**, **Multiply (x)**, **Divide (/)**, or **Convolve (*)**. The orange symbol on the insertion icon shows the current choice; clicking the main part of the button inserts with that composition. The same choice is available in Function details and the advanced Add dialog and is remembered. All roles are available in every model, including for the first function. Background marking is available for Add and Subtract only. For automatic or graphical peak initialization, Add uses data minus the existing model, Subtract uses model minus data, Multiply uses data/model, and Divide uses model/data. Non-finite ratios are excluded. Automatic convolution inserts a kernel centred on the sampled domain; built-in area-normalized peaks start with unit area. All initial estimates remain editable.

The Functions group uses the shared function registry and peak parameter roles, including roles supplied by user formulas and plugins. Selecting a function in the normal Add dialog also makes it the next toolbar choice. The Functions group remembers its own manual-point and background options separately for each function. It does not inherit options from the normal Add dialog or its older shared preferences. Changing the point option during peak placement cancels the pending peak without adding anything; already completed peaks remain. Switching to an automatic peak in the selector also waits for the next **Add** click.

The **Function details** button (a function card with a curve and formula), inside the Functions group, replaces the old toolbar Add component button. It follows the selected function and opens a reference with its formula, description, default parameters, bounds, units, derived quantities and initialization instructions. Its **Function** selector also lets you choose any other registered function, including formulas and plugin functions. This selector and the toolbar selector stay synchronized; the reference updates immediately. Browsing does not add a component and cancels unfinished graphical placement without removing components already added. **Copy formula** copies the displayed expression; **Add function** opens the advanced Add component dialog with that function selected. The Model menu and Ctrl++ still open that dialog directly. Plugin formulas are displayed when supplied by the provider.

In the Functions group's point placement, right-click and Return/Enter commit the same points as **Finish**, provided the minimum has been reached; otherwise placement stays

active. **Undo point** removes the most recent point. In the regular Add dialog's point placement, right-click continues to remove the nearest point.

8.5 Adding a peak with the pointer

In the toolbar Functions group, choose a peak, enable **Initialize with points**, and press **Add function**:

1. click the intended center;
2. drag horizontally to either half-height edge;
3. release to set the initial full width.

A green region previews the width. If no horizontal width is selected, CurveMole uses a default based on 5 percent of the x span, local point spacing, and floating point precision.

For built-in peaks, CurveMole converts the graphical height and FWHM into the native area and width parameters. New custom peaks should declare explicit semantic roles for their position, intensity, and width parameters in Function Builder. Older custom functions without role metadata retain compatibility through conventional names such as `center`, `x0`, `fwhm`, `sigma`, `gamma`, `area`, `amplitude`, or `height`.

Construction options and initial parameters

Add function asks for construction options when the selected function declares them. For **Polynomial**, choose its degree (0-50) before adding it when **Initialize with points** is disabled. In point mode, options declared as derivable from points are not requested beforehand: N distinct polynomial points define degree N-1. The current degree appears in the placement instructions. Removing a point reduces the degree, and moving a point recalculates the coefficients. Cancel leaves the model unchanged. The last confirmed construction options are remembered per function and shared with the advanced Add dialog. Background and point-initialization choices still apply. Functions without construction options retain immediate insertion.

In the advanced **Add component** dialog, the parameter table provides **Initial value**, **Minimum**, **Maximum**, and **Fixed** for every parameter. Construction options update the table; changing polynomial degree preserves edits to coefficients that remain present. Limits apply to the subsequent fit as well as initialization. Use `-inf` and `inf` for unbounded limits within the function's allowed domain. Invalid values are highlighted and prevent insertion until corrected.

Restore defaults restores the selected configuration's parameter values, bounds, and fixed states. **Estimate from data** estimates values using usable, unmasked points of the active spectrum, respecting bounds and leaving fixed parameters unchanged. This is a standalone component estimate against the spectrum, not a fit of the complete composed model. Inspect and edit the estimate before adding. With manual-point initialization disabled, **OK** inserts exactly the displayed configuration, including for peaks and splines; no later automatic or graphical estimate replaces it. Enabling manual-point initialization disables the table and its estimation controls because the selected graph points determine those values.

Developers and plugin authors can declare `FunctionDefinition.configuration_specs` using `FunctionOptionSpec` (integer, number, boolean, text, or choice). An option's `from_points(x, y)` callback declares how graph points determine it. Only options without that callback are requested in point mode. `FunctionDefinition` can also declare `point_initializer(x, y, metadata)` and `minimum_points`; the same rules are used for previews and insertion, including custom and plugin functions. The same controls, validation and component metadata are used by quick and advanced insertion. Polynomial initialization solves on a scaled domain, then converts to the existing `c0 + c1*x + ...` coefficient convention. Numerically unreliable point sets block Finish with an

explanation; previews of inferred functions stay within the selected point interval to avoid misleading extrapolation.

8.6 Initializing any function from manually selected points

In Add component, enable **Initialize from manually selected points on the graph**. This is selected by default for Linear and Cubic spline and can be used for other built-in, custom or plugin functions. For a custom formula, Function Builder can set **Use manual-point initialization by default**.

Click explicit points representing the function you want to create. A live green function preview updates as points are added or removed; it is an initial estimate, not a fit to the measured spectrum. Linear requires at least two points. Other non-spline functions normally require at least two points and at least as many points as parameters. Point-derived configurations can declare their own minimum; Polynomial starts with two distinct x positions and adapts its degree as points are added or removed. Cubic spline requires at least two distinct x positions.

The placement bar offers **Add points**, **Delete points** and **Move points**. Delete points removes the point you click; clicking empty space does nothing. Move points lets you click and drag a point to a new x/y position, with a live function preview. Drag empty space to pan and use the mouse wheel to zoom. **Undo point** reverses the last addition, deletion or completed move. These edits affect the function's initialization points, leaving the experimental data unchanged.

Use **Finish** (or double-click in Add points mode) to commit after the minimum number of points. Deleting below that minimum disables Finish until enough points are restored or added. In Quick Add, right-click or Enter also finishes; in the advanced Add component flow, right-click removes the nearest point. **Esc** cancels the whole insertion. The resulting parameters are fixed by default. Use **Unlock all** or individual **Fixed** checkboxes before optimizing them. Points and the corresponding controls are stored with the component and project.

Reopening a function's control points

Right-click a function in **Model and parameters** and choose **Edit function points** (the same command is in **Model**). Saved control points reopen in the placement bar, initially in **Move points** mode. Switch to **Add points** or **Delete points** as needed; **Undo point**, pan, zoom and the live preview remain available. Polynomial degree follows the current point count. This works for every function with saved manual points, including custom and plugin functions. Older graphical peaks without stored points can reconstruct controls from their current shape; normal peak centre/width handles remain available outside this editor. Peak controls follow the current shape after fitting or dragging, and finishing without moving points leaves the function unchanged. Peak edits use the same estimate in the live preview and on Finish.

For a function marked **Background**, the active spectrum temporarily displays without background subtraction, including a display reconstruction of recorded subtractions. Other spectra retain their display options. Non-background functions keep the current background display. Opening the editor or cancelling changes only the view. On **Finish**, a background that was already subtracted is recalculated with its new function, replacing its existing subtraction rather than subtracting it twice. Grouped backgrounds and later processing are replayed together. The function change and updated subtraction form one **Undo**/**Redo** operation; original imported arrays remain unchanged. A background that has not yet been subtracted is only updated as a model function.

Finish updates the selected function in place, preserving its identity, name, composition and background role, as one **Undo**/**Redo** operation. Existing bounds, fixed states and links

for retained parameter names are preserved; deleting a parameter referenced by another function is rejected. **Cancel** or **Esc** discards the draft; changing spectrum/function or starting another edit also cancels it. The previous display options and viewport return when the editor closes. Control points are saved with the project and remain editable after reopening it.

Drawing a cubic-spline background

After choosing **Cubic spline**, enable **Add as background** and keep manual-point initialization enabled:

1. left-click background nodes on the plot;
2. inspect the live green preview, which updates after each click;
3. use **Undo point** if necessary;
4. finish with **Finish** or double-click after at least two nodes; right-click removes the nearest node;
5. press **Esc** to cancel the entire placement.

Nodes are sorted by x. Clicking again at effectively the same x updates that node instead of creating a duplicate. Exact duplicate x positions are not permitted.

8.7 Graphical editing after placement

Select a peak component to display its handles. Drag the central target to change center and area-derived height. Drag a side line to change FWHM. For a Voigt peak, changing FWHM graphically scales `sigma` and `gamma` by the same ratio, preserving their current relative contribution.

For a function created from manual points, selecting it displays its original control points. Unlock the relevant parameters before dragging a point in x and y. For a spline, unlock that node's y parameter: dragging also updates its stored x position and keeps nodes sorted. For other functions, at least one parameter must be free; CurveMole refines those free values against the moved control points while retaining fixed and linked values. Each completed drag is undoable. Exact numeric parameter edits are available in the table.

For normal peak centre/width handles, fixed values cannot normally be changed by a drag. Hold **Ctrl** while dragging to change the stored value while leaving the parameter fixed. Manual-point handles instead require unlocking the relevant parameters. Linked values cannot be dragged independently because an expression controls them.

8.8 Automatic peak suggestions

Choose **Model > Find positive peaks...**. Despite the menu label, the first dialog can search positive, negative, or both signs. The next dialog selects any registered function classified as a peak, initially preferring the peak selected in the Functions group. CurveMole then:

- subtracts the median as an initial baseline;
- estimates noise through a robust median absolute deviation;
- uses a default prominence of the larger of three noise estimates or 1 percent of the y range;
- estimates width at half prominence;
- sorts suggestions by prominence.

Choose how many suggestions to add. Every suggestion is initialized using the selected peak function's explicit parameter roles or, for legacy functions, recognized parameter names. Suggestions are initial estimates, not a scientific decision about the number, identity, or line shape of peaks.

8.9 Duplicating, disabling, deleting, and reordering

Duplicate a component to create an adjacent copy with a new internal identifier. Disable a component to compare models without losing its settings. Deletion and reordering are undoable. In global models the function list order is independent of the mathematical order; use Calculation order to change precedence explicitly.

The **Reorder** button sorts the function list in ascending parameter order (peak `center`, or X position, by default), including peaks of different types. It also renumbers automatically named functions within each type to match that order. IDs, links and custom names are preserved; custom-named functions are sorted too. Functions without a finite value for their sorting parameter keep their list positions, and equal values retain their relative order. The adjacent **Advanced reorder rules...** button selects another sorting and numbering parameter for each function type and remembers those rules. Both the list order and automatic names are restored by Undo and reapplied by Redo. **Reorder changes presentation only:** parameter values, links, fit results and the mathematical composition order remain unchanged, including multiplication, division and convolution. The display order is saved with the project and copied with the model. Use Calculation order to change precedence in every model.

8.10 Copying a fit to other curves

Model > Copy fit copies selected information from the active curve. Use **Select all** or **Deselect all** when choosing many target curves. The dialog initially shows **Active series**; **All series** exposes the entire project, grouped under colored series headings. Only targets in the displayed scope are copied.

Copy fit to next, beside Copy fit, opens the same options with the next spectrum in the source series already checked. After a successful copy, that spectrum opens automatically in the plot and parameter panel. Cancelling the dialog leaves the active spectrum unchanged. The action is unavailable at the end of the series. Copying remains undoable. **Functions to copy** lists every source function with its role; check only those you want, or use Select all functions / Clear functions. Disabled functions can also be copied and retain their disabled state. Turning off Background hides and excludes background functions without forgetting their checks.

With **Component structure** checked, each target model is replaced by the checked functions, preserving their composition mode and calculation order. With structure off, only their original corresponding target positions are updated; skipping a source function does not shift another function into its place. Other target functions remain unchanged. Relative links to omitted functions must still resolve; otherwise the entire copy is rejected, including masks and ranges. Include the referenced function or disable Internal links. Undo restores the complete target state, including copied masks and ranges. Options include:

- component structure;
- current or best values;
- bounds and fixed states;
- internal links;
- background components;
- masks;
- fit ranges.

Masks and fit ranges are excluded by default because x grids and excluded regions may not correspond scientifically. When copying masks, point transfer uses the configured x tolerance. Review every target model after copying.

9. Parameters, constraints, and links

9.1 Parameter table

The table columns are:

Column	Meaning
Parameter	Parameter name; parameters with a relation show its indicator
Value	Current value or successful fitted value
±1 sigma	Covariance-based standard error when available
Analysis - / +	Lower/upper distances from this spectrum's saved result for the project-wide display method; click this heading to choose a method, with a dash if its interval is unavailable
Fixed	Exclude the parameter from optimization
Lower	Lower bound; blank means negative infinity
Upper	Upper bound; blank means positive infinity
Link	Graphical **Set link...** control; the relation and its source are shown in readable form

Press **Set link...** for the target parameter. Its name appears beside the expression: for a centre link, the dialog shows `center =` followed by the expression field. **Equal to source** is the default. The field shows the selected source explicitly and is read-only for quick relationships; **Advanced expression** makes it editable. Use **Remove link** to remove the relation and make the parameter independent again.

An **Equal to source** link determines the value instead of fitting it independently. Its Fixed cell shows **Linked**, and attempting to fix it explains that you must remove the link first, then set the value and select **Fixed**. Lock all and parameter copying enforce the same rule. Clear Fixed before adding an equality link to a fixed parameter. Numeric Lower/Upper limits and inequality links can still constrain a fixed value. These rules apply to every project.

Choose a source spectrum

The **Source spectrum** menu starts with **Same spectrum as this parameter**. Below it, **Specific spectrum (fixed when copied)** groups spectra under the same series headings and in the same order as the project's spectrum list. Each choice explains what happens when the functions are copied:

Source spectrum choice	After copying to another spectrum
Same spectrum as this parameter (default)	Use the corresponding source function in the destination spectrum
A spectrum under **Specific spectrum (fixed when copied)**	Keep referring to that exact spectrum, even if it is the current spectrum

For example, link Gaussian2's `center` to Gaussian1's `center` in Spectrum 1. After copying the functions to Spectrum 2, **Same spectrum as this parameter** uses Gaussian1 in Spectrum 2. Selecting Spectrum 1 explicitly under the fixed spectra instead keeps Gaussian2 in Spectrum 2 linked to Gaussian1 in Spectrum 1. Copy behavior is saved for each inserted reference, so one advanced expression can combine relative and fixed references without losing their meaning. Undo/Redo, parameter copying and sequential model propagation preserve those choices.

Build an expression from several parameters

In **Advanced expression**, **Source function** and **Source parameter** select a parameter to insert. Click **Add** to insert its readable reference at the expression cursor, then type arithmetic operators, numbers and parentheses as needed. You can insert any number of references and use the same parameter more than once. Selecting another spectrum, function or parameter does not change references already inserted; click **Add** to insert the new selection.

You can also type `(Gaussian2.center + Gaussian3.center) / 2` directly, with or without `${...}` around individual references. New names use the selected **Source spectrum**; once recognized, each reference keeps that spectrum and copy behavior when the selectors change. Use `${Left peak.center}` for function names with spaces, or `${Series / Spectrum / Function.parameter}` for a fixed spectrum reference. Unknown names and duplicate matching names are rejected with a readable message; use **Add** to choose the exact source when names are ambiguous. Typed references are saved with stable identities, just like references inserted with Add.

To set Gaussian1's `center` to the mean of Gaussian2's and Gaussian3's centres:

1. open **Set link...** for Gaussian1's `center` and choose **Advanced expression**;
2. keep **Same spectrum as this parameter** and type `(` in the expression;
3. select Gaussian2, select `center`, and click **Add**;
4. type `+`, select Gaussian3 and `center`, and click **Add** again;
5. type `) / 2` and check the displayed `center =` relation before accepting.

The resulting relation reads `(Gaussian2.center + Gaussian3.center) / 2`, with each parameter shown as an inserted reference. CurveMole retains its stable identity even if the function is later renamed or the list is reordered. References to other spectra also show their spectrum context, so equal names in different spectra remain distinguishable. Existing expressions using `${source}` remain supported, but **Add** lets you build a multi-parameter expression without typing internal identifiers.

Choose a quick relationship

Quick relationships use the selected source parameter and show the resulting relation explicitly:

Relationship	Meaning for the target parameter
Equal to source	`target = source`; the target is derived from the source
At least source	`target >= source`; the target remains a free fitted parameter
At most source	`target <= source`; the target remains a free fitted parameter
Similar to source	`source - tolerance <= target <= source + tolerance`; the target remains free within this interval

Relationship	Meaning for the target parameter
Advanced expression	`target = expression`; the target is derived from all references in the expression

For **Similar to source**, enter a positive tolerance in parameter units or as a percentage. An absolute tolerance of `2` means an interval from `source - 2` to `source + 2`. A percentage tolerance of `5` uses `0.05 * abs(source)`, including when the source value is negative. Percentage tolerances must be positive and less than 100.

The source-dependent bounds in **At least**, **At most**, and **Similar** are hard constraints, recalculated throughout fitting as the source changes. They are combined with the target's ordinary Lower/Upper cells and intrinsic function bounds; they do not replace these static limits or impose a soft penalty. The target remains an independent optimizer variable within the allowed interval, while **Equal to source** and **Advanced expression** remove it from the free parameter count. If the allowed interval collapses to a single value, mark the parameter **Fixed** or use **Equal to source**. This also applies to a percentage tolerance around a source fixed at zero, whose allowed interval is `[0, 0]`.

Relations between different spectra require **Global simultaneous** fitting with all referenced spectra included. CurveMole rejects dependency cycles, including cycles involving source-dependent bounds. An edit is validated immediately: invalid expressions, missing parameters, inconsistent bounds and infeasible constraints are reported explicitly. If a constraint becomes infeasible during fitting, the fit reports an error rather than silently ignoring the relation.

9.2 Parameter states

A parameter can be free to fit, fixed, or derived by an equality/expression link. Free parameters can also have lower, upper or interval bounds. Source-dependent **At least**, **At most**, and **Similar** relations add dynamic bounds while keeping the target free; an equality or advanced expression instead derives its value and removes it from the independent optimizer variables. Bounds can be combined with these relationships, and all active limits must be satisfied.

Intrinsic function bounds always remain active. For example, clearing the visible lower cell cannot make a Gaussian `sigma` negative because the function definition sets a positive intrinsic minimum.

9.3 Parameter paths

The canonical path is:

```
curve_id.component_id.parameter_name
```

A link refers to a path inside `${...}`. For example:

```
${curve_ab12.component_cd34.center}
```

or a derived relation:

```
${curve_ab12.component_cd34.center} + 12.5
```

The graphical **Set link...** dialog creates these paths automatically, so normal GUI use does not require knowing curve or component identifiers. The canonical expression is still shown as a tooltip and remains available to advanced workflows, Python code, and machine-readable exports. For a link between different spectra, select both spectra and use **Global simultaneous** mode; CurveMole blocks incompatible fit modes with an explicit message.

9.4 Link evaluation

Expression links and source-dependent bounds form a directed dependency graph. CurveMole resolves the graph before fitting, detects cycles, and reevaluates the relations at each model evaluation. Equality and advanced-expression targets are derived values rather than independent optimizer variables. If covariance is available, their standard errors are propagated numerically from free parameters. Nonlinear link gradients use scale-aware three-point differences and Richardson extrapolation before propagating covariance. The small cubic-link reference now agrees with its analytic derivative; these remain local uncertainties (Appendix J.3). Targets with **At least**, **At most**, or **Similar** relations remain independent optimizer variables constrained by their current allowed intervals.

9.5 Safe expression language

Formulas and links are parsed as a restricted expression tree. They are never passed to unrestricted Python `eval` or `exec`.

Allowed arithmetic includes `+`, `-`, `*`, `/`, power `**`, remainder `%`, unary signs, comparisons, boolean combinations, and conditional expressions. Constants are `pi`, `e`, and `inf`.

Allowed functions are:

```
abs, sqrt, exp, log, log10,
sin, cos, tan, arcsin, arccos, arctan,
sinh, cosh, tanh,
erf, erfc,
minimum, maximum, clip, where, heaviside
```

Attribute access, imports, indexing, comprehensions, arbitrary function calls, file access, and network access are not part of the expression language.

10. Fitting

10.1 Pre-fit checklist

Before pressing **Ctrl+F**:

1. confirm the active and selected curves; use **Select all** / **Deselect all** when appropriate;
2. inspect invalid and masked points;
3. choose a physically defensible background and peak count;
4. provide reasonable initial values;
5. add justified bounds for poorly determined parameters;
6. inspect links for cycles or unintended dependencies;
7. check that each selected model contains at least one free parameter.

10.2 Fit modes

Single / independent

Each selected curve is fitted as an independent problem. Results are then combined for reporting. Parameter links between different independently fitted curves cannot act as a simultaneous cross-curve constraint; use Global mode for that purpose.

CPU processes (independent fits) in the advanced settings runs multiple spectra in separate processes, up to the selected count and the available CPUs. Both built-in functions and user-defined formulas support this, including mixed models. Formula definitions are reconstructed in each worker; fitted parameters, covariance and completed-spectrum results are returned to the main workspace. Executable plugin functions and plugin solvers retain their single-process execution path.

Sequential

Sequential mode is a propagating refinement. Choose an **Initial source spectrum** whose model and parameters have already been inspected and approved. The source is not refitted. Its model is cloned to the next selected spectrum, fitted there, and the result becomes the source for the following spectrum. Selected spectra before the chosen source are excluded; target spectra do not need models in advance.

Use **Functions excluded from copying** to select any source functions to leave out. Functions marked as background are excluded by default; uncheck one to copy it. Functions already present on each target retain their values, metadata, constraints, and order during copying; the fit still respects their fixed/free state. A function with the same internal identity is kept rather than duplicated. Distinct functions remain distinct even if their names match. Target-local functions are not carried into later spectra. The selection is also preserved when resuming a paused sequence. This is independent of ignoring functions in the parameter-change pause monitor.

For newly copied functions, parameter values and structure are propagated. The dialog separately controls whether to preserve bounds, fixed/free state, internal links, background tags, enabled/disabled state, and composition/grouping. References inserted with **Same spectrum as this parameter** are remapped to corresponding target parameters; explicitly selected fixed spectra keep their original references. Cross-spectrum links and source-dependent bounds require Global simultaneous fitting rather than independent or

sequential fitting. Structural metadata required by the function, such as spline nodes and custom-formula metadata, is always retained.

Two optional safeguards can pause a sequence:

- normalized residual RMSE worsens by both a selected factor and a minimum fraction of the signal scale; defaults are 2.5 times and 2 percent;
- a free parameter changes by more than a selected normalized amount; the default is 75 percent.

Specific source functions can be excluded from the parameter-change trigger while remaining fully copied, fitted, and included in the residual. If fitting or a safeguard pauses the sequence, the affected spectrum becomes active and a persistent **Continue sequential fit** button appears in the status bar. Inspect or edit that spectrum, then continue; the accepted current model becomes the new propagation source. CurveMole does not silently skip a paused spectrum. The warning explains the affected spectrum, reason, measured change and threshold, and next action. **Last pause reason**, beside the sequential controls, reopens the latest warning while the sequence is paused, including after a manual fit used to correct the paused spectrum. The button disappears when the sequence completes or is terminated. The message is saved with the project and is replaced when a later sequential fit pauses.

Global simultaneous

All selected residual arrays are concatenated into one optimization problem. Models may differ between curves. Parameter links can connect curves and express shared, offset, scaled, or otherwise related values. Source-dependent **At least**, **At most**, and **Similar** bounds can also connect spectra; include every referenced spectrum in the global fit.

10.3 Per-spectrum weights

The Fit dialog uses the same **Active series / All series** scope for independent, sequential and global fits, with colored series headings. Hidden spectra are excluded from the plan; the sequential source chooser follows the same scope. Weights and checkbox choices are retained while switching scope. Function selection and enabled/background checkboxes are described in Section 5.4.

The Fit dialog accepts a positive numeric spectrum weight for every selected curve. If s_j is the spectrum weight, all point residuals from spectrum j are multiplied by $\sqrt{s_j}$.

Scale each spectrum to equal numerical contribution additionally divides the residual vector of each curve by the square root of its number of usable points. This prevents a densely sampled curve from dominating only because it contains more rows. It does not normalize signal amplitude or uncertainty and is never enabled silently.

10.4 Local least squares

The default is local nonlinear least squares using SciPy. CurveMole automatically selects:

- Levenberg-Marquardt for an unbounded ordinary least-squares problem;
- trust-region reflective when bounds or a robust loss require it, or when a global fit with at least 16 free parameters has a sufficiently sparse dependency structure. This avoids a dense residual-by-parameter matrix; equality and relational links between spectra remain part of the problem.

Built-in optimizers use dimensionless coordinates based on parameter and data scales; numerical Jacobians, automatic DE limits and bound diagnostics use those scales rather than a floor of one physical unit. Positive built-in widths have a float64 normal minimum (about `2.225e-308`), rather than machine epsilon. Explicit bounds remain preferable for difficult searches. Extreme overflow, underflow, nonidentifiability and poor starting values still require scientific judgment. Optimizer coordinate tolerances, including explicit `x_scale`, apply after this normalization; supplied Nelder-Mead simplex vertices remain physical parameters.

The graphical dialog exposes the maximum number of evaluations and confidence level. New fit plans default to 1,000 evaluations and use a step tolerance that permits a successful early stop when parameter changes become numerically negligible. During local optimization, the status progress is updated and the plotted trial model is refreshed every 20 evaluations. These refreshes preserve the user's current zoom and do not replace the stored parameters unless the fit completes successfully. Advanced algorithm options, the Python API and YAML settings expose tolerances, scaling, local method, random seed and Differential Evolution controls. The solver menu also offers explicit TRF, Dogbox, and LM, plus Nelder-Mead, Powell, and L-BFGS-B. The latter three minimize the selected loss directly; they can be useful for difficult models but may take more evaluations. LM needs unbounded parameters and `linear` loss. Hover over a solver to see its intended use.

10.4.1 Advanced algorithm options and Reset default

The default remains **Local least squares (automatic)** with **linear** loss. **Advanced algorithm options** shows only the controls relevant to the chosen solver. It also contains **Use data errors / weights in the fit**, enabled by default; this choice applies to independent, sequential and global fitting and replica refits. Numeric fields accept scientific notation such as `1e-8`; tooltips explain each option.

Algorithm	Configurable controls
Automatic, TRF, Dogbox, LM	`ftol`, `xtol`, `gtol`, `x_scale`; automatic mode also offers the local method
Differential Evolution	Generations, population multiplier, mutation minimum/maximum, recombination, relative/absolute tolerance, seed and automatic search limits; least-squares controls apply to local refinement
Nelder-Mead	Iteration limit, `xatol`, `fatol`, adaptive coefficients and optional initial simplex
Powell	Iteration limit, `xtol`, `ftol`
L-BFGS-B	Iteration limit, `ftol`, projected-gradient tolerance, line-search steps and correction history

Maximum evaluations remains configurable for all built-in solvers. For Differential Evolution it limits the local refinement; generations/population control the global search. Numerical Jacobian probes and L-BFGS-B numerical gradients may require additional model calls beyond the solver's reported evaluation budget.

An iteration limit of `0` retains the minimizer's automatic limit. An optional Nelder-Mead simplex is a JSON matrix with N+1 rows and N columns in the fitter's free-parameter order (selected curve, component and parameter order, including referenced link dependencies). Leave it blank for normal automatic initialization. Vertices must satisfy bounds. LM requires unbounded parameters and linear loss.

Reset default restores all algorithm/loss parameter fields, evaluation budget and confidence level. It preserves the selected algorithm and loss, fit mode, spectrum

selection/weights and model parameters. Defaults include 1,000 evaluations, `ftol=1e-10`, `xtol=1e-8`, `gtol=1e-10`, `x_scale=jac`, and loss scale 1. DE defaults are 400 generations, population multiplier 15, mutation 0.5-1, recombination 0.7, relative tolerance 0.01, absolute tolerance 0 and seed 1729. Nelder-Mead and Powell tolerances default to `1e-4`; L-BFGS-B uses `ftol=2.220446049250313e-9`, `gtol=1e-5`, 20 line-search steps and 10 corrections. These are starting configurations, not guarantees for every dataset.

10.5 Robust losses

Available losses are:

- `linear` for ordinary least squares;
- `soft_l1`, which smoothly weakens the effect of large residuals;
- `huber`, squared near zero and nearly linear for large residuals;
- `cauchy`, which strongly suppresses very large residuals.

Hover over the loss options in the Fit dialog for short descriptions. Losses alter the penalty assigned to residuals, not the mathematical peak shape.

Robust loss can reduce the influence of large residuals, but it is not a substitute for understanding outliers, detector artifacts, or an incomplete model. AIC, AICc, and BIC require linear loss and a supported Gaussian noise model: active interval-error arrays (including equal half-widths) and arbitrary spectrum balancing also make them unavailable. Their likelihood uses point precision and counts any estimated noise scale. Robust-loss covariance uses a sandwich estimate based on the original weighted-residual Jacobian and the chosen loss's influence function; a singular sensitivity makes that estimate unavailable. CurveMole recommends explicit resampling.

For `soft_l1`, `huber` and `cauchy`, the **Loss scale (f_scale)** field appears. It must be positive and sets the transition between small and large residuals. It is hidden for `linear`, where it has no effect. The scale is expressed in the residual units passed to the optimizer: signal units without uncertainty weighting, standardized residual units with `sigma_y`, further modified by spectrum weights and equal-contribution scaling. The default of 1 is not an automatic noise estimate; choose a scale appropriate to the data. Reset default restores it to 1.

10.6 Differential Evolution initial search

Choose **Differential Evolution + local refinement** when a local initial estimate is not sufficient. Differential Evolution searches globally, then passes its best point to the local least-squares solver.

When a free parameter lacks a finite bound, CurveMole creates one around its current value. By default, the search extends 50% of the absolute current value below and 50% above it. At zero, a scale based on the plotted data is used. Edit the separate **Below** and **Above** percentages in the Fit dialog; existing explicit bounds always take precedence. These temporary search limits do not change the parameter table or the saved model. Inspect them conceptually before interpreting the result: a global search cannot discover a solution outside its search interval. It is usually much slower than a well-initialized local fit.

10.7 Cancellation and failed fits

Use **Fit > Cancel running task** to request cancellation. The request is checked during residual evaluation and resampling. If a built-in fit or uncertainty task does not respond within five seconds, CurveMole retires it and releases the workspace. Calculations own detached inputs; late results from a retired task are ignored. Previous valid results remain available. An uncooperative executable plugin may continue computing in its detached daemon thread, but cannot keep the project locked or prevent the application from closing. Save and autosave can capture the current project while a calculation is active. Finalization keeps progress below 100% until the task has actually finished.

If the solver raises an error or does not converge, CurveMole restores the last valid parameter values instead of committing an incomplete optimizer vector. The curve is marked Failed, the error is written to the Log, and the last successful result remains separate from the failed attempt.

10.8 Common causes of failure

- no free parameters;
- fewer than two usable points;
- initial value outside bounds;
- non-finite model values;
- division by zero in a composed model;
- a missing or cyclic link;
- a linked value outside its bounds;
- incompatible static and source-dependent bounds, or an infeasible dynamic relation;
- an invalid or overly narrow Differential Evolution search interval;
- an underdetermined model;
- inadequate initial values or maximum evaluations.

11. Results, statistics, and diagnostics

11.1 Visual results

The main plot shows data, total model, individual component curves, and residuals. The parameter table shows fitted values and standard errors. A curve state changes to Fitted only after solver success.

The full numeric result is stored in the project and written to exports. The GUI does not provide a single comprehensive on-screen results table; use the analysis bundle for archival inspection.

11.2 Reported global statistics

Let N be the number of usable observations and k the number of independent free parameters. CurveMole reports:

Statistic	Definition or interpretation
Degrees of freedom	$N - k$
RSS	$\sum_i (y_i - f_i)^2$ without point weighting
RMSE	$\sqrt{\text{RSS}/N}$
Chi-square	Sum of squared weighted residuals
Reduced chi-square	Chi-square divided by $N - k$ when positive
Descriptive R^2	$1 - \text{RSS}/\text{TSS}$
AIC	$-2 \log L_{max} + 2K$ for a supported Gaussian noise model
AICc	$AIC + 2K(K+1)/(N-K-1)$ when $N > K+1$
BIC	$-2 \log L_{max} + K \log N$

Here k counts free model parameters and K additionally counts any fitted noise scales. Information criteria include Gaussian normalization constants.

The descriptive R^2 can be misleading for baselines, constrained fits, or models without an intercept. It is included as a familiar descriptor, not a universal model quality score.

11.3 Per-curve statistics

Each curve output records the original usable indices, x, observed y, fitted y, residual, weighted residual, N, RSS, RMSE, and descriptive R^2. The global result combines all curves in the plan.

11.4 Covariance and correlation

For ordinary least squares, covariance is derived from the fit Jacobian. If all curves provide `sigma_y` or declared confidence-interval Y errors and weighting is enabled, the uncertainty is treated as absolute by default and the covariance is not rescaled by residual variance. Otherwise residual variance scales the covariance.

CurveMole uses a pseudo-inverse when the Jacobian is rank deficient and reports a warning. Covariance is unavailable when degrees of freedom are not positive. A warning is

also generated when at least one free-parameter correlation has absolute value 0.95 or greater.

Symmetric normal intervals respect static bounds and any global limits implied by the relations. For **At least**, **At most**, or **Similar** with a free source, the source and target can move together: the target's marginal confidence interval is not clipped to a neighborhood that freezes the source at its fitted value. Reported effective lower/upper limits describe the allowed range at the optimum; confidence intervals use the applicable static/global limits instead of these conditional limits. A parameter at a bound is flagged because its true uncertainty is generally asymmetric and not adequately summarized by a symmetric covariance interval.

These are local normal intervals, not exact Student-t intervals for a small linear sample with an estimated variance. Linked errors use scale-aware gradients; bound diagnostics use local parameter scales and floating-point spacing. See Appendix J.1/J.3 for the audited regression cases.

11.5 Residual diagnostics

Open **View > Diagnostics** after a fit. The summary reports:

- Durbin-Watson statistic;
- number of residuals with absolute standardized value at least 3;
- lag-1 autocorrelation;
- warnings for potential outliers and autocorrelation beyond an approximate 95 percent white-noise band.

The analysis bundle exports the autocorrelation sequence for every fitted curve. A structured residual pattern usually indicates that the model, background, weighting, or data treatment deserves review.

12. Explicit uncertainty analyses

Open **Fit > Uncertainty Analysis** only after a successful fit. These calculations run in the background, can be cancelled, and may be substantially slower than the baseline fit. **Fit covariance** displays the uncertainty already calculated by the fit without running replicas. Monte Carlo, residual bootstrap and block bootstrap default to **Adaptive - stop when intervals stabilize**. **Fixed count** remains available with an initial count of **200 replicates** (not optimizer iterations). Selecting it shows a notice: fixed count does not check interval stability. Prefer Adaptive; use fixed count for tests or when a quick analysis is needed. The optimizer's default evaluation budget is unchanged.

Replica timeout (seconds) limits each Monte Carlo or bootstrap replica to **60 seconds** by default and can be changed before running an analysis. The timer starts when that replica begins calculating, excluding time waiting in the queue, and resets for every replica. It applies in fixed and adaptive modes, with one or multiple CPU processes. Built-in functions and custom formulas run in isolated workers, allowing CurveMole to stop even an unresponsive replica. Executable plugin functions and custom fitter subclasses use cooperative cancellation for the replica deadline; the task cancellation safeguard described in section 10.7 also applies to these analyses.

If a replica exceeds the limit, the entire active analysis stops and is marked **Uncertainty analysis failed (timeout)** for the selected method, with an amber status and a warning icon beside the spectrum. The results panel also shows an explicit timeout warning. The existing fit and previous completed analyses remain available; spectra already completed in the batch keep their results. A later successful analysis clears the failure tag. The timeout is recorded under **Technical details** and saved with results.

Ordinary failed replicas produce one warning for the spectrum, showing failed attempts and successful replicas used for the intervals. Even a single failure is reported, since excluding unsuccessful fits may bias intervals. This warning does not mark every parameter as **Attention**: parameter assessments retain their own diagnostics and may remain **OK**, meaning that those checks flagged no local issue. The spectrum warning still applies to the analysis as a whole. If the adaptive limit is also reached, both warnings are shown. Too few successful samples and parameter-specific diagnostic issues continue to be reported in assessments.

For Monte Carlo and both bootstrap methods, **Replicate mode** offers **Fixed count** or **Adaptive - stop when intervals stabilize**. Adaptive mode defaults to:

- 500 successful replicates for the first interval estimate;
- 200 additional successful replicates per checkpoint;
- less than 5% endpoint change, measured against each interval's current width;
- three consecutive stable checks;
- a hard limit of 10,000 attempts, including failed fits.

All five settings are editable. Each checkpoint recomputes percentile intervals using all successful replicates accumulated so far. Both endpoints of every free parameter must satisfy the tolerance; an unstable check resets the consecutive counter. With these defaults the earliest stable stop is at 1,100 successful replicates. Failed fits do not count toward the initial sample or subsequent blocks, but do count toward the maximum attempts. If the limit interrupts a block, the final intervals still use all successful replicates, and the result explicitly reports that stability was not reached. Identical zero-width intervals are treated as numerically stable; changing to a zero-width interval is not a stable check.

The limit applies separately to each independent spectrum, or once to a joint global fit. The report records the actual successes, failures, attempts, stopping reason, settings and checkpoint history, including when the project is saved and reopened. Reaching the limit

before stability is flagged in the assessment, in a prominent results warning, and beside the spectrum with an amber status and warning icon. The intervals are retained, but the required stability has not been demonstrated. Warnings follow the method selected in the analysis panel and survive save/reopen; switching to another method shows that method's status. This is a numerical stability criterion, not a guarantee of statistical accuracy or model validity. Covariance and profile likelihood do not use adaptive replicate counts.

Delete selected analysis from all spectra removes the method currently chosen in this panel's **Method** field from the entire project. It removes its reports, saved samples and timeout warnings, updates the spectrum tags and available display methods, and preserves all other methods, spectra and fitted parameters. **Undo** and **Redo** restore or remove it. The button is disabled in read-only projects and while a background task is running. The panel scrolls when its controls and results exceed the available height. The deletion applies to the shared selected method, independently of **Run on**. Both the analysis panel and Model and parameters show this same method. Save the project to persist the removal. Previously saved project files are not rewritten until saved; other archives and exported files are not deleted by this button.

CPU processes sets the maximum number of independent spectra analyzed in parallel, with remaining spectra queued. The default is up to four processes, limited by the available CPU count. Built-in functions and custom formulas support this parallelism for both selected spectra and all fitted spectra, in fixed and adaptive modes. A single spectrum or joint global analysis can parallelize its replicates instead; profile scans can parallelize their grid points. Analyses involving executable plugin functions use one process. Raising the process count can increase memory use, and it does not change the requested replicate count. The actual worker process count is recorded under **Technical details**.

Run on selects the active spectrum, spectra selected in the project tree, or all fitted spectra. Each independent spectrum uses its own last successful fit, even if another spectrum was fitted more recently. Switching the active spectrum updates the report shown in the panel. Results are saved separately for each spectrum and method. The spectrum list's **Uncertainty analysed** tag refers to the method currently chosen in **Method**. Switching from bootstrap to Monte Carlo removes the completed tag from spectra without a saved Monte Carlo analysis; switching back restores the bootstrap tag. A saved analysis of the selected method is marked outdated after data/model changes. Hover over the state to see the selected method and all saved analyses. Changing the method preserves the selection and does not rerun or delete any analysis. In **Model and parameters**, click **Analysis - / +** in the parameter-table header to open the **Displayed uncertainty** menu. Its checked entry selects which saved analysis supplies the coloured **Analysis − / +** column for every spectrum. The choice is universal within the project: switching spectra or using **Show all functions** does not change the selected method. The selector lists methods recorded for at least one spectrum in the project. A spectrum without the selected analysis shows a dash in the analysis column; it does not fall back to another method. The sole available method is displayed automatically, including after reopening older projects without a saved display preference. If several analyses are available, the selector remembers one project-wide choice after saving and reopening. Finishing a background analysis preserves a different method selected while it was running. Older per-spectrum preferences migrate to the first available saved choice in spectrum order. Fit covariance analyses use their recorded confidence limits. When no analysis is available anywhere in the project, clicking the column heading explains that an analysis must be run first and offers **Open Uncertainty Analysis**, which opens that panel. The menu is attached directly to the error-column heading; the uncertainty panel's **Method** control chooses the same method to show, inspect or run. Changing either control updates the other and all spectrum tags. Selection does not run an analysis: use **Run explicit uncertainty analysis** to calculate it. If the chosen method has no recorded result, errors remain blank. The header menu marks it as **no recorded result** when other saved methods are available; it does not silently substitute another method. Available saved methods can still be selected. With **Show all functions**, the empty-state link activates the spectrum owning the displayed function before opening its analysis panel. The project-wide selector remains available when several functions are selected and parameter rows are hidden. The adjacent **±1σ** column always retains the

original fit error. The analysis column shows the distances below and above the fitted value separately. It shows a dash when a parameter has no valid interval or the interval excludes the original fitted value; inspect the assessment for details. The parameter CSV retains `parameter_err` for the fit error and adds `parameter_analysis_err_minus`, `parameter_analysis_err_plus`, and the selected method when applicable. Saved analyses remain selectable after reopening, including reports from older project formats and projects opened read-only. When data/model changes make an analysis outdated, its intervals are shown in amber with **Recorded:**. They refer to the earlier fitted value, given in the tooltip, and are not uncertainties of the current parameter values. Refit and rerun the analysis to obtain current intervals. Current-parameter CSV errors exclude outdated analyses; the uncertainty report retains the historical results and their assessment. In **File > Quick export: function parameters**, **Also export covariance matrices** is off by default. When selected, a sibling `*_covariance` folder contains an `index.csv` and labelled matrices: the fit covariance and, separately, sample covariance for the selected bootstrap or Monte Carlo analysis when raw samples are available. Global fits retain cross-spectrum terms in one joint matrix. Profile likelihood provides intervals, not a sample covariance matrix. Running an uncertainty analysis never changes the original fit covariance. Older projects may need a new fit for spectra whose individual baselines were not recorded. Profile likelihood accepts one active spectrum at a time; coupled global fits are analyzed together when resampling is requested.

All methods display a table with spectrum/function names, parameter, original fitted value, confidence interval endpoints and assessment. Click the assessment to read its explicit reasons, including the names and correlation coefficients of strongly correlated parameters. Intervals are not symmetric +/- errors and their midpoint is not necessarily the original fitted value. Technical paths, seed and the baseline timestamp remain available under **Technical details**. The confidence level is configurable for resampling/profile; covariance shows the recorded fit's confidence level. Changing the method recalls its most recent recorded analysis.

Assessment	Meaning
OK	No diagnostic issue is flagged by these checks; this does not establish practical adequacy
Attention	Bound contact, strong correlation, unstable interval endpoints, an open profile scan, outdated results, or another stated diagnostic caveat
Critical	No usable interval, too few successful replicas, rank-deficient covariance, or an interval spanning most of the allowed range
Fixed	Parameter was fixed; its uncertainty was not estimated

These are deterministic diagnostic heuristics, not universal scientific thresholds: strong correlation means absolute r >=0.95; coverage of at least 80% of a finite allowed parameter range is flagged; fewer than 20 successful replicas is critical, and fewer than 200 is flagged for unstable percentile endpoints. Correlations use empirical samples for resampling when available, otherwise the fit covariance. A parameter touching a bound is not necessarily physically wrong. Click the assessment to see why it is flagged; no color certifies the correctness of the physical model. Analyses describe their recorded baseline; editing data or models requires refitting before resampling. Reports remain inspectable but are marked outdated when appropriate. Failed-replica counts alone do not change a parameter's assessment. A parameter can therefore be **OK** while its spectrum has a failed-replica warning. Read both the parameter assessment and the spectrum-level warning before interpreting the interval; **OK** does not mean that every replica succeeded or that the analysis has been scientifically validated.

12.1 Parametric Monte Carlo

This method generates synthetic y values from the fitted model plus independent normal noise using imported absolute `sigma_y`, or a two-sided Gaussian approximation using the lower/upper Y error half-widths and their declared confidence. The latter transforms negative/positive standard-normal draws with the respective scale; it preserves the declared central interval and has its median at the fitted prediction. Every selected curve must provide valid Y errors on the fitted rows. Noise is still generated from these errors when weighting is turned off in the Fit dialog. The chosen approximation and input confidence are recorded in the analysis settings.

Each synthetic data set is refitted using the local solver. The result records the requested, completed, and failed replicates, seed, free parameter paths, samples, empirical intervals, and failure messages.

12.2 Residual bootstrap

Residuals from fitted rows are centered, sampled independently with replacement, added to the fitted model, and refitted. With Y errors, residuals are first standardized using each observation's lower/upper scale, then reconstructed with each destination row's appropriate scale. This accommodates the supplied relative noise levels without adding measurement noise a second time. Masked rows do not supply residuals. This assumes exchangeability of the standardized residuals and can be inappropriate when residuals remain serially correlated or heteroscedastic. Inverse-variance weights use the equivalent relative scales `1/sqrt(weight)`, so they produce the same bootstrap as equivalent sigma values. Arbitrary direct residual multipliers do not declare a noise variance and retain empirical resampling. Supplied noise scales are used for resampling even when the fit's data-error weighting is disabled; they must be positive and finite on fitted rows.

12.3 Block bootstrap

The block bootstrap resamples contiguous circular residual blocks. With Y errors it resamples standardized residuals and restores the destination row's noise scale, as in residual bootstrap. Enter a block length or choose Automatic. The automatic estimate uses the first lag at which the absolute residual autocorrelation falls below e^{-1}, subject to practical limits.

Block resampling is often preferable for spectra or kinetic traces with local correlation, but the block length remains a scientific modeling decision.

12.4 Profile likelihood

Choose one free parameter that is not fixed or equality-linked. CurveMole fixes it across a finite grid, refits all remaining free parameters, and compares a Gaussian likelihood ratio with the baseline. With known absolute Y noise scales it uses the chi-square difference. With an unknown common noise scale it profiles that scale using $N \log(Q/Q_{min})$, where Q is the weighted residual sum of squares. This likelihood-ratio formula is invariant to a common rescaling of residual units; the numerical optimization remains subject to the scaling limits in Appendix J.1. Robust losses, active interval-error arrays (even equal lower/upper half-widths) and arbitrary spectrum balancing are not supported by this Gaussian profile; use bootstrap for those cases. Interval arrays ignored with data-error weighting disabled do not trigger that restriction. A zero-residual baseline without known noise cannot estimate the noise scale and is rejected with an explanation. Convolution is evaluated on the full regular grid before masked rows are selected. The default grid spans approximately

three covariance standard errors on either side, or a fallback span when no standard error is available, while respecting bounds.

The default uses 31 grid points and a one-parameter chi-square threshold. The grid size, confidence level and lower/upper scan limits can be changed in the panel; blank limits select the automatic range, and explicit limits must respect parameter bounds. Failed grid points are counted. The desktop profile currently requires a single-spectrum baseline; it does not profile a global multi-spectrum fit. Profile threshold crossings are refined by bracketed refits. `confidence_regions` contains every connected region detected by the scan; `interval` selects the region containing the fitted estimate and never bridges rejected gaps. Disconnected sets are marked **Critical**. `warnings` reports incomplete scans or unrefined endpoints; inspect every region and extend the scan where needed. A finite grid can still miss a narrow remote mode. The default Gaussian-location example now agrees with the analytic 95% interval, approximately +/-1.959964 standard errors.

12.5 Reproducibility and failed replicates

The default seed is 1729 unless changed through programmatic settings. Resampling outputs record configuration and up to the first 100 failure messages. Do not report an empirical interval without also reporting how many replicates completed and failed. Selecting **Intervals, assessments and matrices** in the export dialog also writes `uncertainty/parameter_assessments.csv` for the recorded analyses, including spectrum/function names, parameter paths, fitted values, interval endpoints, confidence level, method and reasons. Covariance/correlation matrices remain available in the same export option. This CSV contains parameter assessments, not the spectrum-wide failed-replica counts. Keep the `.fitproj` project, or select a project copy in the analysis bundle, to retain the recorded counts, failure messages and adaptive configuration alongside the parameter assessments.

New fit results record `provenance`: original model/constraint snapshots, input content hashes, ordered parameters, initial solver coordinates, scales, resolved solver bounds/method, settings, seeds, function/plugin fingerprints, CurveMole source identity, Python/NumPy/SciPy versions and backend/thread details. Profile and resampling outputs also retain their baseline identity and actual analysis configuration. Historical results created before this feature have no reconstructible original state. Keep original data and scripts: hashes identify inputs but do not store a separate copy of every historical dataset.

13. Projects, saving, and recovery

13.1 The `.fitproj` format

A `.fitproj` is a ZIP container with:

- a versioned JSON manifest;
- original arrays stored as NumPy `.npy` payloads;
- uncertainty and weight arrays when present;
- mask arrays;
- transformation operands;
- binary fit-output, covariance and raw uncertainty-sample arrays;
- models, links, custom functions, results, history, and UI/export state;
- SHA-256 checksums for every binary payload.

The current project schema is 4. Schema 3 introduced NumPy result payloads and a separate streamed fit history; schema 4 also preserves independent-fit covariance and correlation as block matrices. This reduces memory while preserving recorded fits and sample values. Schemas 1 through 3 remain readable. Newly saved projects require a reader supporting schema 4 block matrices; keep a backup when sharing with older installations. Model files remain at schema 2. The stable workflow format is schema 2; 1.2.0 adds workflow schema 3 for batch imports and preprocessing while retaining support for legacy schema 1/2 recipes.

NumPy arrays are loaded with `allow_pickle=False`. CurveMole projects never require executing a Python pickle. The format is documented in project-format.md.

13.2 Atomic saves

CurveMole writes a temporary sibling file, validates the archive, and then atomically replaces the destination. A failed save should therefore not partially overwrite a previous valid project.

Opening confirms the current project's save/discard choice before reading the candidate. Validation, data reading and digest checks run in a cancellable worker while the interface processes events. Editing and another Open/New operation are blocked during that read. An error or cancellation leaves the previous project and its lock intact. A detected file change during opening asks for another try; external changes after opening are checked again before a coordinated save.

13.3 Project locks and read-only mode

Opening a project creates a sibling `.lock` file containing process and host information. If another instance already holds the lock, CurveMole opens the project read-only. Use **Save project as** to create an editable copy. A stale lock left by an abnormal termination may need manual inspection and removal after confirming that no other instance is using the project.

13.4 Normal save and portable copy

Save project updates the current project path. **Save portable copy** writes a complete separate `.fitproj` without changing the active project path. Since project data are already embedded, the portable copy is intended for sharing or archiving a snapshot.

Projects are not encrypted. Use suitable filesystem permissions or encrypted storage for confidential data.

13.5 Autosave recovery

Manual Save, Save As and portable-copy writing also run on a background worker. The status bar identifies the file being saved while the plot and controls remain available. Edits made during writing remain unsaved; a completed snapshot does not clear recovery copies for newer changes. Close/Open waits for the save outcome with a responsive interface and asks again if newer unsaved changes remain.

Autosave runs in a dedicated background thread, so compression, disk writing and integrity checks keep the plot and controls available. Each backup represents a coherent snapshot from the start of the operation; edits made while it is being written remain unsaved and are included in a later backup. Only one autosave runs at a time, with overlapping requests combined into the latest requested state. Built-in fits and uncertainty analyses work on detached inputs, so manual saves and autosaves include the current project and results already committed without waiting for the entire calculation. Later completed results become new unsaved changes. Undeclared external background tasks still defer snapshotting until their worker has finished.

Every ten minutes, a modified revision is written to the operating system's CurveMole user cache. An unchanged revision does not create another recovery. The six newest valid, distinct recovery files for a project are retained.

CurveMole offers **Recoverable sessions** at startup for unresolved sessions, including crashes and exits without a successful Save or explicit Discard. The same list is available from **File > Recoverable sessions**. Sessions are grouped by project name with their recovery date and copy count. **Recover** opens the newest valid copy by default; an older retained copy can also be selected. The recovered workspace is unsaved and does not overwrite the original project.

The proposal explicitly names the original project/file and the recovery file it will open. Each project lists named **backup 1** through **backup 6** entries, with dates and the newest valid copy selected. Older copies without stored names use the internal project name or an identifier to keep unnamed sessions distinct.

Decide later retains the copies and the proposal for the next launch. **Delete selected copies** removes individual copies or selected project groups after confirmation. Saving all current changes successfully or explicitly choosing **Discard** when closing or switching projects removes its recovery copies. Cancelling a dialog or a failed save preserves them. Other projects' copies remain.

The application counts completed recovery files once per minute. Startup recommends periodic manual cleanup to limit disk usage and opens the recovery manager when the total reaches a new multiple of 50: 50, 100, 150, 200 and so on. Each warning is shown once per reached band, remembered across restarts. If several thresholds are crossed between launches, only one warning is shown for the current count. Cleanup to fewer than 50 copies restarts the warning cycle. Copies from multiple projects can be kept indefinitely within each project's six-valid-copy rotation; there is no automatic age expiry. Damaged completed copies can be deleted manually. Abandoned staging files are automatically removed only when their owning process is known to have exited; files still in use are retained.

File > Recent projects lists up to ten successfully opened or saved projects; choose an entry to reopen it, or clear the list from that menu.

13.6 Unsaved changes

An asterisk in the window title marks a dirty project. New Project, Open Project, and Quit ask whether to save, discard, or cancel when unsaved changes exist.

14. Exporting an analysis bundle

14.1 Starting an export

Choose **File > Export analysis bundle** or press **Ctrl+E**. The complete bundle selects every available output type by default, including the outputs offered by the **Quick export** commands. Clear any output types that are not needed. Files that require unavailable results or samples are omitted. The choices are grouped as **Figures**, **Data tables**, **Fit parameters and uncertainty**, **Project and Python**, and **Reports and documentation**. All figure formats and the current-view figure are together in the first group. The option to include full uncertainty samples is beside the project-copy option.

Select a parent directory and use the named-subfolder option to put the export in its own folder. This option is enabled by default; the folder name can be edited. The other save modes remain available:

- export directly into the selected directory by disabling the subfolder option;
- **Create versioned export** to create a timestamped subdirectory;
- **Update existing CurveMole-owned files after confirmation** to update files that CurveMole can verify as belonging to the current project export.

14.2 Detailed fit-results file

`fit_results.csv` associates each spectrum with its model functions and parameters.

Each row represents one parameter of one model component and includes the series, curve and curve ID, component and component ID, displayed function and function ID, enabled/background state, composition operator, parameter name and full parameter path, current resolved value, standard error, confidence interval, bounds, fixed/link state, unit, human-readable value with uncertainty, and derived area/FWHM where defined.

This format makes the association between every data curve, its model functions, and parameter values explicit in a single table.

Values describe the current model. Error and confidence columns can retain recorded fit estimates after a model edit; inspect `curve_state` before interpreting them as uncertainties of the current values. Refit an edited model to obtain current errors.

14.3 Bundle contents

The exported `README.txt` starts with a table of saved files and folders, explaining what each contains and when it is useful. Short sections then explain how to read the data, interpret fit results, and regenerate the plots in Python. The table describes only selected outputs that were actually written.

The main files and folders in a complete bundle are:

File or folder	Contents and purpose
`README.txt`	Start here for a readable guide to this export and the meaning of its data.
`index.csv`	Matches spectra, names, IDs and source files to their exported files.

File or folder	Contents and purpose
`manifest.json`	Exact file inventory and export metadata for scripts or archival checks.
`spectra/`	One numeric spectrum file per curve, including the available fit and component traces; the same output as the spectra quick export.
`data/`	Wide CSV tables of measured data and evaluated model functions.
`fit_results.csv`	Detailed parameter values, errors, constraints and fit state.
`function_parameters.csv`	Wide function-parameter table, as offered by the parameter quick export.
`function_parameters_covariance/`	Available labelled covariance matrices for the parameter export.
`figures/png/`	One PNG image per spectrum and the current-view image when selected.
`figures/svg/`	The corresponding vector SVG figures.
`figures/pdf/`	The corresponding individual PDF figures.
`summary_report.html`	Compact readable project and parameter summary.
`report/summary.pdf`	One summary page per spectrum in a single PDF.
`report/full_reproducibility.html`	Full reproducibility report, including recorded fit settings and project metadata.
`python/`	Sample data, component and curve-metadata tables, JSON fit results and a script to regenerate per-spectrum figures.
`models/`	A reusable `.fitmodel` for each curve's model.
`<project>.fitproj`	Portable project snapshot, optionally including full uncertainty samples.
`fit_settings.csv`	All recorded settings for the last fit.
`uncertainty/`	Available uncertainty intervals, assessments, covariance and correlation matrices.
`diagnostics/`	Available residual autocorrelation diagnostics.
`laboratory_notebook.txt`	Laboratory notebook entries as UTF-8 text.

The numeric data and per-spectrum figures evaluate the current models against the current transformed observations, including models that have not been fitted or have changed since fitting. Their presence does not imply a converged fit. Recorded fit settings, statistics, covariance/correlation matrices, and residual diagnostics describe the last recorded fit and can cover only part of the project. Keep the project copy and inspect recorded curve IDs, timestamps, and fit state when comparing these outputs. The current-view figure additionally follows the display choices described in Section 14.10.

Directories such as `data/`, `python/`, `figures/`, `report/`, `uncertainty/`, and `diagnostics/` are created only when a selected output actually writes a file there. An option whose required fit result is unavailable does not create an empty folder. `index.csv` maps curve IDs and names to their exported files. When the README is selected, `manifest.json` also records the file inventory in machine-readable form.

14.4 Safe overwrite policy

New exports store their ownership list in the project export configuration rather than creating a hidden sidecar file. Existing `.curvemole-export.json` manifests from older versions are still read for backwards-compatible updates.

When **Update existing** is selected, CurveMole refuses to overwrite a colliding file unless it can verify that the file belongs to the current project export. Unrelated files in the directory are preserved.

14.5 Wide data tables

Optional wide CSV files are intended for direct inspection and tools such as Origin or QtiPlot. Depending on data and model, columns include:

- x and y;
- total fit;
- residual as data minus fit;
- total background;
- every enabled component;
- `sigma_y` or point weight;
- mask flag.

Rows remain aligned with the complete curve, including masked and invalid entries.

14.5.1 Exporting optimizer and loss settings

Recorded last-fit settings (CSV) writes `fit_settings.csv`, containing all recorded `FitSettings` fields, including advanced solver parameters, loss and `f_scale`. Each row includes the fit timestamp, mode and curve IDs. It describes the selected last fit result, not unsaved edits in the Fit dialog; fields for inactive algorithms are retained for completeness. No settings file is produced without a fit result. `fit_results.csv` separately contains all model parameter values, bounds, links and available errors. The JSON result includes the same optimizer settings; the full reproducibility HTML report also shows them as a table.

14.6 Python analysis tables and JSON

The Python export keeps sample data, component evaluations, and curve metadata in separate tables:

- `python/observations.csv` has one row per sampled point, identified by `curve_id` and `point_index`. It includes x, measured y, the available fit, residual and background, uncertainty or weight fields, and mask/validity flags.
- `python/components.csv` stores evaluated model components with the same curve and point keys and a component identifier. These are individual function evaluations; multiply, divide, or convolve roles must be respected rather than assuming every component is additive.
- `python/curves.csv` supplies the series/curve names, axis labels and units needed to interpret each curve without repeating them on every sample row.
- `python/replot.py` reads these tables and creates a figure for each curve. Run it with Python and the dependencies described in the exported README. It writes PNG figures to `python/replotted/png/` and SVG figures to `python/replotted/svg/`.

Join samples and components using `curve_id` and `point_index`, not x alone: different observations can share the same x coordinate. Rows retain masked and invalid observations so their original positions remain traceable. Residuals use the data-minus-fit convention. A mask flag is metadata on a sample, rather than a separate numeric quantity to plot. This replaces the earlier `data_tidy.csv` table.

`point_index` starts at zero in current row order; `original_row_index` identifies the imported row. `masked` describes current exclusions, including invalid points, and does not reconstruct which points were included in a historical fit. `valid`, `finite_xy`, and `model_valid` record distinct checks of observations/errors/weights, finite coordinates, and model evaluations respectively.

`python/results.json` records schema version, application version, result metadata, parameters, statistics, warnings, and settings without duplicating large numeric arrays. Its `result` describes the selected last fit, which may cover only a subset of the project curves; `fit_by_curve` and `global_fit_baselines` retain the other recorded fit baselines. These describe recorded fits, while `observations.csv` contains evaluations of the current models. The `.fitproj` copy retains the full project record.

14.7 Reports and reusable files

- `summary_report.html` is a compact human-readable report.
- `report/full_reproducibility.html` includes project metadata.
- `report/summary.pdf` contains one page per curve. Each page identifies that curve and displays its data, available fit/components, and residuals. Curves with different axes or units are kept on separate pages.
- Individual curve figures are saved in `figures/png/`, `figures/svg/`, and `figures/pdf/`, using a consistent Matplotlib style with readable labels and restrained legends. Every spectrum has its own file in each selected format; formats are kept in separate subfolders.
- `models/*.fitmodel` stores each curve's model and custom formulas for programmatic reuse. Preserve the complete `.fitproj` when cross-spectrum relationships or the complete analysis must be restored.
- `.fitproj` is a complete portable project snapshot.

The exported report PDF is an analysis summary, not the CurveMole software manual.

14.8 Quick export: one numeric file per spectrum

Choose **File > Quick export: spectra and fit curves** or press **Ctrl+T** when the result should be immediately replottable as one multi-column table per spectrum. The complete analysis bundle includes these numeric spectrum exports too. The quick export dialog lets you choose any project spectra and any of these options:

- export the measured spectrum with its fitted background subtracted;
- omit masked rows;
- include every enabled individual model function;
- include the combined fitted background;
- include the total fit;
- include residuals as data minus total fit.

The x and spectrum columns are always written. Output names follow `<spectrum>_curvemole.<original extension>`; duplicate names receive a numeric suffix. Available error columns and explicit point weights are included. A numeric `weights_are_inverse_variance` column distinguishes inverse-variance weights (1) from direct residual multipliers (0). Colliding column labels receive a __2, __3, … suffix, preserving all numeric fields. The original delimiter is retained when practical, otherwise the output is tab-delimited. Headers describe axis labels, units, functions, background

state, total fit, and residuals, and never contain whitespace so they remain safe for whitespace parsers. An unknown original extension is preserved.

14.9 Quick export: function parameters

File > Quick export: function parameters writes a wide CSV table with one row per project spectrum and two header rows identifying functions and parameters. The complete bundle saves this table as `function_parameters.csv`. Each parameter has value and fit-error columns; functions with the same displayed name align across spectra, with duplicate names disambiguated. Spectra that are not currently Fitted retain their row but have blank numeric cells. Saved selected-analysis errors are added as separate minus/plus columns, together with the method. **Also export covariance matrices** optionally writes the labelled sibling matrices described in Section 12. The complete bundle includes available parameter covariance exports alongside its full constraints and reproducibility record.

14.10 Quick export: current plot view

Use **File > Quick export: current view with Matplotlib** to save the plot displayed when the command is invoked as a styled SVG, PDF, or PNG figure. It uses the visible curves and the current plot arrangement, axis scales and displayed ranges, rather than combining every curve in the project. This is useful for saving a chosen overlay or a zoomed view. The complete bundle includes the current-view export as `current_view.png`, `current_view.svg`, and `current_view.pdf` in their respective `figures/` format subfolders, alongside the separate figures for every curve.

15. Command-line interface

15.1 General form

```
curvemole [--version] [--json] [--verbose] COMMAND [OPTIONS]
```

Use `--json` for machine-readable output. Normal data go to standard output; logs and errors go to standard error.

15.2 Open the GUI

```
curvemole gui
```

The desktop executable can also receive a project or data path as its first argument.

15.3 List available functions

```
curvemole functions list
curvemole functions list --json
```

15.4 Fit one file

```
curvemole fit spectrum.csv \
  --x wavenumber \
  --y absorbance \
  --function gaussian \
  --background linear \
  --output spectrum.fitproj
```

`--x` and `--y` accept a column name or zero-based index. Optional initial values are `--center`, `--width`, and `--area`. The meaning of `--width` follows the selected function: Gaussian and Voigt use `sigma`, Lorentzian uses `gamma`, and pseudo-Voigt uses `fwhm`.

If no initial peak override is supplied, the CLI uses the strongest automatic positive peak suggestion when available. Background choices are none, constant, or linear.

15.5 Fit a series

```
curvemole fit-series scan_01.csv scan_02.csv scan_03.csv \
  --x 0 --y 1 \
  --function pseudo_voigt \
  --background constant \
  --mode sequential \
  --output series.fitproj
```

Modes are `independent`, `sequential`, and `global`. Use `--global-search` to request Differential Evolution. Missing finite bounds are generated from current parameter values; use a YAML workflow or the Python API to customize percentages or explicit parameter bounds.

15.6 Inspect and validate

```
curvemole inspect analysis.fitproj --json
curvemole inspect raw_data.dat --json
curvemole validate analysis.fitproj
curvemole validate workflow.yml
```

Inspecting a data file reports detected parsing and a preview. Validating a project checks archive structure and checksums. Validating YAML checks the supported workflow schema and required top-level structures.

15.7 Export an existing project

```
curvemole export analysis.fitproj exported_analysis
```

Use `--versioned` for a timestamped subdirectory. Use `--force` only when updating a directory whose CurveMole ownership manifest permits the overwrite.

15.8 Exit codes

Code	Meaning
0	Success
2	Command usage error
3	Data validation error
4	Fit error
5	Project format error
6	Plugin or other CurveMole domain error
10	Unexpected failure

16. Reproducible YAML workflows

16.1 Running a workflow

Appendix G contains the complete workflow schema, nested configuration fields, defaults, examples and limits for authoring workflows. Appendix H explains the downloadable YAML automation audit, which can be attached to an AI conversation to generate workflows using the documented interfaces.

```
curvemole run examples/gaussian_workflow.yml --json
```

Relative paths are resolved from the workflow file's directory. Stable 1.1.0 supports schemas 1 and 2. **In 1.2.0, use `schema_version: 3`** for folder/glob imports, naming templates, shared model selectors, preprocessing and the new output controls. Legacy schema 1/2 recipes remain supported with their original fields. The schema guard makes older executables reject new recipes instead of silently ignoring a processing stage. Appendix G lists every supported field and includes a complete folder workflow.

The beta can import a whole directory, optionally search subdirectories, exclude matching files and sort names naturally (`scan2` before `scan10`). A model can target all curves or a matched subset; `curves: first` is useful for a sequential source model. Preprocessing supports masks, fit ranges and the documented scalar and safe formula transformations. Inputs remain unchanged. The workflow is a fixed pipeline, rather than a general programming language.

Beta validation reports unknown keys and invalid nested structures before reading data. File contents, column availability, plugin functions and convergence still require an actual run. Use `export.project_overwrite: false` to protect an existing project output; this is independent of bundle overwrite/versioning.

Run an existing automation from the desktop

The separate **Automations** toolbar at the bottom contains **Run automation**. On the first click, choose a `.yml` or `.yaml` workflow; CurveMole validates and runs it. Later clicks run the same file, including after restarting CurveMole. The tooltip shows its path. Use the arrow beside Run to **Choose automation** or **Stop automation**. Cancelling the file picker leaves the previous choice intact; if the remembered file is missing, Run asks for another file.

The workflow executes in a separate process and uses its own imported spectra, models and output settings. It does not operate on the currently open GUI project. Completion reports the processed spectra, fit convergence and output paths. Relative paths resolve against the workflow's folder. To keep results, specify `export` in the workflow. Reruns respect its overwrite/versioning settings. Stopping or failing does not roll back files already written. Wait for completion or stop the automation before closing CurveMole.

Python plugins referenced by a desktop automation must first be reviewed and enabled in **File > Plugin Manager**. Their source fingerprints are checked again at launch; a workflow does not implicitly approve executable plugins. This control runs existing workflows; it is not a visual workflow editor.

16.2 Complete example

```yaml
schema_version: 1
name: Gaussian workflow example
series_name: Synthetic spectra

imports:
  - path: gaussian.csv
    config:
      delimiter: ","
      decimal: "."
      header: true
    columns:
      x: x
      y: [y]
      sigma_y: sigma_y
    aliases: [Synthetic Gaussian]

models:
  - curve: Synthetic Gaussian
    components:
      - function: constant
        name: Background
        parameters:
          offset: 0.0
      - function: gaussian
        name: Main band
        parameters:
          area: 3.0
          center: 0.3
          sigma: 1.0
        constraints:
          sigma:
            minimum: 0.001
            maximum: 5.0

fit:
  mode: independent
  curves: [Synthetic Gaussian]
  settings:
    solver: local
    loss: linear
    seed: 1729

export:
  project: gaussian_result.fitproj
  bundle: gaussian_export
  versioned: false
  overwrite: false
```

16.3 Custom formulas in YAML

```yaml
custom_functions:
  - identifier: exponential_decay
    display_name: Exponential decay
    kind: generic
    formula: offset + amplitude * exp(-x / tau)
    defaults:
      offset: 0.0
      amplitude: 1.0
      tau: 1.0
```

```yaml
    bounds:
      tau: [0.000001, 1000000.0]
    derived:
      lifetime: tau
```

Use the custom identifier in a model component exactly like a built-in function.

16.4 Trusted plugins

List local plugin manifest paths under `plugins`. Unattended execution also requires one `--trust-plugin IDENTIFIER` argument for each plugin whose Python source may run:

```
curvemole run workflow.yml \
  --trust-plugin org.example.lineshape
```

Trust is never inferred merely because a manifest is present.

17. Python API

17.1 Stable package-level objects

Appendix F provides the detailed scripting reference: import paths, signatures, domain objects, fitting, uncertainty, persistence and exports. Appendix H explains how to use the self-contained Python automation audit as AI context when generating scripts.

The package root exports GUI-independent classes including `Curve`, `Series`, `Dataset`, `Component`, `Model`, `Parameter`, `Project`, `FitSettings`, `FitPlan`, `FitResult`, `Fitter`, `FunctionRegistry`, and `default_registry`.

17.2 Minimal fit

```
import numpy as np

from curvemole import Component, Curve, FitSettings, Fitter, Model

x = np.linspace(-5.0, 5.0, 1001)
y = 0.1 + 3.0 * np.exp(-0.5 * ((x - 0.4) / 0.8) ** 2)

curve = Curve("Synthetic band", x, y)
model = Model("One Gaussian")
model.add(Component.create("constant", initial={"offset": 0.1}))
model.add(
    Component.create(
        "gaussian",
        initial={"area": 6.0, "center": 0.3, "sigma": 1.0},
    )
)

result = Fitter().fit_single(curve, model, FitSettings(loss="linear"))
print(result.success)
print(result.statistics)
```

The Gaussian expression used to generate y above is height-parameterized, while the CurveMole Gaussian component is area-parameterized. This is why the initial model `area` is not simply the synthetic height.

17.3 Bounds and fixed values

```
peak = model.components[1]
peak.parameters["center"].minimum = -1.0
peak.parameters["center"].maximum = 1.0
peak.parameters["sigma"].minimum = 0.05
peak.parameters["sigma"].maximum = 3.0
peak.parameters["area"].fixed = False

for parameter in peak.parameters.values():
    parameter.validate()
```

17.4 Cross-curve links

```
source_path = model_a.parameter_path(curve_a.id, peak_a.id, "center")
peak_b.parameters["center"].link = "${" + source_path + "} + 5.0"
```

Construct a Global `FitPlan` containing both curve IDs. CurveMole validates the complete graph before optimization.

17.5 Global fit

```
from curvemole.core.fitting import FitMode

plan = FitPlan(
    [curve_a.id, curve_b.id],
    mode=FitMode.GLOBAL,
    settings=FitSettings(loss="linear"),
    spectrum_weights={curve_a.id: 1.0, curve_b.id: 1.0},
    equal_contribution=True,
)

result = Fitter().fit(
    plan,
    {curve_a.id: curve_a, curve_b.id: curve_b},
    {curve_a.id: model_a, curve_b.id: model_b},
)
```

17.6 API compatibility and reproducibility

Use the documented package-root API for reusable scripts; internal modules are implementation details and may evolve. Pin the exact CurveMole version in automated research workflows and record it with results, together with dependency versions. Save the initial script/workflow and inputs before fitting. A saved fitted model is a new starting state: repeating the same seed from that state can change DE's automatic bounds. See Appendix J.7 for the information absent from ordinary fit records and Appendix J.5 for fresh component IDs under strong cancellation.

18. Function Builder and plugins

18.1 Formula Builder

Open **Tools > Function Builder**. Enter:

- an identifier using letters, numbers, and underscores;
- a display name;
- classification as **Peak-shaped function** or **General function**;
- a formula in x;
- optional formulas for derived area and FWHM;
- for peak functions, optional semantic roles for detected parameters.

Background is assigned to components when adding or editing the model; it is not a separate Function Builder classification. **Use manual-point initialization by default** controls the preferred insertion mode for the new function.

CurveMole detects every non-function symbol other than x as a parameter. Newly detected parameters default to 1.0 and are unbounded. After adding the function to the project library, add a component and edit its initial values and bounds in the normal parameter table.

Example:

```
offset + amplitude * exp(-x / tau)
```

Detected parameters are `amplitude`, `offset`, and `tau`.

For a function classified as a peak, the **Peak parameter roles** table can assign at most one of each supported meaning:

- peak position / centre;
- peak height or peak area / integral;
- FWHM, Gaussian sigma, or HWHM/gamma.

These roles tell graphical placement and automatic peak detection how to initialize parameters even when their names are arbitrary. Height and area roles are mutually exclusive. A width role also supplies the standard derived FWHM automatically when a separate FWHM formula was not entered. Leave genuinely unrelated parameters as **Not specified**. An explicitly unassigned role is respected; CurveMole does not guess a meaning from its name.

18.2 Custom function persistence

The first time a formula is added, CurveMole asks the user to create a dedicated `my_curvemole_functions` folder at a chosen location or open an existing folder with that exact name. Each definition is saved there as `<identifier>.curvemole-function.json`, registered in the current application, and also embedded in the project. Formula, classification, derived quantities, and peak roles travel together.

CurveMole remembers the selected folder and loads every valid function from it at each startup. Reusable functions therefore appear in the normal Add dialog, the Functions group selector, and automatic peak-shape selector without requiring a particular project to be open. Copy the folder between computers to move a personal function library; open that folder when CurveMole asks for the library location.

Project-contained definitions remain portable. When a project reopens, CurveMole reconstructs and validates each restricted expression. A malformed user function or library file is skipped and recorded in the Log rather than executed. Formula files use the safe expression system and are distinct from executable Python plugins.

18.3 Local plugin structure

A local plugin consists of a manifest ending in `.curvemole-plugin.json` and a Python module in the same directory. Example manifest:

```
{
  "identifier": "org.example.my-lineshape",
  "version": "1.0.0",
  "api_compatibility": "1",
  "licence": "GPL-3.0-or-later",
  "capabilities": ["functions"],
  "module": "my_lineshape.py"
}
```

The module exposes `register(api)`. Use `api.register(FunctionDefinition(...))` for model functions or `api.add(kind, id, label, callback)` for exporters, importers, fit algorithms, transformations, analyses, commands, workflows, panels, plot layers and lifecycle notifications. Full contracts and runnable examples are in Creating and managing plugins. Built-ins cannot be replaced through this API; contributed entries carry a plugin-specific symbol and identify the providing plugin in their tooltip.

18.4 Trust boundary

Formula Builder expressions are interpreted safely. Plugins are different: a plugin is Python code and can perform any action allowed to the current user account.

CurveMole reads manifests before executing a module and requires an explicit trust decision. Review source, origin, license, and version before trusting. Do not trust a plugin merely because its identifier resembles a known package.

18.5 Plugin Manager

Open **File > Plugin Manager**, choose a permanent directory, and scan. Review a candidate, then explicitly trust and load it. Enabled plugins load again on restart, independently of projects. **Disable** unloads contributions; **Remove** also forgets the installation. Source files and project plugin data are retained. Remove any components using a plugin before unloading it. Close the manager to refresh menus.

Changed local source needs approval again. Callback errors disable the plugin; failed registration rolls back additions. After a crash or forced termination, CurveMole disables persisted plugins before importing their code and explains recovery. Normal closure clears the session marker. The integrated validated-plugin browser can download selected plugins through **Install selected**. A **Plugins** status badge checks at startup and hourly for updates to loaded supported plugins, lets you select updates, and indicates when a restart is needed to activate staged code. Open **Plugin updates...** in the manager or click the badge, then choose **Update selected**. **Help > Check for updates** checks both CurveMole and loaded plugins. Checking for an update does not install it automatically. Disabled plugins are not checked; local-only plugins without a catalog entry need manual updates. Manual ZIP download remains an alternative. Updates replace files in the plugin's existing user-selected folder. If that location cannot be replaced, CurveMole asks for another permanent folder; it does not retain plugin code in an internal application folder.

Cancelling leaves the previous installation in place. The saved updated copy remains selected after disabling, restarting and re-enabling the plugin, even if the original folder still contains an older copy. Running plugin code is replaced on the next load. Set `CURVEMOLE_DISABLE_PLUGINS=1` to skip automatic loading manually. This is failure recovery, not a sandbox for Python code.

18.6 Creating extensions beyond model functions

Use `api.add("exporters", "csv", "Laboratory CSV", export_csv)` to add an exporter. Its callback receives the chosen destination in `context.path` and a project snapshot in `context.project`. Built-in export commands remain available. `fit_solvers` callbacks receive constrained residuals, Jacobian, initial values, bounds, settings and cancellation. CurveMole computes standard statistics from the returned result without a second built-in optimization.

Importers, transformations, actions and workflows commit snapshot edits as one undoable operation on success. Other callbacks observe snapshots. JSON data in `context.data` is scoped to the plugin and saved with the project. The complete plugin author guide provides manifests, exporter and solver code, all callback contracts, limitations and testing instructions.

For a complete exporter, set the manifest's `module` to `my_exporter.py` and its `capabilities` to `["exporters"]`, then create this module beside the manifest:

```
import numpy as np


def export_csv(context):
    if context.active_curve_id is None:
        return "Select a spectrum first."
    curve = context.project.dataset.curve(context.active_curve_id)
    np.savetxt(
        context.path,
        np.column_stack([curve.x, curve.y]),
        delimiter=",", header="x,y", comments="",
    )
    return f"Exported {curve.name}"


def register(api):
    api.add("exporters", "csv", "Laboratory CSV", export_csv)
```

After loading this plugin, choose **File > Plugins: Exporters > Laboratory CSV** (marked with the plugin symbol). CurveMole asks for a destination and then runs the callback. The integrated export commands stay available. The module uses NumPy, already included with CurveMole; no additional dependencies are required.

18.7 Community plugin submissions and validation

To share a plugin, open **File > Plugin manager > Share my plugin on GitHub...**. This opens the submission guide: fork the repository, upload your complete plugin folder at `custom_plugins/by_community/your_plugin/`, then open a pull request to main. For private use, **Choose plugin folder...** installs it on your computer only; loading a plugin does not publish its source. **Browse validated plugins** opens the downloads that have passed the checks.

The catalog starts with the folder selected in Plugin Manager. Choose an existing permanent folder outside CurveMole's internal storage; each download is retained there under `<category>/<plugin>/`. An empty folder field cannot start an installation, and the confirmed destination applies throughout the download batch.

Open **Help > Plugin help** for the manuals of active plugins. These manuals, **Help > Quick Start** and **Help > User manual** open in CurveMole's internal Markdown reader, with formatted headings, tables and images, search, zoom and Back/Forward navigation. Links to local Markdown files stay in the reader; web links open in your browser. Disabled or failed plugins are omitted from the plugin help menu.

The repository's custom_plugins folder contains the submission guide and downloadable plugins, including a TSV exporter, Fityk project importer, cosmic-ray removal, x-axis calibration and ruby fluorescence pressure monitor. These are separately installed extensions, not automatically enabled CurveMole features. Authors submit a folder with manifest, module, licence, README and functional tests through a pull request. GitHub checks manifests, actual registered capabilities, loading/removal and tests on Linux, Windows and macOS. Only a successful main run publishes the validated catalog and tested plugin ZIP files. **Browse validated plugins** and **Install selected** provide the normal installation route inside the manager, with an explicit trust decision. Alternatively, use the stable downloads in the plugin guide or the separate rolling community plugin release, extract the chosen plugin and review/load its manifest in the manager. These direct downloads do not require a GitHub account and are separate from application releases. A source folder or an open PR alone does not indicate validation. Tests demonstrate tested behavior, not complete safety; Python plugins still run with the user's privileges. See the submission guide for required branch protection settings and the limits of validation.

18.8 Acquisition processors and interactive plugin panels

Plugins can register `import_processors` for Automatic folder import (Section 6.4). A processor receives only the detached new acquisition and its settings, runs in a background thread, and returns spectra/models/results to be added as one Undo operation. Enabling a plugin does not start acquisition monitoring.

Plugin panels join the shared dock/tab workspace and can be reopened without losing their current widget state. Supported panel services offer project snapshots, persistent plugin settings and control over the owner's import monitor. A panel may open automatically when its plugin is enabled; monitoring still requires an explicit start. Closing a monitor panel does not stop acquisition. Read each plugin's own README for its scientific assumptions, controls and dependencies.

Plugins can also preview corrections, mark candidates and select measured points directly in the main graph, using the spectrum's normal axes, masks and display settings. Previews leave measured data unchanged until the plugin's explicit accept action. Point selection ends with Done/Esc or when another editing tool is activated. Closing or hiding the panel removes its annotations.

Cosmic-ray removal 0.5.0 uses this shared graph for automatic detection, manual detection and candidate review. Select a candidate in its table or click its coloured marker to highlight the affected interval; teal segments show accepted corrections before **Accept modifications**. **Detect manually** adds candidates by clicking measured samples. **Remove manually** opens the native point editor; its drags and deletions take effect immediately with Undo/Redo. This plugin version requires CurveMole 0.47.1 or later.

19. Troubleshooting

19.1 The application does not start on Linux

Run the AppImage from a terminal to capture the exact error. If Qt reports missing EGL, OpenGL, XKB, or XCB libraries, install the distribution packages listed in Section 3.3. If FUSE mounting fails, use `APPIMAGE_EXTRACT_AND_RUN=1`.

The AppImage is built on Ubuntu 22.04 for broad glibc compatibility. Very old Linux distributions may still require a newer runtime.

19.2 Quick Guide, manual, updates, or issue links do not open

CurveMole attempts to launch the host operating system's opener without exposing the AppImage's private Qt and C++ libraries to it. If the opener still fails, the dialog shows the path or URL to copy manually.

Use these direct addresses:

```
https://github.com/SebRoLENS/curvemole
https://github.com/SebRoLENS/curvemole/releases
https://github.com/SebRoLENS/curvemole/issues
```

19.3 Windows SmartScreen or macOS Gatekeeper warns

Windows and macOS executables are unsigned because CurveMole is a nonprofit project and signing requires developer-paid certification. Confirm the official release source and SHA-256 checksum, then follow the first-launch procedure in Section 3.2.

19.4 Import produces one column

The selected delimiter is probably wrong. Reopen import and compare whitespace, comma, semicolon, tab, or pipe. A decimal comma file commonly uses semicolon as its delimiter. Confirm the header choice and preview before mapping.

19.5 Imported points disappear from fitting

Inspect the Worksheet. A point is excluded when x or y is not finite, `sigma_y` is invalid, a weight is invalid, or the point is masked. Faded markers indicate masks. Use Unmask or undo the transformation that generated missing values.

19.6 A peak appears at the wrong height after placement

Built-in peaks use integrated area, not height. The graphical initializer estimates height relative to the existing model and converts it to area. For overlapping peaks or a poor background, adjust the center target, width handle, and area value before fitting.

19.7 A spline looks unstable

Use fewer, well-spaced nodes. Place nodes where background information exists rather than at every data feature. Graphically placed node y values are fixed by default; unlock only those that the data can independently determine. Nodes created through the API are ordinary free parameters unless explicitly fixed.

19.8 Differential Evolution searches too narrow an interval

Increase the automatic below/above percentages or set physically justified parameter bounds directly. Existing finite bounds override the automatic limits. A linked parameter's resulting value must still satisfy its own bounds.

19.9 The fit is underdetermined or covariance is unavailable

Reduce the number of free parameters, fix known values, add justified links, widen the data range, or improve signal information. More optimizer evaluations cannot create information absent from the data.

19.10 A parameter remains at a bound

First verify the initial value and physical bound. Then inspect correlations and profile likelihood. A bound-active estimate often has an asymmetric uncertainty and may indicate that the data do not locate the optimum within the allowed region.

19.11 A sequential fit pauses

The affected curve becomes active and the status bar displays **Continue sequential fit**. Inspect its propagated model, initial/fitted values, masks, and constraints. The Log and notification identify whether fitting failed, normalized residuals worsened, or a monitored parameter changed beyond its threshold. Correct or approve the model, then continue; this spectrum becomes the new source. For a series with legitimate abrupt changes, adjust or disable the relevant safeguard, or exclude a known discontinuous function from only the parameter-change monitor.

19.12 A project opens read-only

Another instance may hold the sibling `.lock` file. Close the other instance, or use Save As. Remove a lock manually only after confirming it is stale.

19.13 Export refuses to overwrite

Choose a new directory or a versioned export. If updating an existing bundle, keep the project that records ownership of its exported files and explicitly enable update. Older bundles may instead have a `.curvemole-export.json` ownership manifest; retain it. CurveMole will not claim ownership of unrelated colliding files. For the separate numeric spectrum export, an existing destination requires explicit replacement confirmation; symbolic-link destinations are refused even with overwrite.

19.14 A resampling analysis has many failed replicates

The baseline may be near bounds, poorly initialized, or structurally unstable under perturbation. Increase robustness of the model before increasing replicate count. Report completed and failed counts with any interval.

Even one failed replica produces a spectrum-level warning, not an automatic failure of every parameter. Inspect the cause as well as the count: an isolated difficult fit and repeatedly failing parameter regions can have different consequences. Intervals use successful replicas only. Adaptive checks numerical stability of that successful sample; it cannot demonstrate that excluded failures are harmless. If maximum attempts are reached before stability, the recorded intervals remain available with an explicit warning. A replica timeout instead stops the new analysis and retains any previous completed results. See Section 12 for these distinctions.

19.15 An update is detected but cannot be installed automatically

Automatic updating is supported by Linux x86-64 AppImages and Windows x86-64 desktop packages. AppImage and portable replacement requires a writable directory; installed Windows applications use the verified installer. Finish or cancel active tasks. Python/source installations, macOS packages, unsupported architectures, and protected application directories must be updated from the GitHub release page. On Windows, a failed staged update records a helper log in the system temporary directory; include that log when reporting a reproducible updater failure.

19.16 Reporting a problem

Use GitHub Issues. Include:

- CurveMole version;
- operating system and package type;
- exact steps to reproduce;
- expected and observed behavior;
- terminal output or traceback;
- a minimal non-confidential data file if possible.

Review every attachment. Diagnostic material must not contain unpublished spectra, personal information, credentials, or proprietary data unless you intentionally make them public.

20. Reproducible research, privacy, and citation

20.1 Recommended research record

Archive at least:

- exact CurveMole version and its version DOI;
- software RRID for identification in published methods;
- original data and import mapping;
- complete `.fitproj`;
- analysis bundle;
- model functions and component order;
- initial values, bounds, fixed values, and links;
- masks and transformations;
- point uncertainties or weights;
- fit mode, spectrum weights, solver, and loss;
- random seed and replicate counts for uncertainty analyses;
- warnings, failed replicates, and residual diagnostics.

20.1.1 Laboratory notebook

The **Laboratory notebook** button at the end of the toolbar opens project notes and a searchable description browser as a workspace tab. Use **Add description** from the context menu of a series, individual spectrum, or added fit function to create or edit its description. Function descriptions are specific to the spectrum containing that function, including when sequential propagation preserves a component ID.

Descriptions are grouped by series and spectrum. Names follow renaming and moves. When an object is removed, its description remains in **Deleted items** with the last known name and an explicit deleted status. Restoring the same object with Undo restores its live status. Copying a function does not duplicate its notes.

Edits in the notebook are kept in the current project as you type; save the `.fitproj` file to persist them. Notebook contents are also included in portable project copies. A notebook tab restored at startup displays the current project's contents immediately and follows the project when another file is opened. **Save notebook as text** writes the complete notebook as UTF-8. Alternatively, select **Laboratory notebook (TXT)** in **Export analysis bundle**; this writes `laboratory_notebook.txt` and follows the usual export ownership rules. The notebook can be viewed and exported from a read-only project. Editing is unavailable while a background task runs; a paused sequence allows editing normally.

Series headers are bold and blue, with colors adapted to light and dark themes. A note icon beside a series, spectrum, or fit function opens its description directly. Spectrum notes appear beside the spectrum name, independently of the series note. Uncertainty warnings appear in the State column, allowing both icons to remain visible. Clearing the description removes its icon and notebook entry.

20.2 Local processing and privacy

CurveMole performs fitting locally. It has no telemetry and does not upload data, projects, models, results, or formulas. Its automatic startup and hourly version check requests public latest-release metadata from GitHub. When plugins are loaded, the application also

checks the public validated-plugin catalog at startup and hourly. These checks do not upload scientific data. Opening documentation or release notes, downloading an update, reporting an issue, or using Zenodo is otherwise an explicit external action.

Plugins run locally with the user's permissions and can violate these expectations; trust only reviewed plugin code.

20.3 Citation

20.3.1 Research using CurveMole

Cite the exact software version used in your bibliography. This manual describes **CurveMole 1.2.0**, whose reference is:

> Romi, S. (2026). CurveMole: Modular Scientific Curve Fitting (Version 1.2.0) [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.23910620

If you used another release, including a beta, use that release's own version DOI from its GitHub release notes or Zenodo record. Never substitute the current stable DOI for a different version.

20.3.2 Identification in the methods

Use the software name, exact version and version-independent SciCrunch RRID:

> Data were analysed with CurveMole version 1.2.0 (RRID:SCR_029214).

RRID:SCR_029214 identifies the software; the version DOI identifies the archived release. The methods sentence accompanies the bibliographic reference.

20.3.3 Referring to the project in general

Use the **Concept DOI** when discussing CurveMole as a project across versions:

> Romi, S. (2026). CurveMole: Modular Scientific Curve Fitting [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.22113854

The Concept DOI (10.5281/zenodo.22113854) represents all versions. It resolves to the latest Zenodo archive, which may be a beta, and does not identify the version used for an analysis or the latest stable download.

The citation guide explains historical versions and plugin references. Version BibTeX, project BibTeX and GitHub's **Cite this repository** are generated from `CITATION.cff`.

20.4 Author and contact

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20.5 License

CurveMole is released under GPL-3.0-or-later. User data, private projects, results, and unpublished formulas remain under the user's control. Publicly distributed extensions must follow their applicable license obligations.

Appendix A. Keyboard shortcuts

Menu and toolbar shortcuts use plain keys or Ctrl+letter. Contextual editing also uses the special keys listed below. The table uses portable Qt spellings; **Help > Shortcuts...** shows the native labels and actual bindings for this window, including loaded plugins. Native modifier labels can differ on macOS.

Shortcut	Action
Ctrl+N	New project
Ctrl+O	Open project
Ctrl+I	Import data
Ctrl+S	Save project
S	Save project as
Ctrl+E	Export analysis bundle
Ctrl+T	Export one numeric file per selected spectrum
Ctrl+Z	Undo
Ctrl+Y	Redo
Ctrl+A	Add component
Ctrl+F	Open Fit dialog
Ctrl+J	Quick Start
Esc	Cancel graphical peak or manual-point placement
Ctrl while dragging a normal peak handle	Change a fixed graphical parameter without unfixing it
Double-click during manual-point placement	Finish after the minimum number of points
Right-click during manual-point placement	Remove the nearest point
Right-drag / left-drag with Mask active	Mask / unmask an interval
Up / Down with plot focus	Activate previous / next checked spectrum
Left double-click in normal navigation	Frame unmasked experimental points
Right double-click in normal navigation	Frame all experimental points, including masked points
Enter / Return during placement	Finish after the required minimum points
Delete / Backspace in data-point editing	Delete the selected experimental point
F2 in the series/spectra tree	Rename the selected editable item
F4 in the function list	Open the selected function’s composition menu

Keyboard shortcuts for menu and toolbar commands

Open **Help > Shortcuts...** for the **Mouse** and **Keyboard** tabs. The Mouse tab explains navigation, masking, function placement, point editing and selection. The searchable Keyboard tab lists the currently assigned menu, toolbar and plugin shortcuts, together with keys that apply only in a particular editing context.

Every command in the top menus and toolbars has a dedicated keyboard shortcut, including the Background and Initialize with points toggles. Hover a button to see the command name and its shortcut on the first tooltip line; the existing explanation remains below. Menus also display the shortcut. Commands use single keys or Ctrl+letter, without function keys or multi-step sequences. For example: **A** adds a function, **E** reopens its control

points, **F** runs Quick Fit, **Ctrl+F** opens the Fit dialog, **X** cancels the running task, **B** subtracts background, **R** reverts it, **V** shows all data, and **H** opens function details. Standard Ctrl+S, Ctrl+O, Ctrl+Z and Ctrl+Y remain available.

Shortcuts work while the plot or function list has focus. While typing in a text field, spin box or parameter table, plain keys and native copy/paste/undo remain with that editor. Dialogs and popup menus keep their own keyboard handling. Assignments from the previous beta are migrated automatically. Dynamically added commands receive shortcuts too; disabled commands remain unavailable by keyboard.

Appendix B. Built-in identifiers

Identifier	Display name	Parameters
`gaussian`	Gaussian	`area`, `center`, `sigma`
`lorentzian`	Lorentzian	`area`, `center`, `gamma`
`voigt`	Voigt	`area`, `center`, `sigma`, `gamma`
`pseudo_voigt`	Pseudo-Voigt	`area`, `center`, `fwhm`, `eta`
`constant`	Constant	`offset`
`linear`	Linear	`intercept`, `slope`
`polynomial`	Polynomial	`c0` through selected order
`cubic_spline`	Cubic spline	`y0` through last node

Appendix C. File extensions

Extension	Purpose
`.fitproj`	Complete CurveMole project
`.fitmodel`	Reusable model and formula definitions
`.curvemole-function.json`	Reusable safe formula in `my_curvemole_functions`
`.curvemole-plugin.json`	Manifest for an executable Python plugin
`.yml`, `.yaml`	Reproducible workflow
Any text-file suffix	Imported data when the contents form a valid numeric table
`.whl`	Installable Python package
`.AppImage`	Linux desktop package
`.exe`	Windows installer or portable executable
`.zip`	Windows portable package or plugin download
`.dmg`	macOS disk image

Appendix D. Version 1.2.0 limitations

The following boundaries apply to this release:

- local optimizers include nonlinear least squares, Nelder-Mead, Powell, and L-BFGS-B; none guarantees the global optimum for an arbitrary nonlinear model;
- `sigma_x` is stored but not used in optimization;
- multi-source relations use the advanced expression editor, with readable parameter references inserted through the source picker and **Add**;
- fit ranges exist in the core model but do not yet have a complete graphical editor;
- the GUI does not yet provide one consolidated table for every fit statistic;
- Windows and macOS packages are unsigned; Linux release provenance is available through GitHub artifact attestation;
- the interface and manual are currently maintained in English;
- executable plugins run with the user's permissions; they are not sandboxed.

Inspect fit quality and uncertainty for each analysis as described in Section 1.4. Report unclear workflows and reproducible defects through the issue tracker.

Appendix E. Building this manual

The Markdown file you are reading is the only hand-edited manual source. From a source checkout with Pandoc, XeLaTeX, and `latexmk` installed, run:

```
python scripts/build_manual.py
```

The command validates version declarations and local links, then creates:

```
docs/CurveMole_User_Manual.tex
docs/CurveMole_User_Manual.pdf
```

GitHub Actions repeats this process for documentation changes and releases. Release assets use versioned filenames and include both LaTeX and PDF editions. Do not edit the generated files directly; changes will be replaced by the next automated build.

Appendices F and G reproduce the complete Python scripting reference and YAML workflow reference. Keep these copies in sync when editing; the documentation tests compare their contents and execute their complete examples. When a scripting contract changes, review both standalone AI audits as well. Their verification version records an actual implementation review and must not be relabelled as a newly audited release without checking the contracts.

Appendix F. Python scripting reference

Stable API baseline: CurveMole 1.2.0. Current release behavior. This edition includes the November 2026 scientific and data-integrity corrections. Use the documents matching your installed release; these corrections are not implied for older executables.

Applies to Python **3.12+**. This release documents the GUI-independent scientific engine: imports, arguments, return values, side effects and limits. Private names beginning with `_` and GUI classes are outside this scripting contract. Record the installed version and scientific dependency versions with your results.

Scientific corrections (release, 2026-11-02). The seven findings in the latest scientific audit are corrected in this edition: parameter scaling, nonlinear uncertainty derivatives, convolution alignment, compensated summation, profile confidence regions and saved execution provenance. Appendix J describes the changes and their limits. Use the documents attached to your exact release build; these corrections do not apply retroactively to earlier executables.

1. Installation and imports

Install the 1.2.0 wheel in a Python environment, or run `python -m pip install -e .` from a source checkout. A desktop executable is not a general Python interpreter. Run scripts with `python analysis.py`; no Qt event loop or open CurveMole window is needed. A script creates its own objects and does not edit a project currently open in another desktop process.

The complete package-root exports are:

```
from curvemole import (
    Component, Curve, Dataset, FitPlan, FitResult, FitSettings, Fitter,
    FunctionRegistry, Mask, Model, Parameter, Project, Series, Transformation,
    __version__, default_registry,
)
```

Import `FitMode` and `CancellationToken` from `curvemole.core.fitting`, and `SequentialFitPlan` from `curvemole.core.sequential_fit` (also exported by `curvemole.core`). The additional module-level helpers below are not root exports. `FunctionRegistry()` starts empty; `default_registry()` is a shared mutable registry containing the eight built-ins.

Choose Python for loops, conditionals, existing-project editing, transformations, masks, explicit uncertainty analysis, custom export selections, or acquisition integration. YAML provides the narrower import/model/fit/export pipeline.

2. Complete runnable example

This script generates input and writes only to a new output directory:

```
# example: python-end-to-end
from pathlib import Path
import numpy as np
from curvemole import Component, Curve, FitSettings, Fitter, Project
from curvemole.core.export import BundleExportSelection, export_bundle
from curvemole.core.serialization import load_project, save_project
```

```python
def main():
    out = Path("python-analysis-output")
    out.mkdir(exist_ok=False)
    x = np.linspace(-5.0, 5.0, 401)
    rng = np.random.default_rng(1729)
    y = 0.4 + 8.0*np.exp(-0.5*((x-0.7)/0.6)**2)/(0.6*np.sqrt(2*np.pi))
    y += rng.normal(0.0, 0.02, x.size)
    curve = Curve("synthetic band", x, y, sigma_y=np.full(x.size, 0.02))
    project = Project("Scripted analysis")
    project.add_curve(curve)
    model = project.model_for(curve.id)
    background = Component.create("constant", initial={"offset": 0.0})
    background.is_background = True
    peak = Component.create("gaussian", initial={"area": 7.0, "center": 0.5, "sigma":
↪   0.8})
    peak.parameters["area"].set_bounds(0.0, 20.0)
    peak.parameters["center"].set_bounds(-2.0, 2.0)
    model.add(background)
    model.add(peak)
    result = Fitter().fit_single(curve, model, FitSettings(seed=1729))
    if not result.success:
        raise RuntimeError(result.message)
    project.results["last_fit"] = result
    project.snapshot("Python fit", {"result": result.to_dict()})
    project.touch()
    summary = export_bundle(
        project, out / "bundle", result=result,
        selection=BundleExportSelection.none(
            fit_results=True, wide_tables=True, results_json=True, readme=True,
        ),
    )
    saved = save_project(project, out / "analysis.fitproj")
    reopened = load_project(saved)
    print(result.statistics)
    print("Saved", saved, "curves:", len(reopened.curves))
    print("Created", len(summary.created), "bundle files")


if __name__ == "__main__":
    main()
```

`Fitter` updates successful model parameter values/errors and curve state. It does not receive a `Project`, store project result records, add history, or save files: the script does these explicitly. Check `result.success`, `message`, `warnings` and, for sequential work, `paused_curve_id`.

3. Importing files and columns

Import the following from `curvemole.core.importers`:

Callable	Return/purpose
`detect_config(path)`	`ImportConfig`; infer numeric text layout or supported SPE format
`inspect_file(path, config=None, *, rows=30)`	`FileInspection(path, config, columns, preview, warnings)`; preview is a pandas DataFrame
`import_file(path, mapping, config=None)`	`list[Curve]`, including multiple Y columns or X/Y pairs

Callable	Return/purpose
`import_many(paths, mapping, *, config=None, series_name="Imported series")`	`Series` of curves in input order
`ImportPreview(path, config).curves(mapping)`	`list[Curve]`; parse once for repeated preview mappings
`repeated_spectrum_mappings(...)`	`list[ColumnMapping]` for repeated positional groups

`repeated_spectrum_mappings(column_count, *, y_per_x=1, error_kind="none", first_column=0)` accepts the column count and returns repeated positional mappings.

`ImportConfig` fields/defaults: `delimiter=None`, `decimal="."`, `comment_prefixes=("#", ";", "%", "!")`, `header=None`, `encoding="utf-8-sig"`, `skip_rows=0`. With `config=None`, detection runs; an explicit config uses its supplied/default values. Integer column selectors are **zero-based**; string selectors match parsed column names.

`ColumnMapping` fields/defaults are:

Field	Default	Meaning
`x`, `y`	`None`, `[]`	Shared X selector and list of Y selectors
`pairs`	`[]`	Alternative `(x_selector, y_selector)` pairs
`sigma_x`, `sigma_y`	`None`	Standard-deviation columns
`weights`, `variance`, `inverse_variance`	`None`	Alternative weighting representations
`error_y_minus`, `error_y_plus`	`None`	Both interval half-width columns are required
`error_confidence_level`	`0.95`	Central interval confidence, strictly between 0 and 1
`as_separated_files`	`False`	Retained import setting
`spectra`	`[]`	Per-spectrum `ColumnMapping` objects, each with one X and one Y

Choose one Y uncertainty type: `sigma_y`, interval half-widths, `weights`, `variance`, or `inverse_variance`. `sigma_x` can coexist, but X errors do not implement orthogonal-distance regression. Repeated `error_kind` values are `none`, `sigma_y`, `interval`, `asymmetric`, `weights`, `variance`, `inverse_variance`; `interval` uses one half-width for both sides.

```
from curvemole.core.importers import ColumnMapping, ImportConfig, import_file,
 ↪  inspect_file
preview = inspect_file("measurements.csv")
print(preview.columns, preview.warnings)
curves = import_file(
    "measurements.csv", ColumnMapping(x=0, y=[1], sigma_y=2),
    ImportConfig(delimiter=",", header=True),
)
```

Import validity depends on contents, not only suffix. Inspect metadata/warnings, `curve.invalid`, and `curve.effective_mask`. SPE support covers calibrated, one-dimensional, single-ROI WinSpec SPE 2.x, not arbitrary binary/multidimensional SPE data. Imported numeric columns are retained for column formulas.

4. Curves, masks, and transformations

`Curve(name, original_x, original_y, ...)` requires at least two matching rows. Scientific options: `sigma_x=None`, `sigma_y=None`, `weights=None`, `weights_are_inverse_variance=True`, `error_y_minus=None`, `error_y_plus=None`, `error_confidence_level=0.95`. Labels/units default to `x_label="x"`, `y_label="y"`, `x_unit=""`, `y_unit=""`; `source=None`, `metadata={}`, and an autogenerated `id` preserve provenance. Other fields include `visible=True`, `colour="#0072B2"`, `fit_ranges=[]`, `masks={}`, `active_mask="Default"`, `original_columns={}`, `column_labels={}`, `column_axes={}`. Prefer keywords after the first three positional arguments.

Original coordinate/error/column arrays are immutable. Properties `x`, `y`, `columns`, `row_indices`, `current_sigma_x`, `current_sigma_y`, `current_weights`, `current_error_y_minus`, `current_error_y_plus` expose transformed data. Coordinate properties are read-only views: do not assign to `curve.x` or `curve.y`.

Curve method/property	Behavior
`len(curve)`	Current row count
`invalid`	Boolean invalid-data/error flags
`effective_mask`	`True` means excluded; union of invalid rows, **all** masks, and points outside every fit range
`fit_arrays(*, use_data_errors=True)`	`(x, y, residual_scale_or_None, current_indices)`; raises with fewer than two usable rows
`content_hash`	SHA-256 of current scientific data/masks/errors
`add_mask(name)`	Create/select empty named `Mask`; return it
`mask_point(x_value, *, name=None)`	Exclude nearest finite X; return current row index
`mask_interval(lower, upper, *, name=None)`	Inclusive range; return selected row count
`unmask_interval(lower, upper, *, name=None)`	Clear that mask's flags; return changed row count
`clear_mask(name=None)`	Clear one named/active mask
`transfer_mask_from(source, *, tolerance, source_name=None, target_name=None)`	Match masked X positions; return count
`move_points(indices, *, x=None, y=None)`	Record coordinate edits for current row positions
`delete_points(indices)`	Record deletions for current row positions
`apply_transformation(transformation)`	Apply and retain replayable history
`undo_transformation()`, `redo_transformation()`	`bool`: whether an operation was available
`restore_original()`	Clear transformations/redo and recompute original data
`to_metadata()`	Metadata, not a complete numeric serialization

`Mask(name, excluded, ranges=[], id=generated)` stores Boolean exclusions. `active_mask` selects the target for edits; it does not deactivate other masks. `curve.fit_ranges=[(lo,hi), ...]` defines a union of fitting intervals. `row_indices` retains original identities after point deletion.

`Transformation(operation, parameters={}, description="", created_at=generated, operand=None)` is replayable. Prefer checked helpers from `curvemole.core.calculator`; each modifies the curve and returns `Transformation`:

Helper	Supported options
`apply_scalar(curve, operation, value=None)`	`y_add`, `y_subtract`, `y_multiply`, `y_divide`, `x_add`, `x_multiply` require a finite value; `normalize_max`, `normalize_area` need no value
`apply_custom_formula(curve, axis, formula)`	Axis `x` or `y`; only that coordinate variable is exposed; optional matching assignment such as `Y = 2*y`
`apply_column_formula(curve, target, formula)`	Target `x`, `y` or retained column key such as `c3`; access `x`, `y`, one-based `c1`, `c2`, ... and `${original column label}`
`apply_curve_operation(target, operand, operation, *, interpolation="linear", extrapolate=False)`	`curve_add`, `curve_subtract`, `curve_multiply`, `curve_divide`; interpolation `linear` or `nearest`
`apply_background_subtraction(curve, background, *, method, description, parameters=None)`	Subtract supplied array from all rows, including masked rows

Curve arithmetic stores the interpolated operand as a snapshot. Without extrapolation, out-of-range points become invalid; enabled linear interpolation extends endpoint values. Formula output must be finite and match the row count; scalars broadcast. Coordinate/column formulas propagate existing uncertainty using a numerical first-order local derivative with respect to the target axis. Affine scaling is exact; nonlinear propagation is a local approximation. Asymmetric Y sides swap for a negative derivative. Uncertainty in other input columns, correlations, and operand-curve errors are not jointly propagated. Inverse-variance weights follow the same local Y-uncertainty propagation; direct arbitrary residual multipliers retain their declared multiplier semantics. Normalization uses current data, including masked rows.

Nonlinear formula derivatives use local, scale-aware perturbations and Richardson extrapolation. For `z=y**3`, the audited tiny-value example now agrees with `sigma_z=abs(3*y**2)*sigma_y`. Propagation remains a first-order local approximation; input correlations and uncertainties in other columns are not jointly propagated.

```
from curvemole.core.calculator import apply_scalar
curve.mask_interval(1.2, 1.3)
curve.fit_ranges = [(-4.0, 4.0)]
apply_scalar(curve, "y_multiply", 2.0)
curve.undo_transformation()
```

5. Projects, series, and datasets

`Series(name, curves=[], id=generated, metadata={})` offers `add(curve, index=None)`, `remove(curve_id) -> Curve`, `move(curve_id, index)`. `Dataset(series=[])` offers flattened `.curves`, `add_series(series)`, `curve(curve_id) -> Curve`, `series_for(curve_id) -> Series`, `transfer_curve(curve_id, target_series_id, index=None)`, `validate_unique_ids()`.

`Project(name="Untitled", ...)` owns `dataset`, `models` keyed by curve ID, `results`, `fit_history`, `custom_functions`, `custom_formulas`, `notebook`, `ui_state`, `export_config`, `path`, `read_only`, `dirty`, `revision`, and timestamps/version metadata.

Method/property	Result/side effect
`curves`	Flattened curve list
`add_series(series)`	Add series and create missing empty models

Method/property	Result/side effect
`add_curve(curve, series=None)`	Add to specified/default series, create model; return target `Series`
`validate_curve_removal(curve_ids)`	Check IDs and links before deleting; return None or raise
`remove_curves(curve_ids)`	Deduplicate ordered IDs, validate the entire selection, return removed curves
`remove_curve(curve_id)`	Return removed curve; validate remaining link dependencies and remove affected model/result records
`affected_curve_ids(curve_ids)`	Return set including transitive dependents and recorded global-fit coupling
`invalidate_fits(curve_ids, *, preserve=())`	Return affected IDs minus preserve; mark fitted curves Modified/outdated
`model_for(curve_id)`	Return/create curve's model; read-only missing models are temporary and not inserted
`add_component(curve_id, component)`	Add to model, invalidate its fit/dependents and mark project changed
`parameter_map()`	`{full_path: Parameter}` for all models
`resolved_parameter_values()`	Resolve links project-wide into `{path: float}`
`ensure_editable()`	Check write access before mutation; raises on read-only project
`touch()`	Mark dirty, advance revision/timestamp; raises on read-only project
`mark_saved(path=None)`	Mark saved; normal scripts use `save_project`
`snapshot(reason, payload)`	Append history event; does not fit/save
`to_metadata()`	Metadata; serialization writes complete numeric projects

`copy_fit(source_curve_id, target_curve_ids, *, structure=True, values=True, bounds_and_fixed=True, links=True, background=True, masks=False, fit_ranges=False, component_ids=None) -> None` atomically copies selected model aspects across destinations. `component_ids` limits source functions; include functions referenced by links. With `structure=False`, source positions match the same destination positions. Relative links retarget to destination curves; absolute references remain fixed. A successful copy invalidates fitted destination curves and their transitive linked/global-fit dependents; previous result payloads remain as outdated history.

Add a `Series` to the project before passing it explicitly to `add_curve`. The default `ui_state` is `{"autoscale_enabled": True, "autoscale_mode": "all"}`. Direct edits to contained objects do not call `Project.touch()` or refresh old fit records. After edits, use `invalidate_fits([curve.id])`, call `touch()`, refit, and replace stale result records. Invalidation changes fitted states, not result payloads or project dirty state; dependent curves and recorded coupled global fits are included. `preserve` excludes explicitly preserved IDs from that set.

Curve removal rejects any remaining parameter that refers to a removed curve, raising `ConstraintError` before the removal starts; unknown IDs raise `KeyError`. Selecting all mutually linked curves for `remove_curves` is allowed when no remaining curve depends on them. Removal also marks coupled fits outdated and cleans affected stored model/analysis records. Do not delete series/curves through raw list manipulation when dependency validation is required.

6. Functions, components, and models

`Component.create(function_id, *, registry=None, name=None, initial=None, metadata=None, operator="add") -> Component`. **`initial` is the Python keyword**, not `parameters`. Creation uses only valid function parameter specs: missing keys use defaults, unknown initial keys are ignored, and values are clipped to spec bounds. Later explicit parameter validation rejects invalid values.

Built-in ID	Parameters/defaults	Convention/metadata
`gaussian`	`area=1`, `center=0`, `sigma=1`	Signed integrated area; Gaussian standard deviation; FWHM = `2.354820045*sigma`
`lorentzian`	`area=1`, `center=0`, `gamma=1`	`gamma` is HWHM; FWHM = `2*gamma`
`voigt`	`area=1`, `center=0`, `sigma=1`, `gamma=1`	Area-normalized Gaussian/Lorentzian convolution
`pseudo_voigt`	`area=1`, `center=0`, `fwhm=1`, `eta=0.5`	Common FWHM; Lorentzian fraction `eta` in `[0,1]`
`constant`	`offset=0`	Constant value
`linear`	`intercept=0`, `slope=0`	`intercept + slope*x`
`polynomial`	`c0=0` … `cN=0`	`metadata={"order": N}`, default 2, allowed 0-50; ascending powers
`cubic_spline`	`y0=0` …	Required `metadata={"x_nodes": [...]}`; at least two finite increasing nodes

Peak widths have lower bound `np.finfo(np.float64).eps`; area/center may be negative. Two-node splines use `np.interp` linear interpolation with constant endpoint tails. Splines with three or more nodes use natural cubic boundaries and cubic extrapolation. Components are not automatically marked as background: set `.is_background=True`. Other component fields: `enabled=True`, `group=None`, `metadata`, `id`, `name`, `function_id`, `parameters`, `operator`. `to_dict()` / `Component.from_dict(value)` preserve state.

`Model(name="Model", components=[], ...)` offers:

Method	Return/purpose
`add(component, index=None)`	Insert component; reject duplicate IDs
`remove(component_id)`	Return removed component
`delete_components(component_ids: set[str])`	Delete without changing surviving roles
`component(component_id)`	Look up component
`duplicate(component_id)`	Insert/return copy with new component ID
`move(component_id, index)`	Change list order
`set_component_operator(component_id, operator)`	Validate operator/background restrictions
`parameter_path(curve_id, component_id, name)`	Static method: `curve_id.component_id.parameter_name`
`parameter_map(curve_id)`	Full-path parameter mapping
`evaluate(x, *, curve_id=None, values=None, registry=None, components=False)`	Total ndarray; with `components=True`, `(total, {component_id: ndarray})`

Method	Return/purpose
`evaluate_for_curve(curve, indices, *, values=None, registry=None)`	Fit-row model values; evaluate convolution on the full curve before selecting indices
`background(x, *, curve_id=None, values=None, registry=None, component_ids=None, evaluated_components=None)`	Marked/selected background array
`derived_quantities(*, curve_id=None, values=None, registry=None)`	`{component_id: {quantity: float_or_None}}`
`calculation_expression()`	Human-readable composition
`validate(registry=None)`, `validate_composition()`	Check scientific parameter sets/operators
`clone()`	Deep model copy retaining IDs
`to_dict()`, `from_dict(value)`	Model state serialization

Models use global roles: sum enabled `add` and subtract enabled `subtract` components, then apply `multiply`, `divide`, `convolve` to the total. The starting sum uses compensated accumulation independent of random IDs. `operation_order` controls global operations; unspecified operations use role/ID order. Presentation remains separate through `display_order`/`display_components`. Convolution uses full FFT integration with absolute uniform X spacing and a single centered crop for consecutive kernels. A finite, strictly monotonic, uniform grid is required at every scale. For even-length signals, kernels are sampled on an odd grid with an exact central lag, removing the half-sample shift; ascending and descending axes give the same physical center. Kernel coordinates retain the input-domain midpoint convention. Choose a sufficiently wide, finely sampled window: finite-window truncation and discretization remain approximations. Fitting evaluates convolution on the full current grid, then selects residual rows through `evaluate_for_curve`. Use `evaluate(..., components=True)` directly when individual arrays are needed.

For project-linked models, use `model.evaluate(curve.x, curve_id=curve.id, values=project.resolved_parameter_values())`. Without a `curve_id`, evaluation uses the model ID prefix and cannot independently resolve other curves' links. `curvemole.core.models.component_height(component, *, registry=None)` evaluates a peak at its center; `area_for_height(component, height, *, registry=None)` converts height to area for a component having an `area` parameter.

7. Parameters, bounds, and expression links

```
Parameter(name, value, minimum=-math.inf, maximum=math.inf, fixed=False,
          link=None, unit="", standard_error=None, ci_low=None, ci_high=None,
          link_scope="relative", link_reference_scopes=[], link_relation="equal",
          link_tolerance=0.0, link_tolerance_mode="absolute")
```

`set_bounds(minimum=-math.inf, maximum=math.inf)` sets and validates bounds; `validate()` rechecks fields. An invalid `set_bounds` raises and restores both previous bounds atomically. Set the parameter value inside proposed bounds before applying them; failed bounds do not clamp the value. `.is_free` excludes fixed and equality-linked parameters; `.status` reports free/fixed/bounded/linked states. `to_dict()` / `Parameter.from_dict(value)` preserve fields.

Full link paths use **IDs, not display names**, with `${...}` references:

```
source = source_model.parameter_path(source_curve.id, source_peak.id, "sigma")
target_peak.parameters["sigma"].link = "${" + source + "}"
target_peak.parameters["sigma"].validate()
values = project.resolved_parameter_values()
```

`curvemole.core.parameters.resolve_parameter_values(parameters, *, paths=None)` returns `{path: float}`, checking missing references, cycles and bounds. Equality expressions can combine references, e.g. `2*${curve_id.component_id.sigma}+0.1`.

Relations `lower`, `upper`, `similar` keep targets free and require one direct `${path}`. `lower`: target >= source; `upper`: target <= source; `similar`: source +/- tolerance, intersected with static bounds. Similar tolerance must be positive. `link_tolerance_mode="percent"` uses `abs(source)*tolerance/100`, with tolerance < 100. Set target starting value inside the feasible interval. `link_bounds(source_value)` returns intersected bounds or raises `ConstraintError`.

Nonlinear linked-parameter gradients use bound-aware, scale-aware three-point differences and Richardson extrapolation before covariance propagation. The cubic link agrees with `SE(b)=abs(3*a**2)*SE(a)` at the audited small scales. These remain local covariance uncertainties; strong nonlinearity can require resampling.

`link_scope` is `relative` (default) or `absolute` and governs copying. `link_reference_scopes` can specify each reference occurrence's scope in order. `effective_link_reference_scopes()` returns explicit/inherited scopes; `copied_link(source_curve_id, target_curve_id, component_ids=None)` returns a retargeted expression. Cross-curve links require a **global** fit including source dependencies; independent fits cannot optimize another curve in their objective.

8. Safe expression vocabulary

`curvemole.core.expressions.SafeExpression.compile(source)` validates an expression. `.variables` / `.references` are tuples; `.evaluate(variables=None, references=None)` returns scalar/array values. `expression_parameters(source, *, independent="x")` lists inferred parameter names.

Allowed functions are exactly `abs`, `sqrt`, `exp`, `log`, `log10`, `sin`, `cos`, `tan`, `arcsin`, `arccos`, `arctan`, `sinh`, `cosh`, `tanh`, `erf`, `erfc`, `minimum`, `maximum`, `clip`, `where`, `heaviside`. Constants: `pi`, `e`, `inf`. Arithmetic: `+`, `-`, `*`, `/`, `**`, `%`, unary signs. Comparisons, `and`, `or`, `not`, conditional expressions are supported. Use ** for powers; advanced column formulas also normalize ^. Calls accept positional arguments only. Attribute access (`np.exp`), imports, subscripting, comprehensions, arbitrary calls/statements are unavailable. Write `exp(x)`, not `np.exp(x)`.

`curvemole.core.expressions.formula_source(formula, target) -> str` strips an optional leading assignment to the matching target (case-insensitive), while preserving comparisons such as `y == 1` and keyword-free conditional expressions. An assignment to a different target raises `ExpressionError`. Coordinate and column calculator helpers use this parser; it does not execute Python.

The testing interpreter limits source size (16,384 characters), AST complexity (2,048 nodes, depth 64), integer results (4,096 bits), array/broadcast results (10,000,000 elements) and aggregate array work (50,000,000 elements). Excesses raise `ExpressionError`; powers and broadcasting are checked before large results are computed. Trusted Python plugin callbacks are outside these limits.

9. Custom formulas and function registry

Registry methods: `register(definition, *, replace=False)`, `unregister(identifier)`, `get(identifier)`, `values() -> tuple`, `identifiers() -> tuple`, `extend(definitions)`. Duplicate registration without `replace=True` raises `DataValidationError`.

The testing API also provides `copy()` for an isolated registration table, `register_custom(definition, *, replace=False)` to reject formula collisions with builtin identifiers, and `set_project_functions(specifications)` to validate and replace a project's formula layer atomically. Clearing that layer restores the previous definitions

that have not been removed or disabled. Desktop project functions, the library and workflow formulas use the checked path. Explicit low-level `register(..., replace=True)` remains an extension API; avoid mutating the shared default registry when isolation is needed. A registry copy shares evaluator/definition objects; replacing registrations is isolated, while directly mutating a shared definition object is not a deep copy.

```
from curvemole import Component, FunctionRegistry
from curvemole.core.functions import builtin_definitions, formula_definition
registry = FunctionRegistry()
registry.extend(builtin_definitions())
registry.register(formula_definition(
    "exponential_decay", "Exponential decay", "amplitude*exp(-x/tau)+offset",
    defaults={"amplitude": 3.0, "tau": 2.0, "offset": 0.0},
    bounds={"tau": (1e-6, 100.0)},
    derived_formulas={"half_life": "tau*log(2)"},
))
component = Component.create("exponential_decay", registry=registry)
# Supply the same registry to model evaluation and Fitter(registry).
```

`formula_definition(identifier, display_name, formula, *, kind="generic", defaults=None, bounds=None, derived_formulas=None) -> FunctionDefinition` is in `curvemole.core.functions`. Variables except `x` become parameters in sorted order; missing defaults are 1, missing bounds unbounded. Derived formulas use parameter names. `kind="peak"` does not convert arbitrary functions into built-in area profiles.

Executable definitions from the same module use `ParameterSpec(name, default, minimum=-math.inf, maximum=math.inf, unit="")` and `FunctionDefinition(identifier, display_name, kind, evaluator, parameter_specs=(), parameter_factory=None, derived={}, description="", custom_metadata={}, configuration_specs=(), point_initializer=None, minimum_points=None)`. Evaluator protocol: `(x_array, values_mapping, metadata_mapping) -> shape-compatible numeric output`. Methods: `specs(metadata=None)`, `make_parameters(initial=None, metadata=None)`, `evaluate(x, parameters, metadata=None)`, `derived_values(parameters, metadata=None)`. Nonfinite derived values and `ArithmeticError`, `ValueError`, `TypeError` become `None`; other errors, including `ExpressionError` from an invalid derived formula, can propagate.

Construction options are distinct from optimizable parameters: `FunctionOptionSpec(name, label, default, kind="integer", minimum=None, maximum=None, choices=(), description="", from_points=None)` from `curvemole.core.functions`. `kind` is `integer`, `number`, `boolean`, `text`, or `choice`; `.validate(value)` returns the accepted value or raises `DataValidationError`. Numeric options require finite values and inclusive bounds; integers exclude booleans. Choice values must belong to `choices`.

`FunctionDefinition.configuration(metadata=None)` returns a new dictionary with validated supplied/default option values, preserving other metadata keys. `specs` validates configuration before calling `parameter_factory`; component creation stores the configured metadata. This makes polynomial `order` an actual integer construction option, default 2, from 0 through 50, rather than a fitted coefficient or silently converted string.

`has_point_configuration` reports whether any option supplies `from_points`. `configuration_from_points(x_array, y_array, metadata)` infers such option values with each `from_points(x,y)` callback, validates them, then fills other option defaults. An optional `point_initializer(x_array, y_array, configured_metadata)` returns an initial parameter mapping. `minimum_points` supplies host guidance; these direct methods do not enforce the minimum by themselves. Check point count/finiteness and callable presence before using them in scripts.

Built-in polynomial point configuration infers `order = point_count - 1` from distinct X points, rejects degree above 50, and estimates coefficients in the public ascending `c0 + c1*x + ...` basis. Its `minimum_points` is 2. Numerically unreliable interpolation raises `DataValidationError` instead of silently installing bad coefficients:

```
import numpy as np
from curvemole import Component, default_registry
poly = default_registry().get("polynomial")
xp, yp = np.array([0.0, 1.0, 2.0]), np.array([1.0, 3.0, 7.0])
metadata = poly.configuration_from_points(xp, yp, {})
initial = poly.point_initializer(xp, yp, metadata)
component = Component.create("polynomial", metadata=metadata, initial=initial)
# order=2; c0=1, c1=1, c2=1, within numerical precision.
```

Option callbacks and point initializers are Python callables, not executable syntax accepted by YAML or safe formulas. Persist their configured numeric metadata/parameters; installed plugin source supplies custom callback code.

Registration does not persist executable source. Store safe specifications in `project.custom_functions` with keys `identifier`, `display_name`, `formula`, `kind`, `defaults`, `bounds`, `derived_formulas`. This project key differs from YAML custom function specifications, which use `derived`. After loading a project/model in a fresh process, explicitly rebuild/register those formulas; loading data does not execute plugins:

```
from curvemole import FunctionRegistry
from curvemole.core.functions import builtin_definitions, formula_definition
registry = FunctionRegistry()
registry.extend(builtin_definitions())
for spec in project.custom_functions:
    registry.register(formula_definition(
        spec["identifier"], spec.get("display_name", spec["identifier"]),
↪   spec["formula"],
        kind=spec.get("kind", "generic"), defaults=spec.get("defaults"),
        bounds={name: tuple(pair) for name, pair in spec.get("bounds", {}).items()},
        derived_formulas=spec.get("derived_formulas", {}),
    ))
```

Plugin modules expose `register(api)` (the legacy entry point can accept a registry-compatible facade). `api.version == "1"`; `api.register(FunctionDefinition)` adds a function; `api.add(kind, identifier, label, callback, *, description="", auto_show=False)` returns `<plugin_identifier>:<local_id>`; local IDs must be unique across kinds. Plugins cannot replace existing functions. A local plugin consists of a manifest ending `.curvemole-plugin.json` plus a Python module. Minimal manifest:

```
{"identifier":"org.example.tools","version":"1.0.0","api_compatibility":"1",
 "licence":"MIT","capabilities":["workflows"],"module":"tools.py"}
```

`curvemole.core.plugins.PluginManager(registry=None, *, trusted_identifiers=None, storage=None)` offers `discover_local(directory) -> list[PluginCandidate]`, `discover_entry_points() -> list[PluginCandidate]`, `load(candidate, *, trust=False, expected_fingerprint=None) -> PluginMetadata`, `disable(identifier, reason=None)`, `remove(identifier)`. A supplied fingerprint must match the current source before it is loaded; a mismatch raises `CurveMoleError` and requires review/trust of the changed source. Loading requires explicit trust or a trusted identifier; code runs with normal user permissions. `autoload() -> set[str]` synchronously loads enabled installed plugins and returns failed identifiers. `autoload_steps() -> Iterator[tuple[str,bool]]` performs one enabled plugin load per iteration, yielding identifier/success. Disabled or already-loaded plugins are skipped; source fingerprints are rechecked. Setting `CURVEMOLE_DISABLE_PLUGINS=1` suppresses autoload. `load_timings` records elapsed seconds for iterated loads; iterators must be consumed to perform their work. Python entry-point discovery uses group `curvemole.functions`. For a script using trusted plugin functions, load the reviewed candidate, then use the same registry with `Fitter`.

Extension kind	Callback/result
`importers`, `transformations`, `actions`, `workflows`	`callback(context)`; edit detached project snapshot; desktop commits successful return as one Undo step
`exporters`	`callback(context)`; chosen path is `context.path`; own external file write
`analysis`	`callback(context)`; return readable result text
`panels`	`callback(context)`; return new QWidget, requires desktop Qt host
`plot_layers`	`callback(context)`; return list of `{x, y, colour}` line mappings
`hooks`	`callback(event, context)`; events startup/project_refreshed/shutdown
`fit_solvers`	`callback(request)`; return solver result protocol below
`import_processors`	`callback(context)`; process only the new acquisition in a background worker

Desktop context fields are `project` (detached snapshot), `project_id`, `active_curve_id`, `selected_curve_ids` (tuple), `path`, `owner`, and `data` (plugin-owned JSON dictionary saved in `project.ui_state["plugin_data"]`). Empty projects may have no active curve. Read-only callbacks (analysis, exporter, panels, plot layers, hooks) do not commit snapshot changes. General mutating commands conservatively invalidate fits. These host semantics require the extension host; manually invoking a callback does not create a desktop commit.

An import processor instead receives a fresh acquisition project, owner settings and `context.cancellation`; pass that token to fits. It must not access Qt or shared GUI objects. The host merges completed acquisition series/models/results; acquisition UI state/notebook changes are not merged. Plugin loading does not start folder monitoring or execute a workflow automatically.

Solver request fields: `initial`, `bounds` (lower/upper arrays), `residual(vector)`, `jacobian(vector)`, copied `settings`, `cancellation`. Return an object with `x`, `success`, `status`, `message`, nonnegative integer `nfev`, e.g. SciPy OptimizeResult. Respect bounds/evaluation budget, use supplied residual to retain masks/weights/constraints, and check cancellation in long loops. The host recomputes residual/Jacobian and statistics, without applying a second built-in optimizer. Registered solver IDs work in `FitSettings.solver`. Plugin errors quarantine their owner at hosted boundaries; plugins are not sandboxed. This reference does not make arbitrary internal GUI methods stable.

10. Initialization helpers

Import from `curvemole.core.initialization`:

Callable	Return
`find_peak_suggestions(curve, *, sign="positive", prominence=None, distance=None, max_peaks=50)`	Prominence-sorted list; sign positive/negative/both; widths in local X coordinates, including irregular grids; sorts X/Y together and rejects duplicate X
`PeakSuggestion(x, height, fwhm, prominence, sign)`	Center, signed height, FWHM estimate, prominence, sign +/-1
`component_from_suggestion(suggestion, function_id="gaussian", *, registry=None)`	Estimated Gaussian/Lorentzian/Voigt/pseudo-Voigt

Callable	Return
`initialise_peak_component(component, suggestion, *, registry=None)`	Modify/return peak, including recognized custom peak roles
`initialise_spline_component(component, points, *, registry=None)`	Modify/return spline; sorted unique finite `(x,y)` nodes, Y parameters fixed
`estimate_background(curve, function_id="constant")`	Constant or linear estimate; mark background yourself

Suggestions respect effective masking and may be empty. Inspect them before accepting their starting values as physical parameters.

11. Fitting, plans, callbacks, and concurrency

```
Fitter(registry=None)
Fitter.fit_single(curve, model, settings=None, *, cancellation=None, progress=None)
Fitter.fit(plan, curves, models, *, cancellation=None, progress=None,
           on_curve_result=None)
FitPlan(curve_ids, mode=FitMode.INDEPENDENT, settings=FitSettings(),
        spectrum_weights={}, equal_contribution=False)
```

Both fitting methods return `FitResult`; `curves` is an ID mapping or sequence, `models` maps curve IDs to models. `FitMode.INDEPENDENT`, `.SEQUENTIAL`, `.GLOBAL` have string values `independent`, `sequential`, `global`. Legacy `.COPY`/`copy` is accepted by the enum but does not copy structure in `Fitter.fit`; explicitly call `Project.copy_fit` first.

Independent fits use each existing curve/model separately; global fits jointly optimize selected curves and cross-curve links. `Fitter` delegates directly to the canonical propagating sequential implementation. Its **first selected curve is a prepared seed and is not fitted again**. Prepare/fit that curve explicitly before the sequential call. Subsequent curves receive copied source functions by component ID; existing functions with those IDs are retained, not overwritten. An initially empty target model is usually appropriate for full propagation. The models collection must be mutable because targets may be replaced. Prefer `SequentialFitPlan` for the controls below.

`SequentialFitPlan` extends `FitPlan` with defaults: `monitor_residuals=True`, `residual_ratio_limit=2.5`, `residual_nrmse_delta=0.02`, `monitor_parameters=True`, `parameter_change_limit=0.75`, `propagate_bounds=True`, `propagate_fixed=True`, `propagate_links=True`, `propagate_background=True`, `propagate_enabled=True`, `propagate_composition=True`, `ignored_component_ids=()`, `excluded_copy_component_ids=()`, `copied_component_ids=None`, `progress_completed=0`, `progress_total=None`. **Pass `mode=FitMode.SEQUENTIAL`**; the inherited mode default is independent. Anomalies may pause a series. Inspect `paused_curve_id`, correct and explicitly refit that model, then submit a new ordered sequential plan beginning with that corrected curve as its seed, followed by the remaining curves. A plan containing only its seed returns success with zero fits and no curve outputs. There is no `Fitter.resume()`.

```
from curvemole import FitPlan, FitSettings, Fitter
from curvemole.core.fitting import FitMode
ids = [curve.id for curve in project.curves]
plan = FitPlan(ids, mode=FitMode.GLOBAL, settings=FitSettings(loss="soft_l1"))
result = Fitter().fit(plan, project.curves, project.models)
```

`progress(value, text)` receives fraction or `None` and status text. `on_curve_result(curve_id, result)` reports independent spectra, not per-spectrum global/sequential updates. `CancellationToken().cancel()` requests cooperative cancellation; pass the token to fit and handle `FitCancelled`.

`workers>1` enables **independent multi-curve** process parallelism. It does not split global/sequential objectives. Workers reconstruct built-ins and safe formula

functions, not arbitrary executable/plugin definitions; use `workers=1` for those. Put process-launching entry points under `if __name__ == "__main__":`; add `multiprocessing.freeze_support()` if packaging a frozen executable. Notebook kernels may not support spawning processes.

12. Complete FitSettings catalogue

All names are actual constructor keywords; `settings.validate()` checks them. Built-in optimizers use dimensionless coordinates based on parameter and data scales; numerical Jacobians, automatic DE limits and bound diagnostics use those scales rather than a floor of one physical unit. Positive built-in widths have a float64 normal minimum (about `2.225e-308`), rather than machine epsilon. Explicit bounds remain preferable for difficult searches. Extreme overflow, underflow, nonidentifiability and poor starting values still require scientific judgment. Optimizer coordinate tolerances, including explicit `x_scale`, apply after this normalization; supplied Nelder-Mead simplex vertices remain physical parameters.

Field	Default	Meaning/choices
`solver`	`"local"`	`local`, `trf`, `dogbox`, `lm`, `differential_evolution`, `nelder_mead`, `powell`, `lbfgsb`; loaded solver extensions may add IDs
`local_method`	`"auto"`	`auto`, `trf`, `dogbox`, `lm`; local/DE refinement
`loss`	`"linear"`	`linear`, `soft_l1`, `huber`, `cauchy`
`f_scale`	`1.0`	Positive robust-loss residual scale
`max_nfev`	`1000`	Positive function-evaluation budget
`ftol`, `xtol`, `gtol`	`1e-10`, `1e-8`, `1e-10`	Positive least-squares tolerances above machine precision
`x_scale`	`"jac"`	Jacobian scaling or positive scalar
`seed`	`1729`	Integer 0-4294967295
`workers`	`1`	Positive process count
`confidence_level`	`0.95`	Strictly between 0 and 1
`absolute_sigma`	`None`	Automatic covariance error scaling; `True`/`False` overrides
`use_data_errors`	`True`	Apply imported errors/weights; False retains invalid-error rows if X/Y finite
`de_maxiter`, `de_popsize`	`400`, `15`	Positive generations/population multiplier
`de_lower_percent`, `de_upper_percent`	`50.0`, `50.0`	Positive automatic finite search-bound percentages
`de_mutation_min`, `de_mutation_max`	`0.5`, `1.0`	`0 <= min <= max < 2`
`de_recombination`	`0.7`	Between 0 and 1
`de_tol`, `de_atol`	`0.01`, `0.0`	Finite nonnegative DE tolerances
`nm_xatol`, `nm_fatol`	`1e-4`, `1e-4`	Positive Nelder-Mead tolerances

Field	Default	Meaning/choices
`nm_adaptive`	`False`	Adaptive simplex coefficients
`nm_initial_simplex`	`None`	Finite `(N+1) x N` numeric list for active free parameters
`minimize_maxiter`	`0`	Nonnegative iteration limit; 0 automatic
`powell_xtol`, `powell_ftol`	`1e-4`, `1e-4`	Positive Powell tolerances
`lbfgsb_ftol`, `lbfgsb_gtol`	`2.220446049250313e-9`, `1e-5`	Positive L-BFGS-B tolerances
`lbfgsb_maxls`, `lbfgsb_maxcor`	`20`, `10`	Positive line-search/history limits

Auto local method selects TRF with bounds or robust loss, otherwise LM. Explicit LM requires an unbounded ordinary least-squares problem; peak widths are bounded by default. Differential evolution performs a finite search followed by local refinement. Static finite bounds override automatic search percentages. Simplex dimensions must match active free parameters; dynamic coordinates need care.

Y standard deviations scale residuals by `1/sigma_y`. A directly constructed `Curve(weights=...)` defaults to inverse-variance weights and scales by `sqrt(weight)`; set `weights_are_inverse_variance=False` for direct residual multipliers. In contrast, imported `ColumnMapping.weights` already means direct residual multipliers, while `inverse_variance` means square-root scaling and `variance` means inverse-square-root scaling. Interval half-widths use their confidence level and residual side. `spectrum_weights` multiplies residuals by `sqrt(spectrum_weight)`; `equal_contribution=True` also divides each by `sqrt(usable_row_count)`. These choices affect chi-square interpretation. `f_scale=1` does not estimate instrument noise.

13. Results and diagnostics

`FitResult` fields: `success`, `mode`, `message`, `status`, `evaluations`, `parameters`, `curve_outputs`, `statistics`, `warnings`, `settings`, `free_parameter_paths`, `covariance`, `correlation`, `timestamp`, `elapsed_seconds`, `cancelled`, `paused_curve_id`.

`parameters[path]` is `ParameterEstimate` with `path`, `value`, `standard_error`, `ci_low`, `ci_high`, `minimum`, `maximum`, `status`, `link`, `at_bound`, `fixed`, `link_relation`, `link_tolerance`, `link_tolerance_mode`, `global_minimum`, `global_maximum`. Missing errors/limits are `None`, not zero. `.to_dict()` returns a plain mapping.

`curve_outputs[curve_id]` is `CurveFitOutput` with `curve_id`, `indices`, `x`, `observed`, `fitted`, `residual`, `weighted_residual`, `statistics`. Arrays contain only fitted rows. `indices` indexes current rows; `curve.row_indices[output.indices]` recovers original identities. Reported residual is **observed minus fitted**; optimizer residual has opposite sign.

Global statistics: `N`, `k`, `degrees_of_freedom`, `RSS`, `RMSE`, `chi_square`, `reduced_chi_square`, `R_squared`, `AIC`, `AICc`, `BIC`. Per-curve statistics: `N`, `RSS`, `RMSE`, `R_squared`. Gaussian information criteria use the likelihood of the weighted residuals, including its precision normalization. A relative or unknown noise scale adds one likelihood parameter; independent fits retain their separate fitted noise scales. The result records `negative_twice_log_likelihood` and `likelihood_parameters`. AIC/AICc/BIC are unavailable for robust losses, active interval-error arrays (including equal lower/upper half-widths) and arbitrary spectrum balancing. Insufficient degrees or rank may leave covariance unavailable. Matrix rows follow `free_parameter_paths`. Independent-batch matrices may be `BlockDiagonalMatrix`; `np.asarray(matrix)` densifies and can require substantial memory.

`result.to_dict(arrays=False)` omits sampled curve/matrix arrays; `to_dict(arrays=True)` includes them. `FitResult.from_dict(mapping)` restores only outputs with complete arrays. Save full arrays for uncertainty baselines.

New fit results record `provenance`: original model/constraint snapshots, input content hashes, ordered parameters, initial solver coordinates, scales, resolved solver bounds/method, settings, seeds, function/plugin fingerprints, CurveMole source identity, Python/NumPy/SciPy versions and backend/thread details. Profile and resampling outputs also retain their baseline identity and actual analysis configuration. Historical results created before this feature have no reconstructible original state. Keep original data and scripts: hashes identify inputs but do not store a separate copy of every historical dataset.

`Fitter.replay(result, curves, *, verify_environment=True, **fit_keywords)` reconstructs the saved initial plan/models and checks input hashes, function and solver fingerprints. The default also rejects a changed numerical environment. Use `verify_environment=False` explicitly for a comparison across environments. Older results without initial-state provenance raise `FitError`. Replay needs the original matching curves; later transformations can change their content hash. A new fit from a saved optimum is distinct from replay.

`curvemole.core.diagnostics.residual_diagnostics(residual, *, max_lag=None, histogram_bins="auto") -> ResidualDiagnostics` returns standardized residuals, histogram, Q-Q arrays, autocorrelation, Durbin-Watson statistic, outlier indices, warnings; `.summary()` returns compact statistics. `estimate_block_length(residual)` estimates bootstrap block length.

14. Explicit uncertainty analysis

From `curvemole.core.uncertainty`, import `UncertaintyAnalyzer`, `AdaptiveReplicateSettings`, `ResamplingResult`, `ProfileResult`. `UncertaintyAnalyzer(fitter=None, *, replica_timeout_seconds=None)` uses a 60-second per-replica timeout by default. Supply a fitter using the baseline's registry, successful current baseline, matching plan/data, and fitted models.

Three methods return `ResamplingResult` and share arguments `(baseline, plan, curves, models, *, replicates=200, adaptive=None, seed=None, workers=1, cancellation=None, progress=None)`:

- `parametric_monte_carlo(...)`: requires finite positive sigma_y or valid confidence-interval Y errors on fitted rows; weights alone are insufficient.
- `residual_bootstrap(...)`: resamples residuals around the fitted model.
- `block_bootstrap(..., block_length=None, ...)`: residual blocks; None estimates block length for each curve.

Both bootstrap methods standardize sigma and inverse-variance noise equivalently; automatic block lengths use the same standardized residuals. Supplied noise scales remain active for resampling when `use_data_errors=False` disables their use in the fit. Arbitrary direct residual multipliers do not define noise scales. Large global fits with at least 16 free parameters use sparse numerical derivatives when the dependency pattern is less than half full; automatic local mode selects TRF for these problems. Explicit LM and plugin solver callbacks retain a dense Jacobian interface. Covariance reduction uses bounded row chunks; the final parameter covariance still occupies quadratic parameter storage.

```
from curvemole import FitPlan
from curvemole.core.uncertainty import UncertaintyAnalyzer
plan = FitPlan([curve.id], settings=result.settings)
analysis = UncertaintyAnalyzer().residual_bootstrap(
    result, plan, [curve], {curve.id: model}, replicates=200, seed=1729,
)
print(analysis.intervals, analysis.completed, analysis.failed)
```

```
project.results["bootstrap_python"] = analysis.to_dict()
project.touch()
```

Result fields: `method`, `requested`, `completed`, `failed`, `seed`, `parameter_paths`, `samples` (replicate-by-parameter), `intervals` (path to (low,high)), `confidence_level`, `configuration`, `failure_messages`. `to_dict(include_samples=True)` controls stored samples. Inspect failed attempts; `seed=None` uses the baseline settings seed. Replicas use local optimization, including when the baseline used another solver. `replicates` must be positive; replica failures reduce the number of accepted samples, rather than silently counting as successful fits.

`AdaptiveReplicateSettings(initial_successes=500, batch_successes=200, tolerance=0.05, consecutive_checks=3, maximum_attempts=10000)` stops when all percentile endpoints stabilize for the required checks; `.validate()` checks settings. `maximum_attempts` caps attempts.

`profile_parameter(baseline, curve, model, parameter_path, *, lower=None, upper=None, points=31, confidence_level=0.95, spectrum_weight=1.0, equal_contribution=False, workers=1, cancellation=None, progress=None)` returns `ProfileResult` with `parameter_path`, `values`, `delta_chi_square`, `confidence_level`, `interval`, `failed_points`, `confidence_regions`, `warnings`, `provenance`, and `.to_dict()`. Limits must be finite/increasing and inside parameter bounds; at least three points are needed. Equality-linked targets and joint cross-spectrum linked profiles are unavailable; use resampling for the linked global case. Gaussian profiles use a chi-square difference for known absolute noise and $N \log(Q/Q_{min})$ for an estimated common noise scale. Robust losses, active interval-error arrays (symmetric or asymmetric), and arbitrary spectrum balancing are rejected; use bootstrap. Interval arrays that are ignored with `use_data_errors=False` do not trigger that restriction. A zero-residual fit cannot estimate an unknown noise scale: provide known sigma. Convolution preserves the full regular grid before selecting fitted rows.

Profile threshold crossings are refined by bracketed refits. `confidence_regions` contains every connected region detected by the scan; `interval` selects the region containing the fitted estimate and never bridges rejected gaps. Disconnected sets are marked **Critical**. `warnings` reports incomplete scans or unrefined endpoints; inspect every region and extend the scan where needed. A finite grid can still miss a narrow remote mode. The default Gaussian-location example now agrees with the analytic 95% interval, approximately +/- 1.959964 standard errors.

`resampling_batch(method, jobs, *, replicates, option=None, workers=1, adaptive=None, cancellation=None, progress=None, on_result=None)` returns a list for jobs (`baseline, plan, curves_mapping, models_mapping`). Method IDs: `monte_carlo`, `residual_bootstrap`, `block_bootstrap`; option supplies block length. Batch callbacks are `progress(completed_job_count)` and `on_result(job_index, result)`; other uncertainty progress uses `(fraction,text)`.

Analysis calls do not create the desktop's per-curve uncertainty display records. A custom key in `project.results` stores returned data but does not imply GUI selection/integration.

15. Saving projects/models and YAML interoperability

Saved state and full computational replay require different information. Ordinary fit records retain final parameters and settings but do not systematically store the original initial vector, resolved DE search bounds, exact Python/NumPy/SciPy and backend versions, RNG implementation, or source/plugin fingerprints used for that analysis. Keep the original script/workflow, inputs and environment alongside the project. A seed alone is insufficient; refitting the saved optimum can also recompute different automatic search bounds.

Import from `curvemole.core.serialization`:

Callable	Return/behavior
`save_project(project, path, *, include_uncertainty_samples=True, portable=False, update_project_path=True, pre_commit_check=None)`	`Path`; atomic `.fitproj`, creates parent and normalizes suffix
`load_project(path, *, partial_recovery=False, limits=None, cancellation=None)`	Restored `Project`
`validate_project_archive(path, *, raise_on_error=True, limits=None)`	`list[str]` failures; raises by default when nonempty
`capture_project_snapshot(project)`	Detached `ProjectSaveSnapshot`
`save_fitmodel(model, path, *, custom_functions=None)`	`Path` to JSON `.fitmodel`
`load_fitmodel(path)`	`(Model, list_of_custom_function_specs)`
`ProjectLock(Path(path))`	Context manager; inspect `.acquired` before shared-file edits

`pre_commit_check` is an optional callable executed after archive validation and immediately before replacing the destination; raising cancels that replacement. It supports a caller's lock/digest guard and does not itself acquire a lock.

The testing reader and validator accept `limits=ArchiveLimits(...)`, imported from `curvemole.core.archive_validation`. Defaults are 100,000 ZIP members, 1 GiB expanded content, 256 MiB per array and 32 MiB of manifest. Larger trusted loads can opt in to larger limits. `save_project` still validates against the default limits: raising a read limit does not enlarge its save budget. `load_project` optionally accepts a cancellation object with `raise_if_cancelled()`; checks occur during streamed reads, not as a hard interruption of a running NumPy allocation.

`portable=True` strips local paths. Detached snapshots require `update_project_path=False` when saving. Project schema is **4** and model schema is **2**. Workflow schema is **2** in 1.1.0 and **3** from 1.2.0 additions; legacy workflow schemas 1/2 remain accepted for their older fields. Archives contain JSON/numeric arrays, not pickle. Older project schemas remain readable. An otherwise readable project with a newer schema opens read-only; invalid schema numbers or damaged content raise `ProjectFormatError`. Unsupported model-file schemas raise `ProjectFormatError`. `load_project` does not acquire a project lock and `save_project` does not itself honor another process's lock: coordinate shared-file writes. `load_fitmodel` retains IDs and links; retarget links or use `Project.copy_fit` when applying to a different curve.

`curvemole.core.workflow` exports `load_workflow(path) -> dict`, `validate_workflow(value) -> list[str]`, `run_workflow(path, *, trust_plugins=None, registry=None) -> WorkflowOutcome`, `dump_workflow(project, path) -> Path`. Current dumping emits workflow schema 3; new YAML keys require schema 3 so older software rejects them instead of ignoring processing stages. Outcome fields are `project`, `result`, `outputs`. In 1.2.0 it also has `imported_files: list[Path]` and `warnings: list[str]`, both defaulting to empty lists so the original three-field constructor remains valid. Relative paths resolve against the YAML directory. Dumping references source imports/models; it does not embed arbitrary transformed current data or all project state. Version 1.1.0 dump includes saved `curve_ids`, which the loader maps to newly imported curve IDs in model selectors and `${...}` references. Ambiguous display-name aliases are not used as selectors; explicit saved curve identities disambiguate them. Links are resolved project-wide before fitting or export. Save `.fitproj` for full scientific state. The YAML audit details the complete import identity and reference format.

15.1 Beta workflow discovery, selection and preprocessing

The following helpers require **1.2.0** and are module imports, not new package-root exports. They implement the same contract as the YAML runner; using Python is not required merely to enumerate a folder anymore.

From `curvemole.core.workflow_inputs`:

Callable	Return/contract
`resolve_import_paths(item, root)`	`list[Path]`; expands one `path`/`glob`/`directory` import relative to a workflow-directory root
`validate_import_source(item)`	`list[str]` static source-selection errors; does not read files
`validate_name_template(template)`	`list[str]` errors for the restricted naming placeholders
`format_import_name(template, path, root, *, file_index, curve_index, curve_name)`	Generated name; indices are positive one-based file/curve positions

For example, this independently discovers naturally ordered inputs:

```
from pathlib import Path
from curvemole.core.workflow_inputs import resolve_import_paths
files = resolve_import_paths(
    {"directory": "data", "pattern": "*.csv", "recursive": True,
     "exclude": ["*_skip.csv"], "sort": "natural"},
    Path("/absolute/path/to/workflow-folder"),
)
```

A literal `path` is still one file. Batch matching skips symbolic links, supports explicit hidden pattern components, and fails on empty selection unless `allow_empty=True`. Naming placeholders are `stem`, `filename`, `parent`, `relative_path`, `relative_stem`, `file_index`, `curve_index`, `curve_name`; attribute/index access, conversions and format specifiers are rejected. `relative_path`/`relative_stem` are relative to the supplied root.

From `curvemole.core.workflow_selection`, `select_curve_ids(project, aliases, *, curve=None, curves=None, match=None, default_all=False) -> list[str]` selects one identity/name/runtime ID, an explicit list, `curves="all"`, `"first"`, `"last"`, or a case-sensitive name/alias pattern. Only one selector may be supplied; unknown, duplicate or empty selections fail. `validate_curve_selector(specification, *, default_all=False) -> list[str]` checks the corresponding mapping without requiring imported data.

From `curvemole.core.workflow_steps`, `validate_preprocessing(steps) -> list[str]` checks supported stage/operation shapes and safe formula syntax. `apply_preprocessing(project, steps, aliases) -> None` uses the same reversible core calculators and masks as normal scripts, marks the project changed, and invalidates affected fits. All shapes and selections are checked before mutation; a later data-dependent error can leave earlier in-memory edits. The workflow runner writes no outputs before these stages finish.

Beta workflows additionally reject misspelled nested keys and unknown function-specific parameter names, support `models[].curves`/`match` so one model specification can cover a batch, convert nested `columns.spectra` mappings, and provide `export.project_overwrite=False` for project-file protection. Safe YAML formula definitions are now retained in `Project.custom_functions` and emitted by `dump_workflow`; they are not just registered in memory as in 1.1.0. Dumped recipes still do not replay preprocessing, masks, transformed data, fit plans or exports. The YAML reference lists the exact new fields and complete runnable examples.

16. Tables, exports, figures, and notebooks

Import from `curvemole.core.export`:

Callable	Return
`wide_dataframe(curve, model=None, *, registry=None, selection=None, parameter_values=None)`	DataFrame of current data, model, residual, components/errors/mask
`tidy_dataframe(project, *, registry=None, selection=None)`	Long DataFrame of all curves
`analysis_dataframes(project, *, registry=None)`	`(observations, components, curves)` DataFrames; identity `(curve_id, point_index)`
`parameter_dataframe(project, result=None, *, registry=None)`	Parameter/derived-value DataFrame
`fit_settings_dataframe(result), uncertainty_dataframe(project)`	Settings/recorded uncertainty DataFrames
`export_dataframe(frame, path, *, delimiter=",")`	`Path` to numeric table
`export_function_parameters(project, path, *, include_covariance=False)`	`Path`; one row per spectrum with parameter/error columns
`export_parameter_covariances(project, parameter_path)`	`list[Path]`; covariance directory derives from parameter-file path; requires per-curve fit records
`value_with_error(value, error)`	Formatted value/error string
`export_bundle(project, directory, *, result=None, delimiter=",", versioned=False, overwrite=False, include_uncertainty_samples=True, selection=None, current_view=None)`	`ExportSummary`: directory and the `created`, `updated`, `unchanged`, `preserved_external` path lists

`ExportSelection` defaults all fields True except `original_x=False`: `x`, `y`, `total_fit`, `residual`, `background`, `components`, `sigma_y`, `weights`, `mask_flag`. Pass it to table functions; for linked curves, `wide_dataframe` needs `parameter_values=project.resolved_parameter_values()`.

`BundleExportSelection` defaults every field True: `fit_results`, `fit_settings`, `wide_tables`, `tidy_table`, `results_json`, `fitmodel`, `project_copy`, `main_plot_png`, `main_plot_svg`, `html_summary`, `html_reproducibility`, `pdf_summary`, `uncertainty`, `diagnostics`, `readme`, `laboratory_notebook`, `spectra`, `function_parameters`, `parameter_covariances`, `current_view`. `.none(**enabled)` enables only named items; `.to_dict()` / `.any_selected()` inspect selection. Some items require results, recorded analyses or a GUI-provided view. Pass `result` explicitly or set `project.results["last_fit"]`. `versioned=True` uses a timestamped subdirectory. Default export refuses collisions; `overwrite=True` updates CurveMole-owned files and preserves external files. Keep the project's ownership metadata when rerunning.

`curvemole.core.spectrum_export` provides `SpectrumExportOptions(subtract_background=False, unmasked_only=False, include_background=True, include_components=True, include_total_fit=True, include_residual=True)`, `spectrum_export_dataframe(project, curve_id, *, options=None, registry=None) -> DataFrame`, `export_spectra(project, directory, curve_ids, *, options=None, registry=None, overwrite=False) -> list[Path]`. Numeric exports use 17 significant digits and stage conversions before publishing. Existing files require explicit overwrite; symbolic-link destinations are refused. Multi-file filesystem failures can still leave a partial set of published files. Numeric spectrum exports preserve explicit weights and write `weights_are_inverse_variance` as 1 (inverse variance) or 0 (direct multiplier). Wide and quick tables disambiguate all colliding

headers with `__2`, `__3`, … . Returned DataFrames also expose `attrs["column_roles"]` for the allocated X/Y names (and original X in wide exports), so callers need not infer axis identity from display labels.

`export_figure(curves, models, path, *, registry=None, dpi=300, width=9.0, height=5.5, transparent=False, include_components=True, include_residuals=True, include_error_bars=True, parameter_values=None, theme="light") -> Path` creates a standalone figure. Width/height are inches; theme is `light` or `dark`. The `core.export` wrapper forwards these keywords to `core.figure_export`.

`generate_html_report(project, path, *, result=None, title=None, notes="", full=False, registry=None) -> Path`; `generate_pdf_report(project, path, *, result=None, title=None, registry=None, curves=None) -> Path`. `curves=None` selects all project curves; pass an iterable for a subset. Bundle selections avoid supplying plotting internals. Renderers create standalone figures without importing pyplot or changing the backend; a script using pyplot independently can choose `Agg` before importing it.

`project.notebook.set_description(project, kind, object_id, text, curve_id="")` sets descriptions with exact kind names `series`, `spectrum`, `function`. For a spectrum use its curve ID as `object_id`; for a function use the component ID and pass its curve ID separately. Free project notes go in `project.notebook.notes`; call `project.touch()` after assigning them. `curvemole.core.notebook.export_notebook(project, path) -> Path` exports notes and descriptions.

17. Background edit and transaction helpers

These module helpers are for integrations that already manage edits explicitly; they do not create GUI Undo commands or automatically mark/save a project.

`curvemole.core.background_edit.updated_background_history(curve, model, component, *, values, registry)` returns a detached replay-validated `list[Transformation]`, or `None` when that component is not a recorded subtracted background. It replaces existing matching subtraction operands in their original positions, including grouped backgrounds, rather than subtracting twice. Supply current project-resolved values and the same function registry. The input curve/history are not changed by this call. All required background functions must be identifiable; invalid background values or an X-axis change after subtraction raise `DataValidationError`. Subsequent processing/deletions are replayed before a history is returned. A caller must separately commit the returned history using its own transaction and invalidate affected fits.

`curvemole.core.state.copy_result_maps(results) -> dict` copies mutable outer result-map containers and nested per-method map containers. Completed numerical result payloads remain **shared**; it is not a deep clone of arrays/FitResult. `restore_model(target, saved, refs=None) -> None` restores a saved model state while retaining existing component/parameter objects matched by component ID and parameter name. `refs`, when supplied, maps component ID to `(component_object, parameter_mapping)`; missing objects are deep-copied from the saved model. This is useful for undo/rollback with live edit references. Neither helper validates links, invalidates fits, calls `Project.touch`, or writes files.

18. Other low-level utility contracts

These are module utilities, not additional YAML operations or a general desktop job scheduler. Normal scripts should use `Fitter`/`UncertaintyAnalyzer` for worker management.

`curvemole.core.process_pool.managed_process_pool(pool, *, drain=None)` is a context manager yielding an owned executor, draining optional telemetry while shutting it down

and terminating remaining workers within bounded teardown. `bounded_results(pool, inputs, submit, cancellation, stop, *, limit, tick=None)` yields `(input_index, result)` as futures complete, with at most `max(1,limit)` pending submissions. `submit(pool, input_value)` returns a Future; cancellation is checked while waiting and optional `tick()` runs each cycle. On incomplete exit it sets `stop` when supplied and cancels pending futures. Output order is completion order; use the yielded index to recover input order.

`curvemole.core.release_channels.ReleaseVersion(major, minor, patch, final, beta=0)` is an ordered frozen value object. `ReleaseVersion.parse(text)` accepts `1.1.0`, `v1.1.0`, `1.1.0-beta.2`, `1.1.0b2` and returns `None` for invalid version syntax. Properties `numbers`, `text`, `label` expose numeric tuple, normalized tag text and readable label. `default_channel(version)` returns `beta` for a parsed prerelease and `stable` otherwise. `select_release(payload, channel)` selects the greatest eligible version from a release mapping/list and returns its mapping or `None`. `channel` is stable/beta; stable excludes prereleases and both exclude drafts or invalid tags. Invalid payload/channel raises `ValueError`. These helpers perform no network access, application update or release publication.

19. Errors and requesting a script from AI

From `curvemole.core.errors`: base `CurveMoleError`, `DataValidationError`, `ExpressionError`, `ConstraintError`, `FitError`, `FitCancelled` (FitError subclass), `ProjectFormatError`, `PluginTrustError`. Lookups can raise `KeyError`, filesystem operations `OSError`. Fits may return `success=False` without raising; inspect both errors and result diagnostics.

Specify input layout, selectors, units, model IDs, starting values/bounds, outputs, uncertainty method and installed CurveMole version. Ask AI to use only documented APIs, state assumptions, check success, retain warnings, preserve input files, and provide a run command. Supply a few non-sensitive representative rows when needed; repository source is unnecessary for this documented surface.

> Using the attached CurveMole 1.2.0 Python audit, write `analysis.py` for Python 3.12+. CSV has named columns `x`, `signal`, `sigma`. Fit constant background plus an area-normalized Gaussian near 520 with positive sigma, mask 530-532, report fitted values/errors/warnings, and save a new `.fitproj` plus numeric results in a new output directory. Use only documented APIs, guard the script entry point, check convergence, and state scientific assumptions.

For unsupported tasks, AI should name the missing capability rather than invent methods. This document audits API/automation capabilities; it is not a security certification or scientific validation of AI-generated analysis.

Appendix G. YAML workflow reference

Stable API baseline: CurveMole 1.2.0. Workflow schema: 3; legacy schemas 1 and 2 remain accepted. Checked on 2026-11-02. This reference documents CurveMole 1.2.0, including folder/pattern imports, templates, shared selectors, preprocessing, validation and output controls. Use `schema_version: 3` for these workflow fields. Schemas 1 and 2 retain their legacy fields; older executables can reject newer recipes. This is the runner contract, not every desktop action.

Current release behavior. This edition includes the November 2026 scientific and data-integrity corrections. Use the documents matching your installed release; these corrections are not implied for older executables.

Scientific corrections (release, 2026-11-02). The seven findings in the latest scientific audit are corrected in this edition: parameter scaling, nonlinear uncertainty derivatives, convolution alignment, compensated summation, profile confidence regions and saved execution provenance. Appendix J describes the changes and their limits. Use the documents attached to your exact release build; these corrections do not apply retroactively to earlier executables.

W1. What a workflow can do

A workflow imports data into one new project and one series, constructs models, optionally runs one fit plan, and optionally saves a project and an analysis bundle. In 1.2.0, imports can select individual files, glob patterns or folders; optional preprocessing prepares the curves before model construction. The scientific engine is the same engine used by Python and the desktop application. YAML is a declarative recipe, not Python code or a general task scheduler.

The execution order is fixed:

1. Parse and check the YAML root and supported schema.
2. Register custom formula functions, then load explicitly trusted plugins.
3. Create a fresh project and series; expand each import specification and import its files in the selected order.
4. In 1.2.0, apply optional preprocessing, then select curves for each model.
5. Add components/constraints, remap declared curve identities in links, and resolve the complete parameter graph (even when no fit is requested).
6. Run the single optional fit plan.
7. Save `export.project`, then write `export.bundle`, when requested.

The 1.2.0 preprocessing stage supports masks, fit ranges and the documented scalar/formula transformations. There are no YAML stages for background subtraction, curve-to-curve calculator operations, repeated fits, conditional branches, loops, live folder monitoring, opening an existing `.fitproj`, explicit Monte Carlo/bootstrap/profile analyses, or graphical interaction. Loading a plugin registers its extensions; it does not automatically invoke a plugin command, transformation, import processor, or workflow. Use the Python API for operations outside this contract. A plugin can register a fit solver usable through `fit.settings.solver`.

W2. Run, validate, and inspect

Install the exact downloaded CurveMole wheel in a Python 3.12+ environment. For the current operations install the `1.2.0` wheel; older wheels may support only a subset of this interface. Then run:

```
curvemole --version
curvemole inspect spectrum.csv --json
curvemole functions list --json
curvemole validate workflow.yml --json
curvemole run workflow.yml --json
```

`python -m curvemole` is an alternative to the `curvemole` executable. The global `--json` option may also appear before the subcommand. `run` accepts only the workflow path, repeated `--trust-plugin IDENTIFIER` options, and `--json` beyond the global options; it does not accept fit or export overrides.

In 1.1.0, validation checks only the outer schema and container types. In 1.2.0, validation also checks supported keys and nested shapes in imports, selectors, preprocessing, components/constraints, fit settings and exports, so misspelled keys are reported before execution. It does **not** read the input files, execute plugins, establish physical model suitability or prove fit convergence. Column availability, registered functions, links and file collisions still need an actual run. Run the recipe on representative input and examine the results before batch use.

Normal JSON output from `run` has these keys:

```
{"project": "Example", "success": true, "statistics": {}, "outputs": [],
 "imported_files": [], "warnings": []}
```

`success` and `statistics` are `null` when no fit ran. `outputs` contains paths to saved projects and created/updated bundle files; it omits unchanged bundle files. The actual statistics object contains the fit statistics. In 1.2.0, `imported_files` records resolved input paths in import order and `warnings` reports skipped empty selections, for example when `allow_empty: true`. These two additional JSON keys are absent in the 1.1.0 baseline. `run` returns exit code 0 when it returns normally, **even if `success` is false**. Batch callers must check both the process status and the JSON `success` value. A failed fit can still produce exports for inspection.

Exit code	Meaning
0	Command returned normally; inspect fit success separately
2	Command-line usage error
3	Data/import failure; failed YAML validation reported as a result
4	Raised fit failure
5	Project archive failure
6	Other CurveMole domain error, including plugin/workflow errors
10	Unexpected failure, including some malformed nested YAML or file collisions

For desktop use, select **Run automation** in the bottom **Automations** toolbar. The first run asks for a `.yml` or `.yaml`; subsequent runs remember that file. The adjacent menu provides **Choose automation...** and **Stop automation**. The runner starts a separate process with its own fresh project and imported data. It does not fit or change the currently open desktop project. Set `export` to retain results. Stopping kills the worker; files already written remain. Only one automation runs through this control at a time. The desktop application reports nonconvergence separately from execution failure.

W3. Complete runnable recipe

Save the following as `workflow.yml` beside the repository's `examples/gaussian.csv` (or download that example into the same directory).

```yaml
# example: yaml-end-to-end
schema_version: 2
name: Reproducible Gaussian fit
series_name: Reference spectra

imports:
  - path: gaussian.csv
    config:
      delimiter: ","
      decimal: "."
      header: true
    columns:
      x: x
      y: [y]
      sigma_y: sigma_y
    aliases: [reference]

models:
  - curve: reference
    name: Background and Gaussian
    components:
      - function: constant
        id: baseline
        name: Baseline
        is_background: true
        parameters: {offset: 0.0}
      - function: gaussian
        id: peak
        name: Main band
        parameters: {area: 3.0, center: 0.3, sigma: 1.0}
        constraints:
          sigma: {minimum: 0.001, maximum: 5.0}

fit:
  mode: independent
  curves: [reference]
  settings:
    solver: local
    loss: linear
    max_nfev: 1000
    seed: 1729
    use_data_errors: true

export:
  project: outputs/gaussian.fitproj
  bundle: outputs/analysis
  versioned: true
```

Initial numbers must be appropriate for the actual data. Peak `area` is signed integrated area, not peak height; `sigma` is a Gaussian standard deviation. `is_background` marks a component for background-specific exports; it is not automatically inferred from `function: constant`.

W4. YAML and path rules

The document root must be a mapping. Use a single YAML document, spaces for indentation, and literal field names from this reference. YAML is read with `yaml.safe_load`. Use YAML booleans (`true`, `false`), `null`, integers and numeric values rather than quoted versions of them. For example, `fixed: "false"` is a nonempty string and is converted to true by the runner. Prefer ordinary decimal notation such as `0.00000001` when a parser might read an exponent as a string. YAML `.inf`/`-.inf` can represent unbounded limits; `null` cannot replace a numeric bound.

All relative input paths, plugin manifest paths, project output paths and bundle paths resolve against the **workflow file's directory**, regardless of the terminal working directory. Absolute paths are also accepted. ~ and environment variables such as `$HOME` are not expanded. The legacy `path` remains a literal single-file path: put wildcard expressions in the `glob` field or use `directory`/`pattern`. Quote Windows paths with single quotes or use forward slashes. Extensions alone do not determine whether the text importer can read a file; supported calibrated WinSpec SPE 2.x spectra also use the importer.

The validator rejects unknown keys in supported workflow blocks instead of silently accepting typos. Function-specific initial-parameter and constraint names are checked at execution after the function configuration/plugins are available. The 1.1.0 runner can ignore some unused keys, so validation there is weaker. Validation is not a substitute for inspecting imported curves, resolved parameters, convergence and output contents.

W5. Top-level fields

Key	Type/default	Contract
`schema_version`	Required integer	Beta uses 3; legacy 1 and 2 accept only their older fields, with current global composition semantics
`name`	String; workflow filename stem	New project name
`series_name`	String; `Workflow data`	Name of the single imported series
`custom_functions`	List; `[]`	Formula function definitions; registered before plugins and models
`plugins`	List of manifest paths; `[]`	Local `.curvemole-plugin.json` files; explicit trust required
`imports`	Required nonempty list	File specifications, expanded in order
`preprocess`	List; `[]`; 1.2.0+	Ordered curve preprocessing before models/fitting
`models`	List; `[]`	Model/component specifications attached to imported curves
`fit`	Mapping; absent	One fit plan. Absent or `{}` skips fitting
`export`	Mapping; `{}`	Optional project and/or bundle output

For independent/global fitting, each selected curve needs a usable model. In sequential mode, the first curve instead needs a prepared source model whose components propagate to later targets. Multiple model entries targeting the same curve add to the existing model. The runner creates new curve IDs on every import, even for identical inputs.

W6. Import specification and column mapping

Each `imports` item supports:

Key	Type/default	Contract
`path`	One source selector	Literal single-file input; stable baseline supported
`glob`	One source selector; 1.2.0+	File pattern relative to the workflow directory, or an absolute pattern
`directory`	One source selector; 1.2.0+	Folder to scan; exactly one of `path`, `glob`, `directory`
`pattern`	`"*.csv"`; directory only; 1.2.0+	Pattern inside the selected directory
`recursive`	`false`; batch sources; 1.2.0+	Search directory descendants or enable explicit ** in a glob
`exclude`	`[]`; batch sources; 1.2.0+	Exclusion pattern list
`sort`	`natural`; batch sources; 1.2.0+	`natural`, `lexical`, or `none`
`allow_empty`	`false`; batch sources; 1.2.0+	Accept a valid scan with no matching files
`alias_template`	Absent; 1.2.0+	Generate human-readable names for imported curves
`curve_id_template`	Absent; 1.2.0+	Generate stable workflow identities for links/selectors
`columns`	Required mapping	Passed to `ColumnMapping(**columns)`
`config`	Optional mapping	Passed to `ImportConfig(**config)`; omit the whole key to auto-detect
`aliases`	List of strings; `[]`	Replaces imported curve names, in output order
`curve_ids`	List of strings; `[]`	Stable workflow identities mapped to newly imported runtime curve IDs, in output order

Use distinct aliases for imported curves you intend to select by name. Curve selectors in `models`, `fit.curves`, and `fit.spectrum_weights` resolve a declared workflow identity or unambiguous alias/name first, and an actual runtime curve ID second. Repeated imported names are removed from the alias lookup instead of silently choosing one: give those curves distinct `aliases` or `curve_ids`.

`curve_ids` does not assign the real curve IDs. Each listed string becomes a stable workflow identity that the runner maps to the newly generated runtime ID. Identities must be unique across all import items; duplicates raise an error. Use simple nonempty strings without dots, such as `sample_a` and `sample_b`, so canonical parameter references can split the identity from component/parameter names. These identities can select curves and,

unlike display aliases alone, can occur inside `${identity.component_id.parameter_name}` links. The runner remaps only the first dotted segment of each reference, not arbitrary text or component display names. Identity entries take precedence over aliases with the same spelling; avoid accidental collisions.

For one file, both lists follow the order returned by the importer. In batch imports the lists apply once to the flattened curves from all selected files, not repeatedly to each file. Choose `aliases` or `alias_template`, never both; likewise choose `curve_ids` or `curve_id_template`. Shorter explicit lists leave remaining entries at import defaults; extra entries are unused. For repeatable linked workflows, provide a `curve_ids` entry for every referenced curve and explicit component `id` values. Aliases remain optional human-readable labels.

W6.1 Folder and pattern selection (1.2.0)

Use exactly one source selector per import item. `path: data/scan.csv` remains a literal file; `glob: "data/*.csv"` selects a pattern; `directory: data` with `pattern: "*.csv"` scans a folder. A directory scan with `recursive: true` includes matching descendants. A glob with `recursive: true` enables explicit **, such as `glob: "data/**/*.csv"`; a plain `*.csv` glob remains top-level. Using ** requires `recursive: true`.

`natural` ordering compares numeric portions numerically (`scan2` before `scan10`), case-insensitively, then uses the original relative path as a tie-break. `lexical` sorts relative POSIX path strings case-sensitively. `none` keeps scan order, which can depend on the filesystem and should not be used when sequence order matters. Matching regular files are deduplicated within an import item; separate import items may intentionally import the same file again.

`exclude` patterns without a slash match file basenames at any depth; patterns with path components match the rooted relative path and can use **. Batch pattern matching is case-sensitive on every supported operating system. Scans skip symlink files/directories and hidden entries unless a hidden path component was explicitly requested. A directory/base that is itself a symlink is rejected; a literal `path` may refer to a file symlink. A missing explicit `directory` is an error even when `allow_empty` is true. An existing directory/glob with no retained files is an error unless `allow_empty: true`; allowed empty selections are reported in outcome warnings.

`alias_template` and `curve_id_template` accept these fields:

Field	Meaning
`{stem}`	Input filename without its final suffix
`{filename}`	Input filename including suffix
`{parent}`	Immediate parent-directory name
`{relative_path}`	Input POSIX path relative to the workflow directory
`{relative_stem}`	That relative path without its final suffix
`{file_index}`	One-based selected-file index within this import item
`{curve_index}`	One-based curve index returned from this file
`{curve_name}`	Curve name produced by the importer

Only these plain fields and escaped braces are accepted. Attribute/index access, conversions and format specifiers are rejected. Template-generated identities must be unique, nonempty and suitable for canonical parameter paths; use simple strings without dots, for example `batch_{file_index}_{curve_index}`. Index-based identities are repeatable for a fixed file selection/order; adding or removing a file may change them. For enduring scientific identities, use meaningful explicit `curve_ids` or a suitable filename-derived template. Duplicate identities fail.

W6.2 Complete folder-fit recipe (1.2.0)

Place `scan2.csv` and `scan10.csv` in `data/` beside the workflow. Each has the same `x,y,sigma_y` layout as `examples/gaussian.csv`; copying that example twice provides a runnable trial. A file such as `calibration_skip.csv` is excluded.

```yaml
# example: yaml-folder-batch
schema_version: 3
name: Folder Gaussian fits
imports:
  - directory: data
    pattern: "*.csv"
    recursive: true
    exclude: ["*_skip.csv"]
    sort: natural
    alias_template: "{stem}_{curve_index}"
    curve_id_template: "batch_{file_index}_{curve_index}"
    config: {delimiter: ",", header: true}
    columns: {x: x, y: [y], sigma_y: sigma_y}
preprocess:
  - curves: all
    transformations:
      - {operation: y_multiply, value: 1.0}
    mask_intervals: [[4.5, 5.0]]
    fit_ranges: [[-4.0, 4.0]]
models:
  - curves: all
    components:
      - {function: constant, id: background, is_background: true}
      - function: gaussian
        id: peak
        parameters: {area: 3.0, center: 0.3, sigma: 1.0}
        constraints:
          sigma: {minimum: 0.001, maximum: 5.0}
fit:
  mode: independent
  settings: {solver: local, loss: linear, seed: 1729}
export:
  project: outputs/batch.fitproj
  project_overwrite: false
  bundle: outputs/batch-analysis
  versioned: true
```

Natural ordering imports `scan2.csv` first. Each selected curve receives its own model objects; `curves: all` avoids listing every imported filename in `models`. The project guard refuses to replace `outputs/batch.fitproj` on rerun; choose a new path or deliberately change `project_overwrite` after reviewing what to preserve. The bundle has its own independent overwrite/versioning rules in W12. Keep outputs outside the input scan tree, or exclude their paths, so a recursive rerun does not accidentally import previously exported numeric tables.

`config` fields and dataclass defaults:

Key	Default	Meaning
`delimiter`	`null`	Whitespace when explicitly null; otherwise delimiter string such as `","`, `";"`, or `"\t"`
`decimal`	`"."`	Decimal separator, typically `"."` or `","`

Key	Default	Meaning
`comment_prefixes`	`["#", ";", "%", "!"]`	Leading comment markers after left whitespace is stripped
`header`	`null`	Truthy means first retained row supplies column names; false/null means no header when config is explicit
`encoding`	`utf-8-sig`	Text encoding; decoding replaces invalid characters
`skip_rows`	`0`	Number of physical leading rows to skip before comments/blank filtering

Omitting `config` performs delimiter, decimal, header and leading-row detection. Supplying `config: {}` constructs the defaults above, including whitespace and no header; it does **not** request full auto-detection. Specify `header: true` to select columns by a file's heading names.

`columns` fields and defaults:

Key	Default	Meaning
`x`	`null`	Shared X selector; required with a nonempty `y` unless using `pairs`
`y`	`[]`	Y selector list; produces one curve per Y column
`pairs`	`[]`	Explicit `[[x1, y1], [x2, y2]]` list; overrides shared x/y
`sigma_x`	`null`	Optional X uncertainty selector
`sigma_y`	`null`	Optional Y standard-deviation selector
`weights`	`null`	Optional residual-multiplier selector; objective contribution is multiplier squared times residual squared
`variance`	`null`	Optional Y variance selector, converted to `sigma_y = sqrt(variance)`
`inverse_variance`	`null`	Optional inverse-variance selector
`error_y_plus`	`null`	Upper Y error interval selector; requires `error_y_minus`
`error_y_minus`	`null`	Lower Y error interval selector; requires `error_y_plus`
`error_confidence_level`	`0.95`	Confidence represented by the error intervals; strictly between 0 and 1

Key	Default	Meaning
`as_separated_files`	`false`	Separate-spectrum naming/column-role checks; not an output-file operation
`spectra`	`[]`	Beta: nested per-spectrum mappings, converted to `ColumnMapping` objects

Selectors are exact heading strings or **zero-based integer** positions. `x: 0` selects the first column; `x: "0"` asks for a heading literally named `0`. `y` must always be a list, even for one curve. In no-header files the displayed names are `Column 1`, `Column 2`, etc., while integer positions remain zero-based.

Choose at most one Y-error representation from `sigma_y`, `weights`, `variance`, `inverse_variance`, or the paired asymmetric interval fields. Both asymmetric fields can select the same column for symmetric intervals. `sigma_x` may coexist with a Y-error representation, but the fit is a Y-residual least-squares fit; importing `sigma_x` does not enable orthogonal-distance regression. `weights` values multiply residuals directly; `inverse_variance` values multiply residuals by their square root. `sigma_y` divides residuals by its standard deviation; asymmetric confidence intervals are converted to Gaussian tail scales using `error_confidence_level`, choosing the upper/lower scale according to the fitted prediction's side of the observed value. Nonfinite X/Y, nonfinite or nonpositive Y-error scales, and nonfinite or negative weights are excluded through the curve's effective mask. Zero-weight rows remain available in the worksheet but are excluded from fitted observations, N and degrees of freedom when weights are the active Y-error representation. Imported X errors do not determine the fit mask. Check retained point counts in results.

In 1.2.0, `spectra` accepts a list of nested mappings, each describing its own X/Y and error columns. Use it instead of shared `x`/`y` or `pairs` when spectra in the same file have different error layouts. For example, `columns: {spectra: [{x: 0, y: [1], sigma_y: 2}, {x: 3, y: [4], sigma_y: 5}]}`. The stable 1.1.0 loader does not convert these nested mappings; there, use several explicit import entries for different error mappings.

Examples of alternative mappings:

```yaml
# Several curves sharing one X axis and uncertainty column:
columns: {x: 0, y: [1, 2], sigma_y: 3}
aliases: [sample_A, sample_B]

# Independent X/Y pairs in the same file:
columns:
  pairs: [[0, 1], [2, 3]]
  as_separated_files: true
aliases: [scan_A, scan_B]

# Decimal-comma data with explicitly skipped instrument metadata:
config: {delimiter: ";", decimal: ",", header: true, skip_rows: 4}
columns: {x: wavelength, y: [intensity]}
aliases: [measurement]
```

These snippets replace the corresponding fields within an import item; they are not complete workflow documents.

W7. Models, components, and function catalogue

Each `models` entry supports:

Key	Default	Meaning
`curve`	One selector	One workflow identity, unambiguous alias/name or actual curve ID
`curves`	Beta alternative selector	`all`, `first`, `last`, or a nonempty list of identities/names/IDs
`match`	Beta alternative selector	A name/identity wildcard pattern; must select at least one curve
`name`	Existing model name, initially `Model for <curve name>`	Model label
`components`	`[]`	Components to append in order
`operation_order`	`[]`	Component IDs ordering Multiply/Divide/Convolve operations

Each component supports:

Key	Default	Meaning
`function`	Required	Exact registered function identifier
`name`	Function display name	Human-readable component label; never a parameter path
`id`	Generated `component_…` ID	Explicit stable component ID, unique within its model
`parameters`	Function defaults	Mapping of exact parameter names to initial numeric values
`metadata`	Function configuration defaults	Function-specific options validated by the function definition; notably polynomial/spline settings
`operator`	`add`	`add`, `subtract`, `multiply`, `divide`, or `convolve`
`enabled`	`true`	Disabled components do not contribute to model evaluation
`is_background`	`false`	Marks background components for relevant operations/exports
`constraints`	`{}`	Parameter-specific bounds/fixed/link rules

Built-in function identifiers and parameters:

Identifier	Parameters and defaults	Special rules
`gaussian`	`area: 1`, `center: 0`, `sigma: 1`	`sigma > 0`; FWHM = `2.354820045 * sigma`
`lorentzian`	`area: 1`, `center: 0`, `gamma: 1`	`gamma > 0`; gamma is HWHM; FWHM = `2 * gamma`

Identifier	Parameters and defaults	Special rules
`voigt`	`area: 1`, `center: 0`, `sigma: 1`, `gamma: 1`	Both widths positive; area-normalised convolution
`pseudo_voigt`	`area: 1`, `center: 0`, `fwhm: 1`, `eta: 0.5`	`fwhm > 0`; `eta` in `[0, 1]`; eta is Lorentzian fraction
`constant`	`offset: 0`	Constant function
`linear`	`intercept: 0`, `slope: 0`	`intercept + slope*x`
`polynomial`	`c0 … cN: 0`	`metadata: {order: N}`; default N=2, allowed integer range 0-50
`cubic_spline`	`y0 … yN: 0`	`metadata.x_nodes` supplies at least two finite strictly increasing X nodes

Peak areas and centers are unbounded by default. Positive built-in width lower bounds are the float64 normal minimum (about `2.225e-308`). `Component.create` clamps initial values to the function's default bounds; explicit constraint bounds are subsequently validated against the resulting value. Set sensible positive initial widths and physically appropriate bounds explicitly. All eight functions are usable as any component role. Construction options are distinct from fitted parameters: polynomial `metadata.order` must be an actual integer from 0 through 50 (not `2.0`, a string, or a Boolean). Function-declared option defaults are inserted by `Component.create`; declared option types/ranges are validated before parameter construction. Python/plugin point-initialization callbacks are not YAML stages. Two spline nodes use linear interpolation between the nodes and constant endpoint values outside that interval; three or more use a natural cubic spline with extrapolation.

Global composition first sums all Add functions and subtracts all Subtract functions. Multiply, Divide and Convolve then act on that complete sum. Their fallback order is Multiply, Divide, Convolve, with component ID as a tie-breaker; use explicit IDs and `operation_order` for a chosen order. This is not a nested expression language. The legacy `composition_mode` key is ignored; schemas 1, 2 and 3 use the current global roles. YAML list position does not create parentheses. Convolution uses full FFT integration with absolute uniform X spacing and a single centered crop for consecutive kernels. A finite, strictly monotonic, uniform grid is required at every scale. For even-length signals, kernels are sampled on an odd grid with an exact central lag, removing the half-sample shift; ascending and descending axes give the same physical center. Kernel coordinates retain the input-domain midpoint convention. Choose a sufficiently wide, finely sampled window: finite-window truncation and discretization remain approximations. Masks and fit ranges select rows after full-grid convolution.

Add/Subtract accumulation uses compensated summation independent of random IDs. Explicit IDs remain useful for links and operation ordering, rather than for controlling summation roundoff.

For example, this model evaluates `(constant + gaussian) * response`:

```yaml
models:
  - curve: reference
    operation_order: [response]
    components:
      - {function: constant, id: baseline, parameters: {offset: 0.1}}
      - {function: gaussian, id: band, parameters: {area: 3, center: 0.3, sigma: 1}}
      - function: linear
        id: response
        operator: multiply
        parameters: {intercept: 1, slope: 0}
        constraints:
          intercept: {fixed: true}
          slope: {fixed: true}
```

W7.1 Shared selectors and preprocessing (1.2.0)

Models, preprocessing steps and the fit plan share curve selection. Use one of `curve: identity`, `curves: [identity_a, identity_b]`, `curves: all`, `curves: first`, `curves: last`, or `match: "sample_*"`. A pattern selects matching curve names and declared aliases/identities using case-sensitive shell matching; it does not scan autogenerated runtime UUID strings. All/pattern selection follows project order; an explicit list retains its order. Unknown selectors, duplicate resolved curves or empty selections raise errors. `all`, `first` and `last` are values of `curves`, not implicit substitutions inside canonical parameter links. A curve literally named `first` remains selectable with `curve: first` or `curves: [first]`.

A model entry requires a selector. Preprocessing and fitting default to all curves when no selector is supplied. Every selected model is constructed independently; explicit component IDs may repeat across different models, while each model's IDs remain unique. For a sequential fit, use `models: [{curves: first, ...}]` to prepare the source only. The first sequential curve is not automatically fitted; subsequent compatible targets receive propagated components and retain any deliberately existing target-local functions (see W10).

Each `preprocess` entry permits selectors plus the following operations. Steps run in listed order after imports and before model construction. Within a step, the order is transformations, clearing masks, masking intervals, unmasking intervals, then assigning fit ranges. Intervals therefore refer to the transformed X axis.

Key	Contract
`transformations`	Ordered list of supported scalar or formula operations
`clear_masks`	Boolean; true clears existing named masks on selected curves
`mask_intervals`	List of finite `[lower, upper]` pairs; inclusive masks
`unmask_intervals`	List of finite pairs; clear matching active-mask exclusions
`fit_ranges`	List of finite pairs; replaces the fitting-interval union

Scalar transformation `operation` values are `y_add`, `y_subtract`, `y_multiply`, `y_divide`, `x_add`, `x_multiply`, `normalize_max`, `normalize_area`. The first six require a finite numeric `value`; normalizations need no value. Division by zero and invalid normalization inputs fail. `custom_formula` requires `axis: x` or `axis: y` plus `formula`; `column_formula` requires `target` and `formula` and can use retained `c1`, `c2`, ... columns. Both use the restricted expression vocabulary in W9, never arbitrary Python. For example:

```yaml
preprocess:
  - match: "sample_*"
    transformations:
      - {operation: x_add, value: 0.25}
      - {operation: custom_formula, axis: y, formula: "2*y"}
    mask_intervals: [[531.0, 532.0]]
    fit_ranges: [[500.0, 550.0]]
```

Unmasking one named mask does not override exclusions in other masks or invalid numeric rows. Fit ranges select a union of intervals; they do not delete data. Transformations preserve original arrays and are retained in the saved project. Background subtraction and curve-to-curve operations still require Python or the desktop. Static validation checks shapes before preprocessing; execution can still fail for actual data, formula domains or unavailable retained columns.

W8. Parameter constraints and links

For a component's `constraints`, keys must be existing parameter names:

```yaml
constraints:
  center: {minimum: -1.0, maximum: 2.0}
  sigma: {minimum: 0.001, maximum: 5.0, fixed: false}
  area: {fixed: true}
```

Rule	Default when omitted	Contract
`minimum`	Retain the function parameter's lower bound	Numeric inclusive lower bound
`maximum`	Retain the function parameter's upper bound	Numeric inclusive upper bound
`fixed`	`false`	Exclude the parameter from optimisation when true
`link`	`null`	Restricted mathematical expression with canonical parameter references
`link_scope`	`relative`	`relative` or `absolute`; copy behaviour, not expression shorthand
`link_reference_scopes`	`[]`	Per-reference copy behaviour; if nonempty must match expression reference occurrences
`link_relation`	`equal`	`equal`, `lower`, `upper`, or `similar`
`link_tolerance`	`0.0`	Nonnegative finite tolerance; must be positive for `similar`
`link_tolerance_mode`	`absolute`	`absolute` or `percent`; percentage must be less than 100

The initial parameter must lie inside its final minimum/maximum interval, and minimum must not exceed maximum. Equality-linked parameters are computed from their source expression; they are not free. Bound-linked parameters remain free unless fixed. `lower` means target at least source; `upper` means target at most source; `similar` keeps target within source ± tolerance. Percentage tolerance uses `abs(source)*percentage/100`. Dynamic and static bounds are intersected; empty intersections and link cycles are errors. Non-equality links require one direct `${curve_id.component_id.parameter_name}` reference, without arithmetic. Equality links may combine several canonical references in a formula.

Stable links on fresh imports: declare `imports[].curve_ids` to give every referenced curve a stable workflow identity, and explicit component `id` values. The runner replaces the identity prefix in each `${...}` reference with the actual imported curve ID before resolving links. For example, with `curve_ids: [sample_a]` and component `id: peak`, `${sample_a.peak.center}` is a valid reference. A display alias alone is not expanded inside a parameter expression. Do not substitute component names for IDs or invent a template language. The entire project parameter graph is resolved before fitting or exporting; unknown references, invalid links and cycles therefore fail even when `fit` is absent. Use global mode when the linked source and target need joint optimization; a global plan does not create links automatically.

W8.1 Complete linked global-fit example

Save this as `linked.yml` beside `gaussian.csv` from the repository example. It imports that file twice to keep the demonstration immediately runnable; for a real replicate analysis, replace the second path with the second measurement. Both peak centers are fitted through the source parameter. Background offsets, areas and widths remain separate.

```yaml
# example: yaml-linked-global
schema_version: 2
name: Linked Gaussian centers
imports:
  - path: gaussian.csv
    config: {delimiter: ",", header: true}
    columns: {x: x, y: [y], sigma_y: sigma_y}
    aliases: [Reference spectrum]
    curve_ids: [sample_a]
  - path: gaussian.csv
    config: {delimiter: ",", header: true}
    columns: {x: x, y: [y], sigma_y: sigma_y}
    aliases: [Second spectrum]
    curve_ids: [sample_b]
models:
  - curve: sample_a
    components:
      - {function: constant, id: background, is_background: true}
      - function: gaussian
        id: peak
        parameters: {area: 3.0, center: 0.3, sigma: 1.0}
        constraints:
          sigma: {minimum: 0.001, maximum: 5.0}
  - curve: sample_b
    components:
      - {function: constant, id: background, is_background: true}
      - function: gaussian
        id: peak
        parameters: {area: 3.0, center: 0.3, sigma: 1.0}
        constraints:
          center: {link: "${sample_a.peak.center}"}
          sigma: {minimum: 0.001, maximum: 5.0}
fit:
  mode: global
  curves: [sample_a, sample_b]
  settings: {solver: local, loss: linear, seed: 1729}
export:
  project: outputs/linked.fitproj
  bundle: outputs/linked-analysis
  versioned: true
```

Run `curvemole validate linked.yml`, then `curvemole run linked.yml --json`. Inspect `success` and the saved result parameters. The two resolved center values must agree, and the target estimate is equality-linked rather than free. This example requires CurveMole 1.1.0: 1.0.1 did not remap workflow curve identities.

`link_scope` and `link_reference_scopes` govern retargeting when a model is copied through the corresponding Python/desktop operations. They do not change how a reference is resolved during this YAML runner.

W9. Custom formula functions

Define formulas before using their identifier as a component `function`:

```yaml
custom_functions:
  - identifier: exponential_decay
    display_name: Exponential decay
    formula: "amplitude * exp(-x / tau)"
    kind: generic
    defaults: {amplitude: 1.0, tau: 2.0}
    bounds:
      tau: [0.001, 1000.0]
    derived:
      half_life: "tau * log(2)"
```

Required keys are `identifier` and `formula`. `display_name` defaults to the identifier; `kind` defaults to `generic`; `defaults`, `bounds` and `derived` default to empty mappings. `kind: peak` classifies a function as a peak but does not rename parameters or automatically establish graphical parameter roles. Formula variables other than `x` become parameters in alphabetical order; unspecified defaults are 1.0 and unspecified bounds are unbounded. Each bounds entry is a two-number `[minimum, maximum]` list. `derived` maps result names to scalar parameter formulas; derived formulas have no X array. A nonfinite derived result is reported as unavailable. Invalid symbols or expressions can raise an expression error and fail an operation; use only parameters declared by the main formula and check each derived formula on representative parameter values. In the current release edition, a formula may replace a non-builtin registration, but builtin identifier collisions are rejected. Choose a distinct identifier for each custom function. A workflow without an explicit Python registry uses an isolated registry; a supplied registry remains owned by its caller. Formula definitions are not arbitrary Python and need no executable-plugin approval.

Custom-function persistence: 1.2.0 copies safe formula definitions into `Project.custom_functions`, using the project-format `derived_formulas` key, and `dump_workflow` writes them back as YAML `custom_functions` with `derived`. Saved projects/model exports therefore retain these safe definitions for later sessions. This does not embed executable plugin source. The 1.1.0 runner only registered YAML functions in memory and did not persist their definitions; keep its original recipe and re-register the definitions before refitting that older project in a fresh session.

The exact expression vocabulary is:

- Arithmetic `+`, `-`, `*`, `/`, `**`, `%`; unary `+`, `-`, `not`.
- Comparisons `==`, `!=`, `<`, `<=`, `>`, `>=`; `and`, `or`; conditional `a if condition else b` (also works elementwise).
- Constants `pi`, `e`, `inf`.
- Functions `abs`, `sqrt`, `exp`, `log`, `log10`, `sin`, `cos`, `tan`, `arcsin`, `arccos`, `arctan`, `sinh`, `cosh`, `tanh`, `erf`, `erfc`, `minimum`, `maximum`, `clip`, `where`, `heaviside`.

Use `**` for powers, never `^`. Calls use positional arguments only, for example `where(x > 0, amplitude * exp(-x/tau), 0)`. Attribute access (`np.exp`), subscripts, imports, comprehensions, arbitrary function calls, string literals and Python execution are rejected. Keep formulas finite across the actual X domain.

W10. Fit plan

Key in `fit`	Default	Meaning
`mode`	`independent`	`independent`, `sequential`, `global`, or legacy `copy`
`curves`	All imported curves	Nonempty selector list; 1.2.0+ also accepts `all`, `first`, `last`
`curve`, `match`	Absent; 1.2.0+	Alternative single-curve or wildcard selection; use only one selector

Key in `fit`	Default	Meaning
`settings`	`{}`	`FitSettings` keyword arguments, listed below
`spectrum_weights`	`{}`	Curve selector → finite positive weight
`equal_contribution`	`false`	Divide each curve's weighted residual by the square root of its retained point count

Independent fits use each curve's own model and starting parameters.

Sequential mode uses the first selected curve as an already prepared source; it does not fit that source. The source must have a nonempty model with values already suitable for propagation. Later curves are processed in `fit.curves` order. The preceding model is cloned with its function identities, values, structural metadata, bounds, fixed state, links, enabled/background flags and composition. Source component IDs select which functions continue to propagate. Missing components are added to each target by ID; an existing target component with that ID is retained, rather than overwritten, and target-local components remain. For the usual propagate-and-fit case, define the model only on the source and leave target models empty. A one-curve sequential plan succeeds with no fit outputs because there are no subsequent curves.

Default sequential safeguards are active even for the plain `FitPlan` created by YAML. Residual monitoring pauses when normalized RMSE exceeds the previous value by both a factor of 2.5 and an absolute increase of 0.02. Parameter monitoring pauses at a normalized free-parameter change of at least 0.75; its denominator uses the larger value magnitude or a parameter-specific scale near zero, not simply the starting value. Failures and nonconvergence also pause. Propagation flags, monitoring thresholds, exclusions and durable pause/resume are not YAML fields; use Python `SequentialFitPlan` or desktop sequential controls to configure them. A pause returns `success: false`; the saved project can retain copied/fitted state for review, but the YAML runner cannot interactively continue. Inspect the project or use Python before continuing the sequence. `copy` is an accepted legacy mode but follows the independent execution branch; use `sequential` for actual propagation.

Global fits solve all selected curves together. This alone does not make their parameters shared; sharing requires canonical links using declared stable curve identities as shown in W8. Spectrum weight multiplies residuals by `sqrt(weight)`; the default weight is 1.0. `equal_contribution` is point-count scaling, not automatic noise estimation. Use `use_data_errors: false` to ignore imported uncertainties/weights during fitting, including their exclusion of unusable uncertainty rows; intrinsically invalid X/Y remain excluded.

W11. Complete solver settings and defaults

Only the following built-in settings are passed to `FitSettings`; spell names exactly. An unrecognised setting raises `TypeError`. Do not add GUI-only controls such as propagation thresholds or Monte Carlo replicate counts here.

Built-in optimizers use dimensionless coordinates based on parameter and data scales; numerical Jacobians, automatic DE limits and bound diagnostics use those scales rather than a floor of one physical unit. Positive built-in widths have a float64 normal minimum (about `2.225e-308`), rather than machine epsilon. Explicit bounds remain preferable for difficult searches. Extreme overflow, underflow, nonidentifiability and poor starting values still require scientific judgment. Optimizer coordinate tolerances, including explicit `x_scale`, apply after this normalization; supplied Nelder-Mead simplex vertices remain physical parameters.

Setting	Default	Meaning/accepted values
`solver`	`local`	`local`, `trf`, `dogbox`, `lm`, `differential_evolution`, `nelder_mead`, `powell`, `lbfgsb`, or an enabled registered plugin solver ID
`local_method`	`auto`	`auto`, `trf`, `dogbox`, `lm`; local solve/refinement method
`loss`	`linear`	`linear`, `soft_l1`, `huber`, `cauchy`
`f_scale`	`1.0`	Finite positive robust-loss residual scale
`max_nfev`	`1000`	Positive integer evaluation limit
`ftol`	`0.0000000001`	Positive least-squares cost tolerance, above machine precision
`xtol`	`0.00000001`	Positive least-squares step tolerance, above machine precision
`gtol`	`0.0000000001`	Positive least-squares gradient tolerance, above machine precision
`x_scale`	`jac`	`jac` or a finite positive scalar
`seed`	`1729`	Integer from 0 through 4294967295
`workers`	`1`	Positive integer; multiple processes for independent multi-curve fits
`confidence_level`	`0.95`	Parameter interval confidence, strictly between 0 and 1
`absolute_sigma`	`null`	Auto-select absolute covariance scaling from imported Y-error availability; explicit true/false overrides when data errors are used
`use_data_errors`	`true`	Boolean: use imported errors/weights; false forces residual-scaled covariance
`de_maxiter`	`400`	Positive integer DE generation limit
`de_popsize`	`15`	Positive integer DE population multiplier
`de_lower_percent`	`50.0`	Finite positive percentage for automatic lower search limits
`de_upper_percent`	`50.0`	Finite positive percentage for automatic upper search limits
`de_mutation_min`	`0.5`	Minimum mutation factor; `0 <= min <= max < 2`
`de_mutation_max`	`1.0`	Maximum mutation factor
`de_recombination`	`0.7`	Recombination fraction in `[0, 1]`

Setting	Default	Meaning/accepted values
`de_tol`	`0.01`	Finite nonnegative relative DE tolerance
`de_atol`	`0.0`	Finite nonnegative absolute DE tolerance
`nm_xatol`	`0.0001`	Finite positive Nelder-Mead X tolerance
`nm_fatol`	`0.0001`	Finite positive Nelder-Mead objective tolerance
`nm_adaptive`	`false`	Adaptive Nelder-Mead option
`nm_initial_simplex`	`null`	Finite `(N+1) × N` numeric list for N free parameters; do not include fixed/equality-linked parameters
`minimize_maxiter`	`0`	Nonnegative integer; 0 keeps the minimizer's automatic iteration limit
`powell_xtol`	`0.0001`	Finite positive Powell X tolerance
`powell_ftol`	`0.0001`	Finite positive Powell objective tolerance
`lbfgsb_ftol`	`2.220446049250313e-9`	Finite positive L-BFGS-B objective tolerance
`lbfgsb_gtol`	`0.00001`	Finite positive L-BFGS-B gradient tolerance
`lbfgsb_maxls`	`20`	Positive integer maximum line-search steps
`lbfgsb_maxcor`	`10`	Positive integer stored correction pairs

`local` with `auto` chooses an appropriate least-squares method. Explicit LM is incompatible with bounds and robust loss. Differential Evolution finds an initial solution then locally refines it. Finite user bounds take priority over its automatic search intervals; set scientifically meaningful finite bounds for difficult searches. Minimisers and local methods have different convergence properties; inspect results rather than relying only on solver termination. Multiple-worker independent fits support built-ins and safe custom formulas; executable plugin model functions cannot be reconstructed automatically in worker processes. `workers` is not a general parallelisation switch for every mode.

W12. Output contract and reruns

New fit results record `provenance`: original model/constraint snapshots, input content hashes, ordered parameters, initial solver coordinates, scales, resolved solver bounds/method, settings, seeds, function/plugin fingerprints, CurveMole source identity, Python/NumPy/SciPy versions and backend/thread details. Profile and resampling outputs also retain their baseline identity and actual analysis configuration. Historical results created before this feature have no reconstructible original state. Keep original data and scripts: hashes identify inputs but do not store a separate copy of every historical dataset. For checked initial-state replay use the Python `Fitter.replay` interface. Rerunning YAML or fitting the current optimum is a new calculation.

Key in `export`	Default	Meaning
`project`	Absent	Save the generated project; parent directories are created and suffix becomes `.fitproj`
`project_overwrite`	`true`; 1.2.0+	Allow replacing `export.project`; false refuses an existing project
`bundle`	Absent	Export the standard analysis bundle with its default selections
`versioned`	`false`	Add UTC `export-YYYYMMDD-HHMMSS` subdirectory under the bundle directory
`overwrite`	`false`	Request updating existing bundle files, subject to ownership checks

These are the only YAML export keys. `project_overwrite` is an addition in 1.2.0; the stable baseline always uses its existing direct project-save behaviour. Controls such as delimiter, selected bundle items, portable project copies, saved uncertainty samples, and current-view plot exports are Python/desktop features, not accepted YAML options.

Project and bundle replacement differ. `export.project` saves directly with `save_project`; it can replace an existing project without consulting `export.overwrite`. In 1.2.0, set `project_overwrite: false` to refuse replacement, or choose a distinct project path when a previous file must be kept. `versioned` applies only to the bundle, not to the project filename.

Bundle exports refuse conflicting files unless updating is requested and CurveMole can establish ownership. `overwrite: true` is not permission to replace arbitrary existing files. Each YAML run starts a fresh project and therefore normally lacks the ownership record from a previous run; fixed-directory reruns may still be refused even with `overwrite: true`. A readable bundle manifest is not sufficient proof of ownership. Prefer a new directory or `versioned: true`; timestamp resolution is one second, so avoid simultaneous exports into the same versioned parent. Use Python with the saved project when you need its retained ownership state for a controlled update.

Exports are successive operations, not one transaction. If bundle export fails, a project saved earlier can remain. If no `export` is present, the CLI/desktop worker only retains its in-memory project for that run. `run --json` is a summary; use the saved `.fitproj` or exported result tables/JSON for detailed parameters, per-curve arrays, covariance, diagnostics and warnings. Retain the recipe with the saved results. Safe custom formulas are persisted by current workflows as described in W9; executable plugin source still needs its own archived manifest, module and dependencies.

W13. Trusted executable plugins

List manifest files, not Python modules or plugin directories:

```yaml
plugins:
  - plugins/lab_tools.curvemole-plugin.json
```

For the command line, supply each manifest's **plugin identifier**, not a filename:

```
curvemole run workflow.yml --trust-plugin org.example.lab_tools --json
```

The identifier above is a placeholder; replace it with the identifier declared inside your actual manifest. For the desktop runner, first review and enable the same local plugin through **File > Plugin Manager**. The worker verifies that the approved manifest/source fingerprint still matches at launch. Merely placing a path in `plugins` does not authorise execution. Plugins run Python with the user's permissions. Safe formula functions are a different mechanism.

The workflow does not export plugin source. Keep manifest/source versions and dependencies with the research record. Plugin-specific solver semantics require that plugin's own documentation in addition to this core contract.

W14. Python workflow functions and round-trip limits

The workflow module also exposes:

```
from curvemole.core.workflow import (
    WorkflowOutcome, dump_workflow, load_workflow, run_workflow, validate_workflow,
)

value = load_workflow("workflow.yml")       # dict; checks root/schema
errors = validate_workflow(value)           # list[str]; static checks, no data import
if errors:
    raise ValueError("; ".join(errors))
outcome = run_workflow("workflow.yml")      # WorkflowOutcome
print(outcome.project.name)
print(outcome.result.success if outcome.result else None)
print(outcome.outputs)                      # list[pathlib.Path]
# In 1.2.0: outcome.imported_files and outcome.warnings are also available.
```

Exact runner signature: `run_workflow(path, *, trust_plugins: set[str] | None = None, registry: FunctionRegistry | None = None) -> WorkflowOutcome`. Passing a registry permits caller-provided functions. `dump_workflow(project, path) -> pathlib.Path` writes YAML containing imports and model/component specifications. Current dumping emits schema 3; the stable 1.1.0 dumper emits schema 2. `WorkflowOutcome` has `project`, `result` (`FitResult | None`), and `outputs` (`list[Path]`). Beta adds `imported_files` (`list[Path]`) and `warnings` (`list[str]`) with empty-list defaults, preserving the original three-positional-argument constructor.

`dump_workflow` is a starting recipe, **not a complete project round trip**. It does not embed data arrays, transformed data, masks, preprocessing stages, results, a fit plan, exports or plugin declarations. Beta includes persisted safe custom-function definitions; the stable 1.1.0 dumper did not. It uses original source paths and saved import mappings, or a fallback `data/<curve-name>.csv` path that it does not create. It does not export the original series structure. A curve originally selected from one of multiple Y columns must have its dumped mapping checked: re-importing a stored multi-Y mapping can produce additional curves. Dumped imports include each saved curve ID in `curve_ids`, and model selectors use those IDs; on replay the runner maps them to new runtime IDs and remaps parameter links. This preserves link addressing for correctly matched imports, but does not embed or guarantee the source data. Review each import mapping and identity list, insert any missing plugin definitions and the desired `fit` and `export`, and check constraints before running. Use `.fitproj` for preserving the actual project state.

W15. Using the AI YAML audit

The downloadable **AI YAML automation audit** is a standalone copy of this contract with instructions for an AI author. Upload `ai-yaml-audit.md` to the AI with your requested analysis, software version, file header/column layout, representative numeric rows, initial values/bounds and desired output paths. The AI does not need the CurveMole source for

workflows covered here. An audit is a version-specific implementation inventory and usage contract, not a security certification or a guarantee that generated scientific choices are correct.

Example prompt:

> Use the attached CurveMole YAML audit as the API contract. I am running CurveMole 1.2.0. Create one schema-3 workflow for the attached column layout. Use only documented fields and function names. Ask for missing scientific assumptions. Return the complete YAML, the validate/run commands, the expected outputs, and any limitations. If the task needs capabilities missing from YAML, explain why and request the Python audit rather than inventing keys.

For scripts, supply `ai-python-audit.md` instead. Supply both audits when the task combines Python orchestration and YAML execution. Review the generated paths, models, units, error interpretation and overwrite behaviour; then validate and run on a small representative dataset. Record the actual version, YAML, input data and output with the result. For another CurveMole version, obtain matching audits before asking the AI to adapt a workflow.

Appendix H. Using the automation audits with an AI

H.1 What the two audits are for

The two downloadable Markdown documents are self-contained context packs for creating CurveMole automations with an AI assistant. Each records the interfaces verified against the stable 1.1.0 API and the marked 1.2.0 workflow additions, including exact names, configuration fields, defaults, examples, return values and practical limits. Here, "audit" means an inspection of the implemented automation interface. It does not mean a security certification, an executable script, or an assessment of your scientific model. Neither document requires an AI service or uploads data from CurveMole.

Desired output	Context to attach	Detailed manual reference
Python script, notebook or custom batch logic	Python automation audit	Appendix F
A `.yml` or `.yaml` file for `curvemole run` or the desktop Run automation button	YAML automation audit	Appendix G
Advice on which interface can perform a task	Both audits	Appendices F and G

Python is appropriate for loops, conditional processing, existing-project editing, unsupported transformations, explicit uncertainty analyses or selective exports. In 1.2.0, YAML also supports folder/pattern imports, shared curve selectors, masking, fit ranges and the documented scalar/formula preprocessing. Use schema 3 for those additions. YAML follows a declarative sequence of imports, preprocessing, model construction, one fit and exports. YAML has no general task graph, loop, conditional, Python execution or uncertainty stage. Loading a plugin does not add arbitrary new YAML stages. The desktop Run automation button runs an isolated workflow with its own imported data; it does not edit the open GUI project.

H.2 Download and version matching

For 1.2.0, use the versioned audits attached to the exact release. The exact stable-source editions are also available here:

- Download Python audit
- Download YAML audit

Save each page as a `.md` file. On GitHub, **Download raw file** also saves the Markdown source. In a source checkout they are in `docs/`; Python distributions and desktop packages include them with the bundled documentation. The read links above also work from the offline manual.

Check `curvemole --version` and the CurveMole version declared inside the audit. These URLs are pinned to this stable tag. Keep the matching documents with your analysis; historical editions remain available at their original release tags. If versions differ, ask the AI to flag the mismatch and verify the relevant contracts before producing an automation; do not assume a document for another release is current. These files supplement the current-version manual; they do not alter previously published release assets.

H.3 Give the AI a concrete request

Attach the actual Markdown file to the AI conversation, rather than only a link. Use both when the task may exceed YAML's supported stages. Give the AI the following information, using an anonymized sample or a description when full data are not needed:

1. Installed CurveMole version, operating system, and whether you will run Python, the CLI or the desktop Run automation button.
2. Input paths, delimiter, header/leading rows, x and y column names, units, number of spectra, and uncertainties or weights with their meaning and confidence.
3. Function identifiers or formulas, initial values, bounds, fixed parameters, links, background roles, masks and transformations required by the analysis.
4. Independent, sequential or global fit requirements, weighting and solver choices.
5. Output paths, formats, files that must be preserved, and expected rerun behavior.

Do not let an AI choose undocumented identifiers or silently decide scientific constraints from filenames. If the task needs a plugin, provide that plugin's own API documentation and version: the audits cannot describe every user-installed extension.

A YAML request can be written as:

```
Use the attached audit for CurveMole 1.2.0. Generate a complete schema-3
workflow and the CLI commands to validate and run it. Import *.csv from data/
in natural filename order. Files have a comma separator, a header and x, y, sigma_y
columns; sigma_y means one standard deviation. Apply a constant background and
one Gaussian to all curves and fit independently.
Ask for any missing initial values, bounds or scientific choices. Save the project
and analysis bundle in new paths, with project_overwrite: false. Explain how to
check the JSON success value and imported_files list.
Use only documented fields; flag requirements that need Python instead.
```

A Python request can be written as:

```
Use the attached Python automation audit as the interface reference for CurveMole
1.1.0. Write a complete runnable script, including imports, for the input layout
and model I describe below. Use the documented core API and exact argument names.
Check fit success and warnings, keep input files unchanged, and save a new project
and an analysis bundle. Include installation and execution commands. Ask for
missing scientific choices, state assumptions and flag unsupported operations.
Do not invent convenience methods or assume GUI state is available to the script.
```

Add your actual data layout and analysis choices to either prompt. Ask the AI to identify the reference sections for the APIs or keys it uses. The documents are context for generation; the generated result still needs execution and inspection.

H.4 Review, execute and inspect

For YAML, save the result beside the referenced input files, or adjust the paths. Relative paths are resolved from the YAML file's directory. In the CurveMole Python environment, run:

```
curvemole --version
curvemole validate workflow.yml
curvemole run workflow.yml --json
```

Beta validation checks supported keys, nested structure, selector syntax, fit settings and safe preprocessing formulas. It does not read the input data or prove that columns, function-specific parameters, links or convergence are correct. Review the complete schema in Appendix G and run a small representative case. CLI exit status 0 alone does not establish convergence: read the JSON `success` value. An absent fit produces `success: null`. Runtime exceptions and unsuccessful optimization are different outcomes.

For Python, install the matching CurveMole package in a Python 3.12+ environment and run `python script.py`. Review the script's imports, paths and outputs first, then run it on a small representative dataset. Python's relative paths normally follow the working directory unless the script explicitly anchors them to its own file. Use `if __name__ == "__main__":` when using worker processes. Check `FitResult.success`, `cancelled`, `message` and `warnings`, inspect residuals and parameter bounds, and confirm the saved files contain the intended curves and fit.

For the desktop, choose the YAML file with **Run automation** or **Choose automation**. It uses the same path rules and schema. Review executable plugins in **File > Plugin Manager** before using them. Stopping a run does not remove outputs already written. Project saves and bundle exports have different overwrite rules; see Appendices F and G before rerunning with existing output paths.

Keep the generated script or YAML, the exact audits supplied to the AI, the software version, input data, settings, random seeds and output files together. Correct execution and fit convergence are necessary checks; the scientific interpretation still requires the checks in Section 1.4.

Appendix I. Release integrity and resource limits

Inverse-variance weights follow reversible Y transformations: multiplying Y by `a` divides its inverse variance by `a**2`. Direct arbitrary residual multipliers retain their declared multiplier semantics. Undo restores the original weights. Zero-derivative transformations do not create infinite statistical information.

Per-spectrum numeric exports use 17 significant digits. The Python `export_spectra(..., overwrite=False)` default refuses existing destinations; the GUI asks before replacing files. Symbolic-link destinations are refused even when overwrite is requested. Numeric conversions are staged before publication. A multi-file export can still be partially published if a filesystem commit fails; existing files are never implicitly replaced.

Custom project functions are scoped to the current project. Custom definitions loaded from projects, the library or workflows may not reuse builtin identifiers. The low-level registry `register(..., replace=True)` remains an explicit extension API; a supplied workflow registry is owned by its caller. The default workflow registry and automatic export registry are isolated from the global registry.

Project opening validates and reads in a cancellable worker while the GUI processes events. The default `ArchiveLimits` budget is 100,000 ZIP members, 1 GiB total expanded content, 256 MiB per numeric array and 32 MiB of manifest. Array dimensions are checked from the header before allocating; the current loader accepts at most eight array dimensions. The Python API reader and validator can explicitly receive a larger `limits=ArchiveLimits(...)` for trusted large data. Project saves still validate against the default budget; raising a read limit alone does not enlarge the save budget. A rejected read-only mutation has no side effects; copying a fit invalidates its targets and transitive linked dependents, retaining historical results as outdated.

Safe mathematical expressions accept at most 16,384 source characters, 2,048 AST nodes and depth 64. Integer results are bounded to 4,096 bits; powers are checked before computation. Arrays and broadcasts are limited to 10,000,000 elements, with an aggregate evaluation-work budget of 50,000,000 elements. These limits bound the built-in expression interpreter; they do not sandbox trusted Python plugins or impose a hard operating-system deadline on their code.

Local plugin approval fingerprints include Python helper sources. Flat plugin folders exclude other plugins' entry points unless imported; package plugins also cover their package resources. Extra resources in a flat folder can be declared as a manifest `files` list of relative paths. Dependencies installed outside the plugin source folder remain part of the trusted Python environment. Changed fingerprints require renewed approval. Desktop automatic updates require a valid SHA-256 release digest; otherwise use the release's manual download page.

Appendix J. Scientific audit corrections

The seven findings checked on 2026-11-02 are corrected in this release edition. Earlier release files retain their previous behavior. Independent regression tests compare calculations with analytic references and checked archive replay.

J.1 Units, parameter scales and numerical controls

Built-in optimizers use dimensionless coordinates based on parameter and data scales; numerical Jacobians, automatic DE limits and bound diagnostics use those scales rather than a floor of one physical unit. Positive built-in widths have a float64 normal minimum (about `2.225e-308`), rather than machine epsilon. Explicit bounds remain preferable for difficult searches. Extreme overflow, underflow, nonidentifiability and poor starting values still require scientific judgment. Optimizer coordinate tolerances, including explicit `x_scale`, apply after this normalization; supplied Nelder-Mead simplex vertices remain physical parameters.

The Gaussian reference recovers center 0.4 and sigma 0.8 after coherent X/area rescaling by 1, `1e-9` and `1e-18`, for both local and DE fits. The irregular `[0,1,2,4,8]*1e-18` grid is now rejected by FFT validation.

J.2 Nonlinear formula uncertainty propagation

Nonlinear formula derivatives use local, scale-aware perturbations and Richardson extrapolation. For `z=y**3`, the audited tiny-value example now agrees with `sigma_z=abs(3*y**2)*sigma_y`. Propagation remains a first-order local approximation; input correlations and uncertainties in other columns are not jointly propagated.

The previous `y=1e-9`, `sigma_y=1e-11` cubic error was about `3.66685e-22`; the corrected propagated value is `3e-29`. Inverse-variance propagation uses the same corrected derivative.

J.3 Nonlinear linked-parameter uncertainty

Nonlinear linked-parameter gradients use bound-aware, scale-aware three-point differences and Richardson extrapolation before covariance propagation. The cubic link agrees with `SE(b)=abs(3*a**2)*SE(a)` at the audited small scales. These remain local covariance uncertainties; strong nonlinearity can require resampling.

J.4 Convolution alignment on even-length grids

Convolution uses full FFT integration with absolute uniform X spacing and a single centered crop for consecutive kernels. A finite, strictly monotonic, uniform grid is required at every scale. For even-length signals, kernels are sampled on an odd grid with an exact central lag, removing the half-sample shift; ascending and descending axes give the same physical center. Kernel coordinates retain the input-domain midpoint convention. Choose a sufficiently wide, finely sampled window: finite-window truncation and discretization remain approximations.

The Gaussian signal/kernel case now recovers center 0.25 with 100 or 101 samples and either axis direction, instead of compensating a half-sample offset.

J.5 Cancellation and fresh component IDs

Enabled Add/Subtract terms use Neumaier compensated summation and ordering that is independent of random IDs. All permutations of `1e16+1-1e16` retain 1; the associated fit recovers the small offset 1. Ordinary float64 limits still apply.

J.6 Profile confidence regions

Profile threshold crossings are refined by bracketed refits. `confidence_regions` contains every connected region detected by the scan; `interval` selects the region containing the fitted estimate and never bridges rejected gaps. Disconnected sets are marked **Critical**. `warnings` reports incomplete scans or unrefined endpoints; inspect every region and extend the scan where needed. A finite grid can still miss a narrow remote mode. The default Gaussian-location example now agrees with the analytic 95% interval, approximately +/- 1.959964 standard errors.

In the `y=a**2` reference the confidence set has separate negative and positive regions. Zero is excluded and the assessment is Critical. Failures are reported; failed grid points never join accepted regions across an unknown gap.

J.7 Reproducibility and saved analysis provenance

New fit results record `provenance`: original model/constraint snapshots, input content hashes, ordered parameters, initial solver coordinates, scales, resolved solver bounds/method, settings, seeds, function/plugin fingerprints, CurveMole source identity, Python/NumPy/SciPy versions and backend/thread details. Profile and resampling outputs also retain their baseline identity and actual analysis configuration. Historical results created before this feature have no reconstructible original state. Keep original data and scripts: hashes identify inputs but do not store a separate copy of every historical dataset.

`Fitter.replay(result, curves, verify_environment=True)` reconstructs the saved initial models and plan, checks data and function/solver identity, and rejects a changed numerical environment by default. `verify_environment=False` permits an explicit comparison across environments; it does not promise exact equality. A direct new fit from the saved optimum remains a new calculation. Replay of the audited DE result after archive save/load preserves its initial vector, resolved bounds and final parameters. Fixed/adaptive Monte Carlo and bootstrap comparisons also match between one and two workers in the tested environment. These checks do not promise identical floating-point results or random streams across versions, machines, changed plugins or different thread/backend settings.